CareEdge ANALYTICS & ADVISORY

A subsidiary of CARE Ratings Ltd.

Industry Research Report on Rice Industry in India

16 June 2025

Disclaimer

This report is prepared by CARE Analytics and Advisory Private Limited (CareEdge Research). CareEdge Research has taken utmost care to ensure accuracy and objectivity while developing this report based on information available in CareEdge Research's proprietary database, and other sources considered by CareEdge Research as accurate and reliable including the information in public domain. The views and opinions expressed herein do not constitute the opinion of CareEdge Research to buy or invest in this industry, sector or companies operating in this sector or industry and is also not a recommendation to enter into any transaction in this industry or sector in any manner whatsoever.

This report has to be seen in its entirety; the selective review of portions of the report may lead to inaccurate assessments. All forecasts in this report are based on assumptions considered to be reasonable by CareEdge Research; however, the actual outcome may be materially affected by changes in the industry and economic circumstances, which could be different from the projections.

Nothing contained in this report is capable or intended to create any legally binding obligations on the sender or CareEdge Research which accepts no responsibility, whatsoever, for loss or damage from the use of the said information. CareEdge Research is also not responsible for any errors in transmission and specifically states that it, or its Directors, employees, parent company – CARE Ratings Ltd., or its Directors, employees do not have any financial liabilities whatsoever to the subscribers/users of this report. The subscriber/user assumes the entire risk of any use made of this report or data herein. This report is for the information of the authorized recipient in India only and any reproduction of the report or part of it would require explicit written prior approval of CareEdge Research.

CareEdge Research shall reveal the report to the extent necessary and called for by appropriate regulatory agencies, viz., SEBI, RBI, Government authorities, etc., if it is required to do so. By accepting a copy of this Report, the recipient accepts the terms of this Disclaimer, which forms an integral part of this Report.

Table of Contents

1 Economic Outlook 6
1.1 Global Economy 6
1.2 Indian Economic Outlook 7
1.3 GDP Growth and Outlook 7
1.4 Consumer Price Index 8
1.5 Contribution of the Agriculture Sector to GDP 9
1.5.1 Subsidies for fertilisers 9
1.5.2 Top States- Major Crops Production (2023-24) 10
1.6 Concluding Remarks 11
2 Overview of the Rice Industry 13
2.1 Overview of the Indian Agricultural Industry 13
2.2 Overview of Rice Industry 13
2.3 India's Rice 15
2.4 Growth Potential 19
2.5 Domestic Challenges 20
2.6 Extent of Competition 21
3. Global and Indian Rice Industry 23
3.1 Global Rice Industry 23
3.2 Trend in Indian Rice Industry 25
3.2.1 Trend of Rice Area under Cultivation and Yield 25
3.2.2 Trends in Supply of Rice 26
3.2.3 India's Export Destinations 28
3.2.4 Import of Rice to India 31
3.2.5 MSP in Rice 32
3.3 India Dominating Global Basmati Rice Market 36
3.4 Key players in the Indian Rice Industry 40
3.4.1 KRBL Limited 40
3.4.2 LT Foods Limited 41
3.4.3 Chaman Lal Setia 41
3.4.4 GRM Overseas 42
3.4.5 Sarveshwar Foods Limited 43
4. Aata 45
5. Sooji 47
6. Maida 48
7. Besan 49
8. Salt 50
9. Sugar 51
10. Impact of Pandemic and Other Geopolitical Developments on Indian Rice Industry 54
11. Complexities Hindering the Growth of the Rice Industry in India 58
12. Government Policies 60
13. Threats & Challenges to Issuers and their Products 62
14. Road Ahead 64
15. Competitive Landscape 66

15.1 LT Foods Limited 67
15.2 KRBL Limited 68
15.3 Amir Chand Jagdish Kumar (Exports) Limited 69
15.4 Chaman Lal Setia Exports Limited 70
15.5 GRM Overseas Limited 71
15.6 Sarveshwar Foods Limited 71
15.7 KPIs Comparison 73

List of Charts

Chart 1: Global Growth Outlook Projections (Real GDP, Y-o-Y change in %) 6
Chart 2: Trend in Real Indian GDP growth rate 7
Chart 3: Retail Price Inflation in terms of index and Y-o-Y Growth in % (Base: 2011-12=100) 8
Chart 4: Agriculture sector contribution to GDP (at constant prices) 9
Chart 5: Subsidies provided to the fertiliser sector 10
Chart 6: Rice is a major crop in the Indian agriculture industry in terms of area under cultivation 14
Chart 7: China and India dominate the Global Rice Production during MY 2023-24 and MY 2024-25 23
Chart 8: India is the largest exporter of rice to the world (MY 2024-25) 24
Chart 9: Top Global Importers of Rice with % Share (MY 2024-25) 25
Chart 10: Trend of Rice Area under Cultivation and Yield 26
Chart 11: Rice production in India commands a significant 41% share among other crops during 2024-25 27
Chart 12: IMD Forecasts for All India Rainfall (in LPA) and Yield (Kg/Ha) 27
Chart 13: Rice Production in Major Indian Producing States during 2024-25 28
Chart 14 Basmati and Non-Basmati World Exports from India (in million mt) 29
Chart 15: Middle East Accounting for Larger Share in India's Total Exports of Rice in FY25*, (In Tonnes) 30
Chart 16: Total World Imports to India (in mt) 31
Chart 17: Basmati and Non-Basmati World Imports to India (in mt) 32
Chart 18: MSP for Paddy over Past Years 34
Chart 19: Price Trend of Rice (All India Average) 34
Chart 20: Wholesale Price Index for Rice (Basmati & Non-Basmati Rice) 35
Chart 21: State-level Satellite data and Field-based Basmati Rice Area and Production during Kharif 2023 37
Chart 22: Production Trend of Basmati and Non-Basmati Rice in India (In Tonnes) 38
Chart 23: Basmati Rice: India's Pride on the International Stage 39
Chart 24: KRBL Limited's Domestic and Export Sales (in Rs. Crores) 40
Chart 25: LT Foods Limited's Domestic and Export Sales (in Rs. Crores) 41
Chart 26: Chaman Lal Setia Exports Limited's Domestic and Export Sales (in Rs. Crores) 42
Chart 27: GRM Overseas Estimated Production Domestic and Export Sales (in Rs. Crores) 42

Chart 28 :Sarveshwar Foods Limited (SFL)'s Estimated Production 43
Chart 29:Sarveshwar Foods Limited (SFL)'s Exports 44
Chart 30: Past Trends in Sugar Stock, Production, Consumption, and Exports (in Lakh Tonnes) 52

List of Tables

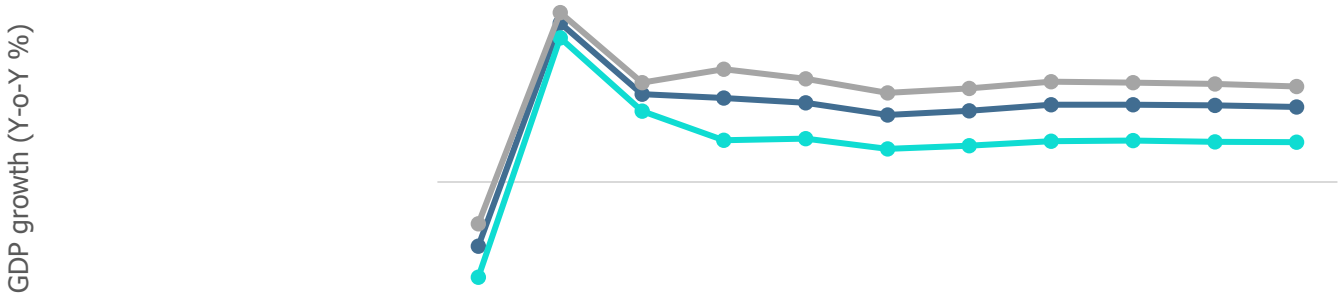
Table 1: GDP growth trend comparison - India v/s Other Economies (Real GDP, Y-o-Y change in %) 6
Table 2: RBI's GDP Growth Outlook (Y-o-Y %) 8
Table 3: Major crops state-wise production 10
Table 4: Primary Basmati Quality Characteristics 17
Table 5: Peer business profiling 66
Table 6: Player Profile 67
Table 7: Financials of LT Foods Limited 67
Table 8: Financials of KRBL Limited 68
Table 9: Financials of Amir Chand Jagdish Kumar (Exports) Limited 69
Table 10: Chaman Lal Setia Exports Limited 70
Table 11: GRM Overseas Limited 71
Table 12: Financials of Sarveshwar Foods Limited 72

1 Economic Outlook

1.1 Global Economy

Global growth, which reached 3.5% in CY23, is estimated to stabilise at 3.3% for CY24 and projected to decrease to 2.8% for CY25. Global trade is expected to be disrupted by new US tariffs and countermeasures from trading partners, leading to historically high tariff rates and negatively impacting economic growth projections. The global landscape is expected to change as countries rethink their priorities and policies in response to these new developments. Central banks' priority will be to adjust policies, while smart fiscal planning and reforms are key to handling debt and reducing global inequalities.

Chart 1: Global Growth Outlook Projections (Real GDP, Y-o-Y change in %)



	CY20	CY21	CY22	CY23	CY24E	CY25P	CY26P	CY27P	CY28P	CY29P	CY30P
World	-2.7%	6.6%	3.6%	3.5%	3.3%	2.8%	3.0%	3.2%	3.2%	3.2%	3.1%
Advanced Economies	-4.0%	6.0%	2.9%	1.7%	1.8%	1.4%	1.5%	1.7%	1.7%	1.7%	1.7%
Emerging Market and Developing Economies	-1.7%	7.0%	4.1%	4.7%	4.3%	3.7%	3.9%	4.2%	4.1%	4.1%	4.0%

Source: IMF – World Economic Outlook, April 2025; Notes: P-Projection, E-Estimated

Table 1: GDP growth trend comparison - India v/s Other Economies (Real GDP, Y-o-Y change in %)

	Real GDP (Y-o-Y change in %)										
	CY20	CY21	CY22	CY23	CY24E	CY25P	CY26P	CY27P	CY28P	CY29P	CY30P
India	-5.8	9.7	7.6	9.2	6.5	6.2	6.3	6.5	6.5	6.5	6.5
China	2.3	8.6	3.1	5.4	5.0	4.0	4.0	4.2	4.1	3.7	3.4
Indonesia	-2.1	3.7	5.3	5.0	5.0	4.7	4.7	4.9	5.0	5.1	5.1
Saudi Arabia	-3.6	5.1	7.5	-0.8	1.3	3.0	3.7	3.6	3.2	3.2	3.3
Brazil	-3.3	4.8	3.0	3.2	3.4	2.0	2.0	2.2	2.3	2.4	2.5
Euro Area	-6.0	6.3	3.5	0.4	0.9	0.8	1.2	1.3	1.3	1.2	1.1
United States	-2.2	6.1	2.5	2.9	2.8	1.8	1.7	2.0	2.1	2.1	2.1

Source: IMF- World Economic Outlook Database (April 2025)

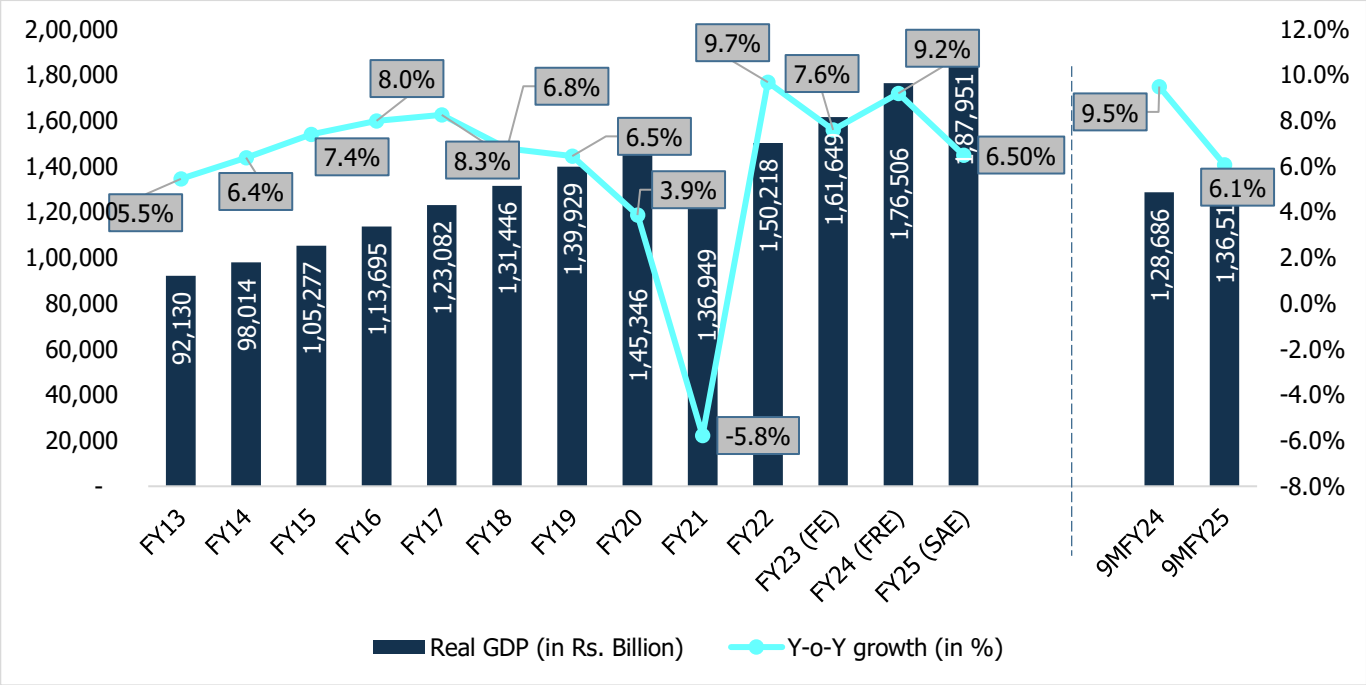
Note: P- Projections, E-Estimated; India's fiscal year (FY) aligns with the IMF's calendar year (CY). For instance, FY24 corresponds to CY23.

1.2 Indian Economic Outlook

1.3 GDP Growth and Outlook

Resilience to External Shocks remains Critical for Near-Term Outlook.

Chart 2: Trend in Real Indian GDP growth rate



Source: MOSPI; Note: SAE- Second Advance Estimates, FRE – First Revised Estimates, FE – Final Estimate

India's real GDP grew by 9.2% in FY24 (Rs. 176,506 billion) which is the highest in the previous 12 years and is estimated to grow 6.5% in FY25 (Rs. 187,951 billion), driven by double digit growth particularly in the Manufacturing sector, Construction sector and Financial, Real Estate & Professional Services. In 9MFY25, GDP grew 6.1% Y-o-Y, with private consumption increasing by 7.6% and government spending increasing by 3.8% Y-o-Y.

GDP Growth Outlook

FY26 GDP Outlook: Real GDP growth is projected at 6.5%, driven by strong rural demand, improving employment, and robust business activity. The agriculture sector's bright prospects, healthy reservoir levels, and robust crop production support this growth. Manufacturing is reviving, and services remain resilient, despite global uncertainties. Investment activity is gaining traction, supported by healthy balance sheets and easing financial conditions. However, risks from geopolitical tensions, global market volatility, and geo-economic fragmentation persist.

Persistent geopolitical tensions, volatility in international financial markets and geo-economic fragmentation do pose risk to this outlook. Based on these considerations, the RBI, in its April 2025 monetary policy, has projected real GDP growth at 6.5% y-o-y for FY26.

Table 2: RBI's GDP Growth Outlook (Y-o-Y %)

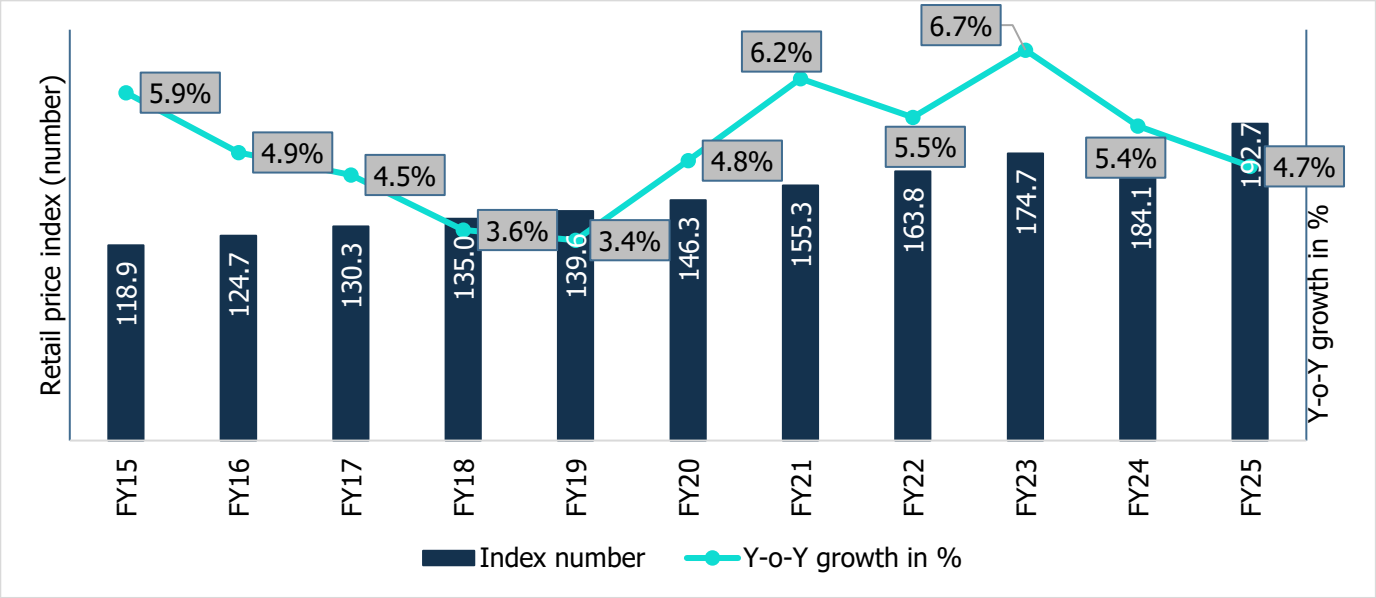
FY26P (complete year)	Q1FY26P	Q2FY26P	Q3FY26P	Q4FY26P
6.5%	6.%	6.7%	6.6%	6.3%

Source: Reserve Bank of India; Note: P-Projected

1.4 Consumer Price Index

The CPI (general) and food inflation in March 2025 over March 2024 (3.34%, provisional) witnessed the lowest Y-o-Y inflation since August 2019. The moderation was driven by the decline of price inflation in Vegetables, Eggs, Meat & fish, Cereals and Pulses and Milk.

Chart 3: Retail Price Inflation in terms of index and Y-o-Y Growth in % (Base: 2011-12=100)



Source: MOSPI

The CPI is primarily factored in by RBI while preparing their bi-monthly monetory policy. At the bi-monthly meeting held in April 2025, RBI projected inflation at 4.0% for FY26 with inflation during Q1FY26 at 3.6%, Q2FY26 at 3.9% and Q3FY26 at 3.8% and Q4FY26 4.4%.

Considering the current inflation situation, RBI has cut the repo rate to 6.00% in the April 2025 meeting of the Monetary Policy Committee.

Further, the central bank changed its stance to be accommodative. While headline inflation moderated during January-February 2025, with a decline in food inflation.

The economic growth outlook for India is expected to remain resilient, but it will require careful monitoring due to depreciation of the Indian rupee in recent months. Additionally, certain key sectors may face headwinds amid hiked tariffs from the US.

The RBI has adopted for a non-inflationary growth with the foundations of strong demand and supply with a good macroeconomic balance. The domestic growth and inflation curve require the policies to be supportive with the volatile trade conditions.

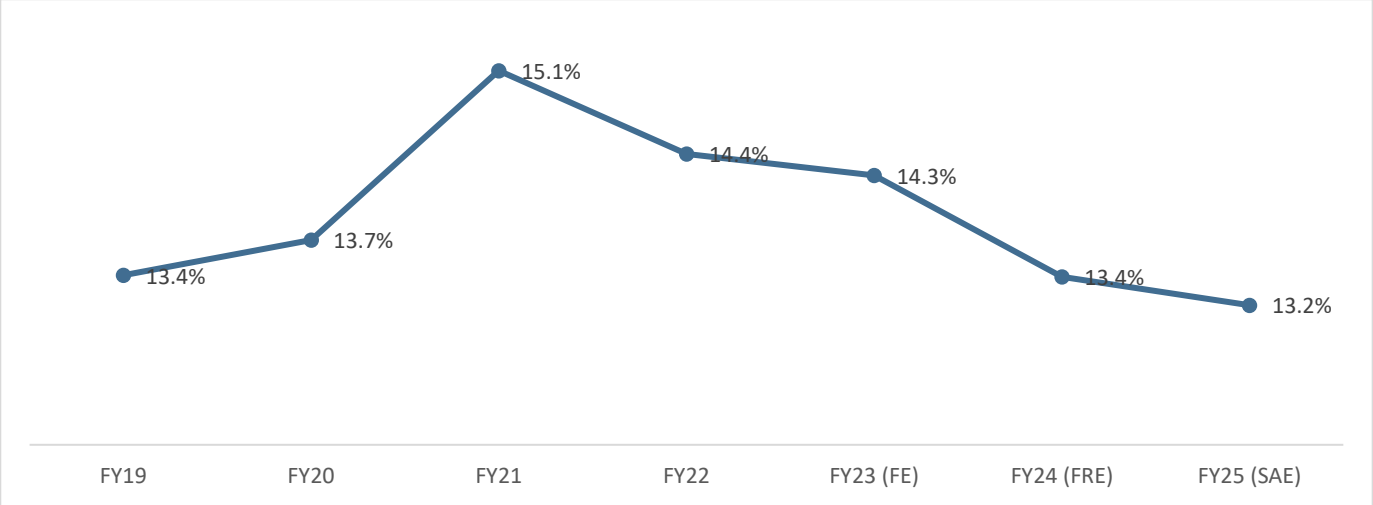
1.5 Contribution of the Agriculture Sector to GDP

Agriculture has been a cornerstone of India's economy since independence, contributing significantly to GDP and supporting the workforce. While its share in GDP is expected to decrease to 13.2% in FY25, agriculture remains crucial for rural livelihoods, global trade, and sectors like food processing and textiles. Despite challenges like climate change, it continues to play a vital role in economic growth, poverty reduction, and social stability. Its growth is supported by government initiatives aimed at enhancing productivity, ensuring food security, and improving farmer welfare.

The Economic Survey highlights key strides in India's agriculture, with foodgrain production is 328.8 million tonnes in FY24 and oilseeds production increasing, reducing dependence on imports. Government initiatives like e-NAM, FPOs, and PMKMY aim to improve agricultural efficiency, support farmer incomes, and provide social security, bolstering the sector's growth.

Additionally, the government promotes sustainable practices through PM-PRANAM and ensures financial stability with schemes like MSP and PMFBY, which cover over 610 lakh hectares in FY24. These efforts enhance productivity, food security, and farmer welfare, contributing significantly to agriculture's role in India's GDP.

Chart 4: Agriculture sector contribution to GDP (at constant prices)



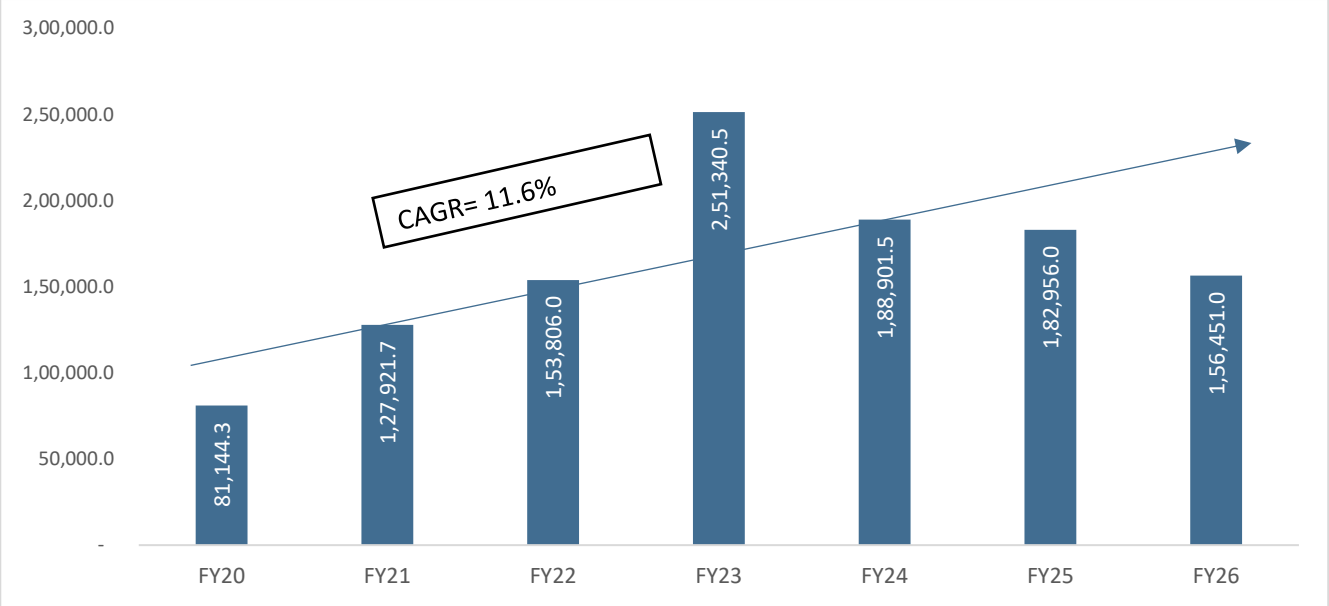
Source: MOSPI; Note: FRE – First Revised Estimates, FE– Final Estimate, SAE- Second Advance Estimates

1.5.1 Subsidies for fertilisers

Fertilisers are essential for enhancing soil fertility, boosting crop productivity, ensuring food security, and supporting the livelihoods of millions of farmers in India. As India is highly dependent on imports to meet its fertiliser and raw material needs, the domestic prices, in turn, are influenced by the world prices. Thus, to avoid the burden of price hikes on farmers and disturbance in agricultural produce, the government provides subsidies on fertilisers to farmers through manufacturers. In the previous year, considering the surge in prices of fertilisers, the government doubled the fertiliser subsidy for the Rabi season (from 1st October 2022 to 31st March 2023) to Rs. 51,875 crores as compared to Rs. 28,655 crores. The subsidy was doubled not only on account of a surge in prices of fertilisers but also due to the geopolitical issues between Russia and Ukraine and logistics issues.

For FY26, the upfront subsidy budget decreased by 13% to Rs. 164,102.5 crore as compared to Rs. 156,451.0 crore in FY25. This will be adequate for the year on account of reduced prices of raw materials and natural gas. It will also aid the urea and complex fertilisers manufacturers to effectively manage their working capital requirement.

Chart 5: Subsidies provided to the fertiliser sector



Source: Union Budget FY25 document

1.5.2 Top States- Major Crops Production (2023-24)

The major crops produced in the top 3 states in India in 2023-24 are:

Table 3: Major crops state-wise production
Food Grains:

Crop	State	Production (Lakh Tonnes)
Rice	Uttar Pradesh	157.2
	West Bengal	151.2
	Telangana	166.3
Wheat	Madhya Pradesh	212.8
	Punjab	177.8
	Uttar Pradesh	354.3
Maize	Karnataka	54.9
	Bihar	46.1
	Madhya Pradesh	43.3
Total Nutri/Coarse Cereals	Rajasthan	80.3
	Karnataka	76.1
	Madhya Pradesh	54.9
Gram	Madhya Pradesh	31.9
	Maharashtra	28.6
	Rajasthan	22.3
Tur	Maharashtra	10.2
	Karnataka	8.6
	Uttar Pradesh	3.8
Total Pulses	Madhya Pradesh	61.8
	Rajasthan	40.0
	Maharashtra	36.3
	Madhya Pradesh	398.4

Crop	State	Production (Lakh Tonnes)
Total Foodgrains	Uttar Pradesh	592.9
	Punjab	325.9

Oilseeds:

Crop	State	Production (Lakh Tonnes)
Groundnut	Gujarat	46.4
	Rajasthan	20.2
	Madhya Pradesh	9.9
Rapseed & Mustard	Rajasthan	59.8
	Uttar Pradesh	18.7
	Madhya Pradesh	17.5
Soyabean	Madhya Pradesh	54.7
	Maharashtra	52.3
	Rajasthan	11.7
Sunflower	Karnataka	0.7
	Haryana	0.3
	Odisha	0.2
Total Oilseeds	Rajasthan	95.7
	Madhya Pradesh	83.7
	Gujarat	71.9
Sugarcane	Uttar Pradesh	2055.6
	Maharashtra	1120.9
	Karnataka	418.1
Cotton	Gujarat	90.6
	Maharashtra	80.5
	Telangana	50.8
Jute & Metals	West Bengal	78.7
	Assam	6.8
	Bihar	9.9

Source: India Budget, Economic Survey

Note: 1. Data for the year 2023-24 is of the 3rd Advance Estimates

2. Cotton Production in Bales, 1 Bale = 170 Kg

3. Jute & Mesta Production in Bales, 1Bale=180 Kg

1.6 Concluding Remarks

India's average crop yield is lower than the global average due to outdated practices, poor irrigation, and climate challenges. However, with technological advancements, better infrastructure, and improved practices, India has the potential to boost production, increase yields, and become more competitive in the global agricultural market.

India, with 14% of the global crop-protection market, is a key player in boosting agricultural productivity. Demand for chemicals is projected to grow from 61,097 tonnes in FY20 to 89,170 tonnes by FY36. The industry is adopting sustainable practices and innovations, driving food security and reducing agriculture's ecological impact, solidifying India's leadership in crop protection. As per the latest forecasts by various agencies including the IMD, the monsoon is expected to be normal this year as well as no impact from El Nino effect is expected.

Global economic growth faces headwinds from geopolitical tensions, volatile commodity prices, high interest rates, inflation, financial market volatility, climate change, and rising public debt. However, India's economy remains relatively

strong, with an IMF forecast of 7% GDP growth in CY24 (FY25 according to the fiscal year), compared to the global projection of 3.2%. Key drivers include strong domestic demand, government capital expenditure, moderating inflation, and improving business confidence.

Public investment is expected to grow, with the government allocating Rs. 11.21 lakh crores for FY26. Private sector investment is also improving, reflected in new projects and capital goods imports. High-frequency indicators suggest the agriculture sector in Q3FY25 grew by 5.6%. Agricultural growth is supported by healthy kharif crop production, higher reservoir levels and better rabi sowing. Additionally, improvement in rural demand owing to healthy sowing, improving reservoir levels, and progress in the south-west monsoon, along with the government's thrust on capex and other policy support, will aid the investment cycle in gaining further traction.

2 Overview of the Rice Industry

2.1 Overview of the Indian Agricultural Industry

Agriculture continues to remain the backbone of the Indian economy

Agriculture remains a cornerstone of the Indian economy, providing the primary source of livelihood for about 58% of the population. India boasts the world's largest cattle herd (buffaloes) and the largest areas planted for wheat, rice, and cotton. It is also the leading global producer of milk, pulses, and spices, and ranks as the second-largest producer of sugar, rice, fruits, vegetables, tea, farmed fish, sugarcane, wheat, and cotton. With the second-largest agricultural land area globally, the sector employs approximately half of the country's population, underscoring the vital role of farmers in sustaining the nation.

The government's initiatives, such as supporting farmer-producer organisations, promoting crop diversification, and enhancing agricultural productivity through the Agriculture Infrastructure Fund, have significantly contributed to the industry's growth. Additionally, the Pradhan Mantri Kisan Samman Nidhi (PM-KISAN) scheme has provided income support to farmers, encouraging alternative income sources and increasing their resilience to weather-related shocks.

A significant portion of India's agriculture depends on a timely and normal monsoon season, critical for ensuring adequate produce. Recognising this, the Government of India remains committed to the welfare of farmers.

As per the Union Budget 2025-26, the Ministry of Agriculture and Farmers' Welfare has been allocated Rs. 1.27 lakh crore, reflecting a 2.9% decrease compared to Rs. 1.31 lakh crore in FY24-25.

In the 2025-26 budget, several key initiatives have been announced to further bolster the agriculture sector:

- **Prime Minister Dhan-Dhaanya Krishi Yojana:** A comprehensive program aimed at enhancing agricultural productivity and ensuring food security.
- **Rural Prosperity and Resilience Programme:** A multi-sectoral initiative designed to boost rural employment, empower women, and incorporate best practices in the rural sector through assistance from multilateral agencies.
- **Mission for Aatmanirbharta in Pulses:** A six-year mission focusing on achieving self-reliance in pulses, particularly Tur, Urad, and Masoor, to reduce import dependence.
- **Establishment of a Makhana Board in Bihar:** Aimed at promoting the cultivation and marketing of makhana, benefiting local farmers and boosting regional economies.
- **National Mission on High-Yielding Seeds:** An initiative to promote the use of high-yielding seed varieties, enhancing crop productivity and ensuring food security.
- **Second Gene Bank:** Establishment of a gene bank housing 10 lakh germplasm lines to preserve genetic diversity and ensure future food and nutritional security.
- **Five-Year Mission for Cotton Productivity:** A program aimed at promoting extra-long-staple cotton varieties, enhancing quality and yield.
- **Kisan Credit Card (KCC) Loan Limit Increase:** The KCC loan limit has been raised from Rs. 3 lakhs to Rs. 5 lakhs, providing greater financial support to farmers.
- **12.7 Lakh Metric Tons Urea Plant in Namrup, Assam:** Establishment of a urea plant to meet the fertiliser needs of the region, supporting agricultural productivity.
- **Framework for Sustainable Harnessing of Fisheries:** Special focus on the Andaman & Nicobar and Lakshadweep islands to promote sustainable fishing practices and boost the local economy.

2.2 Overview of Rice Industry

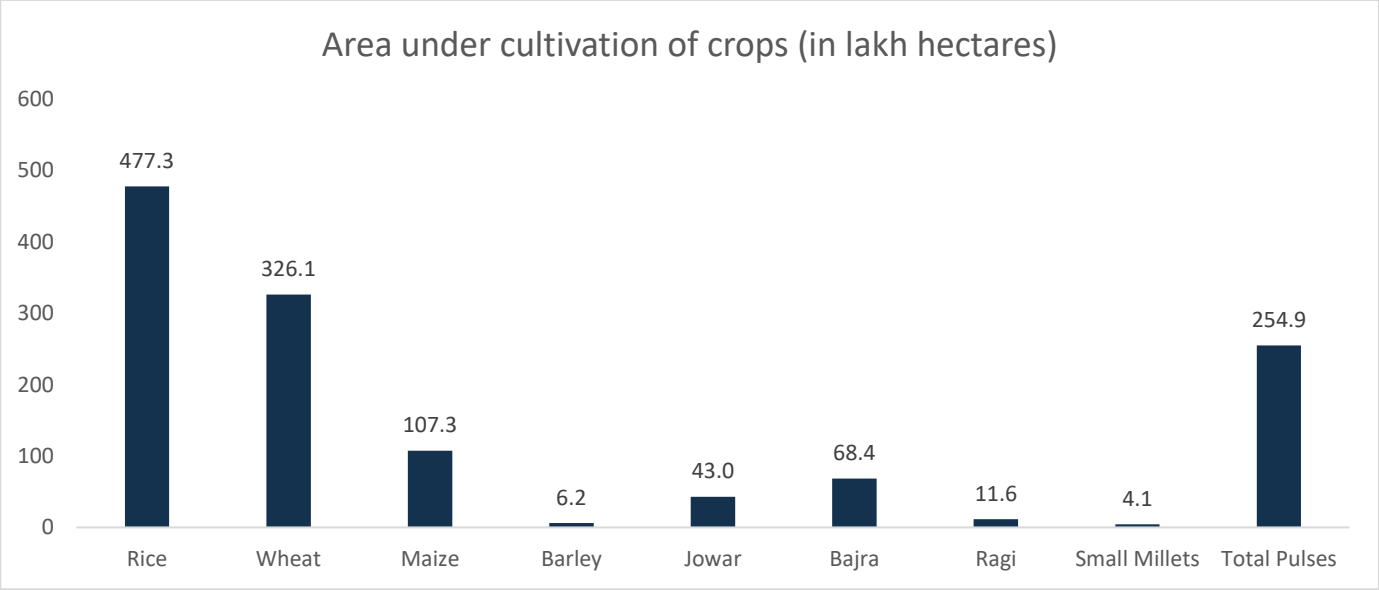
The rice industry in India is a crucial component of the country's agricultural sector, economy, and food security. Rice is the most common staple food in many Asian countries, including India. India is one of the world's largest producers

and consumers of rice. Rice cultivation is widespread across the country, with various regions specialising in different varieties. India cultivates a diverse range of rice varieties to cater to regional preferences and culinary traditions. It forms the primary source of nutrition for millions of people, particularly in states like West Bengal, Odisha, Bihar, Uttar Pradesh, and Tamil Nadu. Rice is consumed in various forms, including steamed rice, biryani, pulao, idlis, dosa, and more. Major rice-producing states include West Bengal, Uttar Pradesh, Punjab, Andhra Pradesh, Telangana, Tamil Nadu, Odisha, Bihar and Haryana.

India is a major player in the global rice market as an exporter. Basmati rice, famous for its aroma and long grains, is one of the most prized varieties, with significant exports. Other popular varieties include Sona Masuri, Kolam, Mogra, Swarna, etc. The government imposes quality standards and regulations to maintain the reputation of Indian rice in the global market and implements various policies and schemes to support rice farmers. These include Minimum Support Prices (MSPs), subsidies on fertilisers and seeds, irrigation schemes, crop insurance, and infrastructure development.

Rice is a water-intensive crop, and water scarcity poses a major challenge. Efficient water management and irrigation practices are essential to ensure sustainability. Efforts are underway to adopt modern agricultural practices, mechanisation, biotechnology, and precision farming techniques to improve rice production, yield, and quality. Research institutions and agricultural universities collaborate with farmers to develop high-yielding varieties, drought-resistant strains, and eco-friendly cultivation methods. There is ample scope to increase the productivity of rice in the country, as about 90% of the cultivated land belongs to marginal, small and medium farmers, which is another constraint in increasing the productivity of rice. The rice industry in India continues to evolve and adapt to changing circumstances. With the adoption of advanced technologies, sustainable practices, and supportive government policies, the Indian rice industry aims to enhance productivity, ensure food security, and maintain its competitive edge in the global market.

Chart 6: Rice is a major crop in the Indian agriculture industry in terms of area under cultivation



Source: Ministry of Agriculture & Farmers Welfare
Data for the year 2024-25 is as per the 2nd Advance Estimates released

India's agricultural landscape is characterised by the cultivation of a wide range of crops, catering to both domestic consumption and international trade. Among these crops, rice holds a significant share of 36.7%, with substantial

acreage dedicated to its cultivation, followed by wheat with a share of 25.1% and total pulses with a share of 19.6%. Apart from rice, India cultivates a diverse array of crops including wheat, pulses, millets, oilseeds, sugarcane, cotton, fruits, and vegetables. The cultivation of these crops varies based on factors such as agro-climatic conditions, soil fertility, water availability, and market demand.

2.3 India's Rice

Different Varieties of Rice:
India is home to numerous varieties of rice, each with its unique characteristics in terms of flavour, texture, aroma, and culinary use. Some of the most popular types of rice cultivated and consumed in India include:

1. Basmati Rice: Known for its distinctive aroma and long grains, Basmati rice is one of the most celebrated varieties in India, which caters to diverse preferences across different regions of the country. It is primarily grown in the foothills of the Himalayas in the states of Punjab, Haryana, and Uttar Pradesh. Primarily grown in the Himalayan foothills of Punjab, Haryana, and Uttar Pradesh, Basmati rice stands out as a highly favoured and desired variety with origins tracing back to the Indian subcontinent. Its cultivation requires meticulous care, involving precise water management, soil preparation, and harvesting techniques to maintain its exceptional quality and aroma.

Basmati rice from India has been granted a Geographical Indication (GI) tag, recognising its unique identity and ensuring protection against counterfeit products in international markets. The GI tag covers Basmati rice grown in specific regions of Punjab, Haryana, Himachal Pradesh, Uttarakhand, Delhi, Western Uttar Pradesh, and Jammu & Kashmir. This designation reinforces India's stronghold over the global Basmati rice market and safeguards its authenticity.

Basmati rice holds immense cultural significance in India, featuring prominently in traditional festivals, celebrations, and religious ceremonies. It is a staple ingredient in classic Indian dishes such as biryani, pulao, and pilaf, revered for their exquisite taste. As a leading producer and exporter of Basmati rice, India dominates the global market with its premium-quality grains. Basmati rice exports contribute significantly to India's agricultural economy, generating substantial revenue and foreign exchange earnings. The demand for Indian Basmati rice extends to international markets including the Middle East, Europe, North America, and Southeast Asia, where it is prized for its superior quality and aroma.

2. Jasmine Rice: The jasmine rice replicates the basmati rice quality in terms of taste and texture. Renowned for their aromatic qualities, both Basmati and jasmine rice are often considered interchangeable options. In India, there's a growing preference for Thai cuisine, and jasmine rice is the preferred choice to accompany these dishes, aligning perfectly with the flavours and aromas of Thai culinary delights. Jasmine rice, also known as Thai fragrant rice, is indigenous to Thailand, where it has been cultivated for centuries. Jasmine rice is a staple ingredient in Southeast Asian cuisines, particularly Thai, where it serves as the foundation for dishes like green curry, stir-fries, and rice bowls. Its fragrant aroma and tender texture complement a wide range of savoury and spicy flavours, making it an ideal accompaniment to various dishes.

3. Mogra Rice: Mogra is among the cheapest rice varieties in the market, known for its affordability, versatility and suitability for everyday meals. While it shares some similarities with premium varieties like Basmati in terms of aroma and texture, Mogra rice is distinct and widely appreciated for its practical use in a variety of dishes. Mogra rice typically features shorter grains compared to Basmati rice. It possesses a mild fragrance, making it a pleasant addition to daily meals without being overly aromatic. Mogra rice is cultivated in several parts of India, including Maharashtra, Gujarat, and Madhya Pradesh, where the climate and soil conditions are favourable. The cultivation involves standard paddy

farming techniques with a focus on efficient water management and timely harvesting to ensure good quality grains. It is versatile enough for various rice dishes, including fried rice, pulao, and khichdi, providing a good balance of flavour and texture. It is widely available across India in local markets and supermarkets.

4. Indrayani Rice: Indrayani Rice is a highly esteemed variety of rice originating from the state of Maharashtra, India. It derives its name from the sacred Indrayani River in Maharashtra, where it has been cultivated for generations. This medium-grain rice variety thrives in the fertile plains of Maharashtra, particularly in regions with rich soil and adequate irrigation. Cultivating Indrayani rice requires careful attention to traditional farming techniques, including proper water management, soil preparation, and harvesting methods, to ensure optimal yield and quality. While this rice primarily refers to a specific variety cultivated in Maharashtra, similar medium-grain rice varieties are grown in other parts of India, each with its distinct flavour profile and culinary uses. This rice is a staple in many Maharashtrian households and is used in a variety of traditional dishes, where it is used to prepare traditional dishes such as Varan Bhaat (dal rice), Masale Bhaat (spiced rice), and Kheer (rice pudding). The rice's versatility and adaptability make it a beloved grain in Maharashtrian cuisine, cherished for its ability to elevate the flavours of various dishes. The cooking time for this variety of rice is very low, which means you need to be pretty quick in turning off the stove, in comparison to other rice varieties.

5. Kolam: Kolam rice, renowned for its unique characteristics and cultural significance, holds a special place in South Indian cuisine. This report provides an overview of Kolam rice, detailing its origins, cultivation, attributes, culinary uses, and cultural importance. Originating from the fertile paddy fields of Kerala, Kolam rice, also known as Kollam or Kaima rice, benefits from the region's tropical climate and ample rainfall. Cultivators use traditional and organic farming methods, including manual harvesting, to maintain their distinctive flavour and texture. Kolam rice is a staple in South Indian cuisine, particularly in Kerala, where it is used in traditional dishes like Malabar biryani, Kerala-style curries, and festive delicacies. Its soft texture and flavour-absorbing qualities make it ideal for rice-based dishes, soups, and desserts. Kolam rice is also integral to religious rituals, social gatherings, and daily meals, symbolising prosperity and hospitality. Notably, its small, spongy grains emit a floral aroma when cooked, making it a popular choice for sweet Pongal and other everyday dishes.

6. Regular White or Polished Rice: Regular white rice, also known as polished rice, is one of the most commonly consumed types of rice worldwide. It undergoes a refining process that removes the husk, bran, and germ, resulting in a milled, white grain. This process extends its shelf life and enhances its cooking qualities. White rice has a polished, bright white appearance, with uniform, medium-to-long grains. It has a mild, neutral flavour that readily absorbs the flavours of other ingredients in a dish. White rice is cultivated in various regions around the world, including Asia, North and South America, and parts of Africa. Major producers include China, India, Indonesia, and the United States.

7. Sugandha: Sugandha Rice is a premium variety of long-grain rice known for its exceptional aroma and high quality. Often compared to Basmati rice, sugandha rice is appreciated for its distinct flavour, fragrance, and versatility in culinary applications. Sugandha rice features long, slender grains that elongate significantly when cooked, similar to Basmati rice. When cooked, sugandha rice grains are fluffy and separate, providing a light and airy texture. Sugandha rice is predominantly grown in the northern states of India, including Haryana, Punjab, and Uttar Pradesh. These regions offer ideal conditions for its cultivation, such as fertile soil and a suitable climate. The cultivation of sugandha rice involves advanced agricultural techniques, including the use of high-quality seeds, proper irrigation, and effective pest management to ensure optimal yield and quality. There is a significant demand for sugandha rice in international markets, particularly in the Middle East, Europe, and North America. Its premium quality and aromatic properties make it a favoured choice among consumers abroad.

8. Parboiled Rice: Parboiled Rice is a partially precooked rice that undergoes a unique processing method involving soaking, steaming, and drying the rice while it is still in its husk. This process alters the rice's texture, nutritional content, and cooking properties, making it a popular choice in many parts of the world. Parboiled rice has a slightly yellowish or amber colour due to the parboiling process, which changes the hue of the rice kernels. Parboiled rice retains more nutrients compared to regular white rice because the parboiling process drives some of the vitamins and minerals from the bran into the grain. It is often richer in vitamins like B6 and B12, and minerals such as magnesium and zinc.

Eligibility of a Rice Variety to be Notified as Basmati

According to the provisions of the Seed Act, 1966, and the recommendations of the Central Sub Committee on Crops, Standards, Notification, and Release of Varieties for Agricultural Crops, established by the Central Seed Committee under Section 3 of the Seed Act, 1966, the following standards have been established to determine and classify a rice variety as Basmati.

According to the Indian Agricultural and Processed Food Products Export Development Authority (APEDA), a rice variety is eligible to be called basmati if it has

Table 4: Primary Basmati Quality Characteristics

S. No.	Parameters	Value
1	Minimum average precooked milled rice length (mm)	6.6
2	Average precooked milled rice breadth (mm)	≤2
3	Minimum length/breadth ratio of precooked milled rice (L/B Ratio)	3.5
4	Minimum average cooked rice, length (mm)	12.0
5	Minimum cooked rice length/precooked rice length ratio or Minimum elongation ratio	1.7
6	Average volume expansion ratio	> 3.5
7	Aroma	Present (Qualitative sensory analysis as Panel Test*)
8	The texture of cooked grain for high integrity (without bursting the surface), non-stickiness, tenderness, good taste and good mouthfeel	Present (Qualitative sensory analysis as Panel Test*)

Source: APEDA

Note 1: The grain sample for analyses will necessarily have to be 'aged' for three months under protected conditions at normal room temperature as a milled kernel

*: As per standardised protocol (Directorate of Rice Research, Hyderabad)

Cultivation of Basmati Rice

Special Status of Basmati

Basmati rice holds a unique position in rice cultivation due to its excellent cooking and eating qualities. Basmati rice is renowned for its distinctive aroma, flavour, and ability to elongate more than twice its original size when cooked. These distinct characteristics result from specific harvesting, processing, and geographical conditions. The GI tag acknowledges the unique nature and origin of the Basmati rice genome, establishing it as a distinct brand and identity linked to its country of origin. Basmati rice is cultivated in the Himalayan foothills of the Indian subcontinent and is

protected under Geographical Indications [1](GI). In India, the GI designation covers seven states: Western Uttar Pradesh, Punjab, Haryana, Uttarakhand, the Jammu division of Jammu and Kashmir, Himachal Pradesh, and Delhi. Seed production of Basmati rice is restricted to these designated GI areas. The cultivation of Basmati rice demands meticulous care and expertise.

Recommended Practices for Satisfactory Yield

To achieve satisfactory yields of Basmati rice, the following practices are recommended:

- Timely Sowing and Transplanting: Ensure adherence to the recommended schedule for better quality rice.
- Avoid Mixing Varieties: Maintain the purity of Basmati varieties by preventing mixing.
- Insect-Pest Management: Regularly monitor insect-pest populations and apply recommended control measures.
- Disease Management: Basmati varieties, particularly Pusa Basmati 1121, are susceptible to foot rot disease. Use only disease-free seeds and apply seed treatments to prevent seed-borne diseases.

Climatic Requirements

Basmati varieties, like semi-dwarf rice varieties, thrive in conditions of prolonged sunshine, high humidity, and assured water supply. Superior cooking and eating characteristics are achieved when the crop matures in relatively cooler temperatures. High temperatures during the grain-filling period can negatively impact the quality features of Basmati rice, such as kernel elongation and the non-stickiness of cooked rice.

The cultivation of Basmati rice demands meticulous care and expertise. Farmers follow traditional methods, respecting the natural rhythms of the land. The sowing, nurturing, and harvesting processes are guided by experience and passed down through generations, ensuring the authenticity and quality of the grain.

Notified Basmati Varieties as per APEDA (August 31st, 2022)

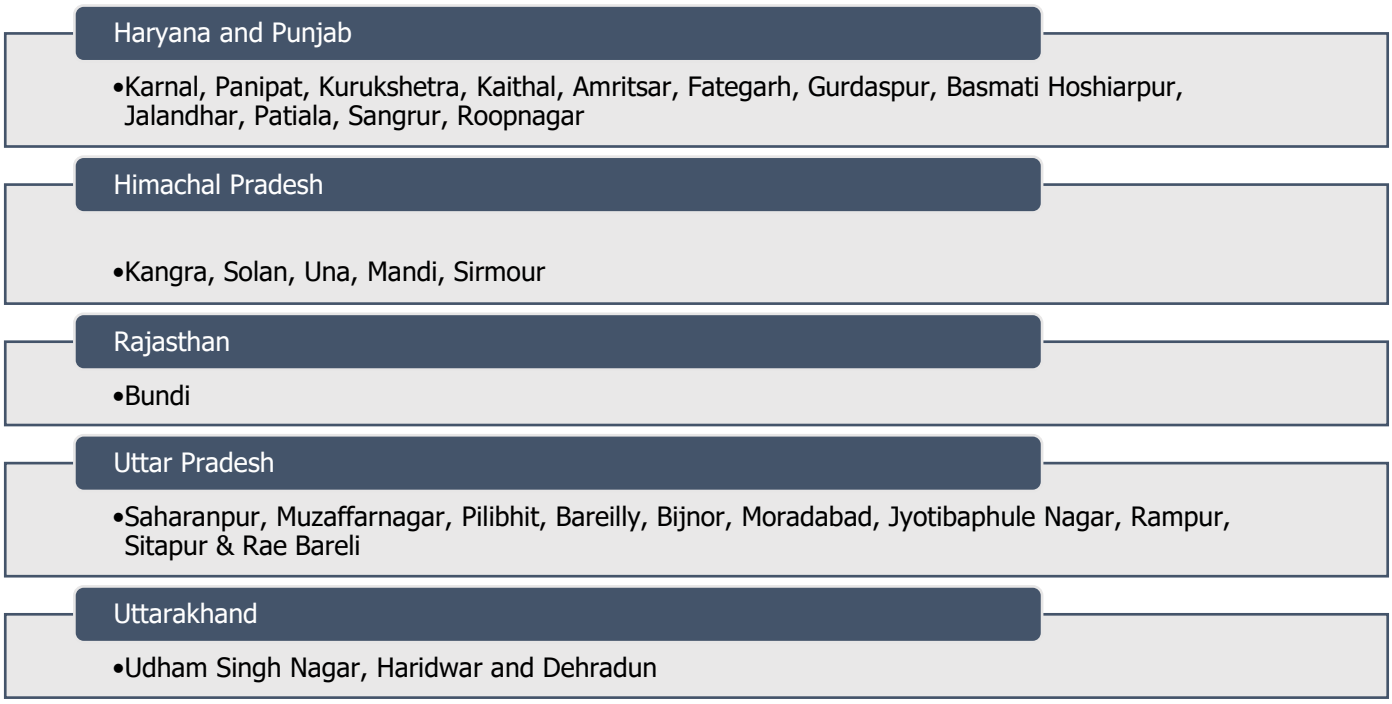
India cultivates various Basmati rice varieties, each with its unique characteristics and flavour profiles. Some popular varieties include Traditional Basmati, Pusa Basmati, 1121 Basmati, 1509 Basmati, etc. These varieties differ in grain length, aroma intensity, cooking qualities, and suitability for different culinary applications.

So far, 45 varieties of Basmati rice have been notified under the Seeds Act, 1966. These are Basmati 217, Punjab Basmati 1 (Bauni Basmati), Basmati 386, Punjab Basmati 2, Punjab Basmati 3, Basmati 370, Haryana Basmati 1, Taraori Basmati (HBC 19), Type 3 (Dehraduni Basmati), Pant Basmati 1 (IET 21665), Pant Basmati 2 (IET 21953), Kasturi, Mahi Sugandha, Basmati CSR 30 (After amendment), Malviya Basmati Dhan 10-9 (IET 21669), Ranbir Basmati, Basmati 564, Pusa Basmati 1, Pusa Basmati 1121 (After amendment), Pusa Basmati 1509 (IET 21960), Pusa Basmati 6 (Pusa 1401), Pusa Basmati 1609, Pusa Basmati 1637, Pusa Basmati 1728, Vallabh Basmati 22, Vallabh Basmati 21 (IET 19493), Vallabh Basmati 23, Vallabh Basmati 24, Pusa Basmati 1718, Punjab Basmati 4, Punjab Basmati 5, Haryana Basmati 2, Pusa Basmati 1692, Jammu Basmati 118, Jammu Basmati 138, Jammu Basmati 129, Jammu Basmati 123, Pusa Basmati 1847, Pusa Basmati 1885, Pusa Basmati 1886, Pusa Basmati 1985, Pusa Basmati 1979, Pusa Basmati 1882, Punjab Basmati 7.

[1] Geographical Indications (GI) are a form of intellectual property rights that identify a product as originating from a specific geographical location, which endows the product with unique qualities, reputation, or characteristics. In the context of rice, GI tags help protect and promote traditional rice varieties that have distinct attributes linked to their place of origin

Rice Cultivating Districts of India

Haryana and Punjab: The Basmati bowl of India



Source: Industry Sources

2.4 Growth Potential

Factors that are driving the growth of the Basmati rice industry are as follows:

1. **Export Opportunities**: India is the largest exporter of Basmati rice globally, with countries like Iran, Saudi Arabia, Iraq, UAE, etc. being major importers. The growing demand for Basmati rice in international markets presents significant export opportunities for Indian farmers and exporters, driving the expansion of Basmati rice cultivation.
2. **Increasing Demand:** Basmati rice enjoys high demand both domestically and internationally due to its various characteristics. With rising incomes, changing dietary preferences, and increasing awareness of healthy eating habits, the demand for Basmati rice is expected to continue growing in India. HORECA (Hotel, Restaurant and Catering), encompassing the hotel and food service sectors, relies on Basmati rice as a staple ingredient due to its widespread popularity among consumers. Restaurants and hotels consistently face high demand for Basmati rice, driving up production and sales within the market. This surge in demand plays a pivotal role in fuelling the growth of the Basmati rice market.
3. **Premium Pricing**: Basmati rice commands premium prices in both domestic and international markets compared to other rice varieties. This premium pricing offers lucrative returns for farmers and incentivises them to expand Basmati rice cultivation.
4. **Geographical Expansion:** Basmati rice cultivation traditionally concentrated in states like Punjab, Haryana, Uttar Pradesh and parts of Bihar. However, there is potential for the geographical expansion of Basmati rice cultivation to other regions with suitable agro-climatic conditions. States like Uttarakhand, Jammu and Kashmir, Himachal Pradesh, and West Bengal etc. are exploring Basmati rice cultivation, further enhancing its growth potential.

5. **Sustainable Practices**: There is increasing emphasis on sustainable agricultural practices, including organic farming and water-saving techniques, in Basmati rice cultivation. Adoption of sustainable practices not only ensures environmental conservation but also enhances the quality and marketability of Basmati rice, contributing to its growth potential.
6. **Growth in Per-Capita Consumption and the Shift towards Packaged Rice:** There has been a noticeable rise in per-capita consumption of rice alongside a notable shift towards packaged varieties. This trend indicates a growing preference among consumers for convenient, pre-packaged rice options. A robust economy and rising income enable more people to opt for quality rice, especially the basmati variety. Moreover, people tend to adapt to changing lifestyles.
7. **Convenience/Online Store**: Convenience stores are designed to provide people with various essential items for daily living, including financial services, food, groceries, fuel, and more. Now, convenience stores worldwide are expanding their inventory to include Basmati products. This trend is driving up sales rates and significantly boosting revenue within the industry. Following the pandemic crisis, there has been a notable surge in the trend of online ordering for food and groceries. Additionally, the proliferation of e-commerce platforms and quick commerce services has further facilitated people in purchasing Basmati rice and other essentials from their homes, ensuring faster deliveries and enhanced convenience.
8. **Government Support:** The Indian government provides various support schemes and incentives to promote Basmati rice cultivation. This includes Minimum Support Prices (MSPs), subsidies on seeds, fertilisers, and irrigation, crop insurance, infrastructure development, and export promotion initiatives. Government support encourages farmers to invest in Basmati rice cultivation and contributes to its growth.
9. **Increased Awareness Through Branding and Marketing:** Growing consumer awareness, driven by branding and marketing efforts, has led to a shift towards organised players in the Basmati rice market. Leading brands are leveraging digital campaigns, packaging innovations, and certifications (such as GI tagging and organic labelling) to build trust and attract consumers. This has boosted demand for branded Basmati rice over unbranded alternatives, enhancing market growth and premiumization.

Overall, the growth potential of Basmati rice in India is promising, fuelled by increasing demand, export opportunities, technological advancements, government support, geographical expansion, brand recognition, and sustainable practices. Continued focus on these factors can further propel the growth trajectory of Basmati rice cultivation in India.

2.5 Domestic Challenges

- **Government Restrictions on Rice Exports:** India has imposed several restrictions on rice exports over the years to balance domestic supply and price stability. In July 2023, the government banned non-basmati white rice exports to control inflation, followed by a 20% export duty on parboiled rice. Additionally, a Minimum Export Price (MEP) of $950 per tonne was introduced for basmati rice to stabilise the domestic market. Earlier, between 2007 and 2011, India had also banned non-basmati rice exports due to concerns over food security. These frequent policy changes create uncertainty for exporters and global buyers, impacting India's position as the leading rice exporter.

- **Availability of Rice from Mandis:** The availability of rice at Mandis fluctuates due to various factors, including seasonal supply, procurement delays, and weather conditions. Erratic monsoons and labour shortages often lead to inconsistent arrivals, affecting mandi prices. Farmers also prefer selling to private traders instead of government agencies due to quicker payments and fewer procurement formalities. Additionally, mandi fees and transportation costs vary across states, leading to regional price disparities that impact supply chain efficiency.

- **Quality Issues:** Quality concerns remain a significant challenge in both domestic and export markets. Indian rice sometimes faces issues related to moisture content, broken grains, and pesticide residues, which can lead to rejections in international markets. Several shipments to the EU and the Middle East have been turned away due to excessive pesticide levels. Additionally, the lack of standardisation and grading mechanisms affects the marketability of Indian rice, making it difficult for exporters to command premium prices.

- **MSP vs. Export Market Prices:** The Minimum Support Price (MSP) for paddy in FY25 was set at Rs. 2,300 per quintal for the common variety. However, global rice prices often remain lower than the MSP, making Indian rice less competitive in export markets. This price disparity forces exporters to either seek government subsidies or operate at lower margins, reducing India's competitiveness against countries like Thailand and Vietnam, where production costs are lower.

- **Storage Issues & Supply Shortages:** Inadequate storage facilities pose a major problem for both farmers and the government. Due to limited warehousing infrastructure, farmers are often forced to sell their produce immediately after harvest at lower prices. Additionally, government warehouses face spoilage and wastage due to improper storage conditions. The lack of efficient procurement and distribution networks leads to supply shortages, impacting both domestic consumption and export commitments.

2.6 Extent of Competition

The rice industry in India faces significant competition domestically and internationally. By focusing on quality, innovation, and strategic policies, India can maintain its position as a leading rice producer and exporter.

Domestic Competition:

The competition in India's rice industry is multifaceted, involving a diverse mix of small, medium, and large players competing on various fronts such as price, quality, product differentiation, and branding. Regional preferences and the export market add further complexity. Success in this competitive landscape depends on innovation, quality assurance, and strategic marketing.

1. Climate Change: Variations in rainfall patterns, heatwaves, and extreme weather events can disrupt rice production in India, impacting its ability to compete in the global market.

2. Fragmentation: The Indian rice market is highly fragmented, with numerous small and medium-scale farmers contributing to overall production. This fragmentation can make it challenging to maintain consistent quality standards and compete with larger players in the market. Thus, the rice industry comprises a wide array of market players, including:

- Small-Scale Farmers: These players focus on local markets and often produce traditional and speciality varieties of rice, catering to regional tastes and preferences.

- Medium-Sized Enterprises: These companies operate at regional levels, supplying rice to both local and neighbouring states. They typically have a mix of traditional and hybrid varieties.

- Large Corporations: Big players like LT Foods (Daawat), KRBL Limited (India Gate), Amir Chand Jagdish Kumar (Aeroplane Rice), Kohinoor Foods Limited (Kohinoor) and Chaman Lal Setia Exports (Maharani) dominate the market with extensive distribution networks, advanced processing facilities, and strong brand recognition.

3. Product Differentiation: The competition is also driven by the variety of rice products available in the market, including Basmati Rice which is known for its aroma and long grains, basmati rice is a premium segment with significant export potential while Non-Basmati Rice, which comes in all different shapes and sizes (i.e. long, short, thick and

slender) with no aroma, includes a wide range of varieties catering to domestic consumption, with lower price points compared to basmati. Moreover, increasing health consciousness has led to a rise in demand for organic, brown, and speciality rice varieties, prompting players to diversify their offerings.

4. State Procurement: The Indian government's Minimum Support Price (MSP) program aims to ensure a minimum income for farmers, but can also create competition between private millers and government procurement agencies for paddy (unmilled rice). Seasonal fluctuations and changes in consumer demand impact pricing strategies.

5. Quality and Branding: Quality and branding play crucial roles in differentiating products in the competitive rice market. Key strategies include quality assurance, which aims to ensure high standards through quality control measures and certifications (e.g., organic certification), building brand loyalty through consistent quality, effective marketing, and strong distribution networks and innovation involving the introduction of fortified, flavoured, or quick-cook rice products to meet changing consumer preferences.

6. Interstate Competition: Rice-producing states in India compete with each other for domestic market share. This can lead to price fluctuations depending on regional surpluses or deficits. Major rice-producing states in India include West Bengal, Uttar Pradesh, Andhra Pradesh, Punjab, Haryana, Tamil Nadu, Odisha, and Bihar.

International Competition:

1. Major Exporters: India is the world's leading rice exporter, supplying over 150 countries. However, it faces competition from other major players like Vietnam, Thailand, and Pakistan, as these countries offer different rice varieties and may have lower production costs in some cases.

2. Trade Policies: International trade policies, including tariffs and export restrictions, can significantly impact the competitiveness of Indian rice in the global market. For example, export restrictions imposed by India to ensure domestic food security can make Indian rice less attractive to international buyers.

3. Competitors: In the Asia Pacific region, which is the largest consumer and producer of rice, countries like India, China, Indonesia, and Bangladesh are major competitors. These countries possess abundant rice resources and actively participate in both domestic and international markets.

4. Quality and Price: Indian rice needs to compete on both quality and price in the global market. While basmati rice enjoys a premium status, non-basmati varieties need to offer competitive pricing and consistent quality to sustain market share.

5. New Rice Varieties: Other rice-producing countries might develop new high-yielding or climate-resistant rice varieties, posing a challenge to Indian rice competitiveness.

3. Global and Indian Rice Industry

3.1 Global Rice Industry

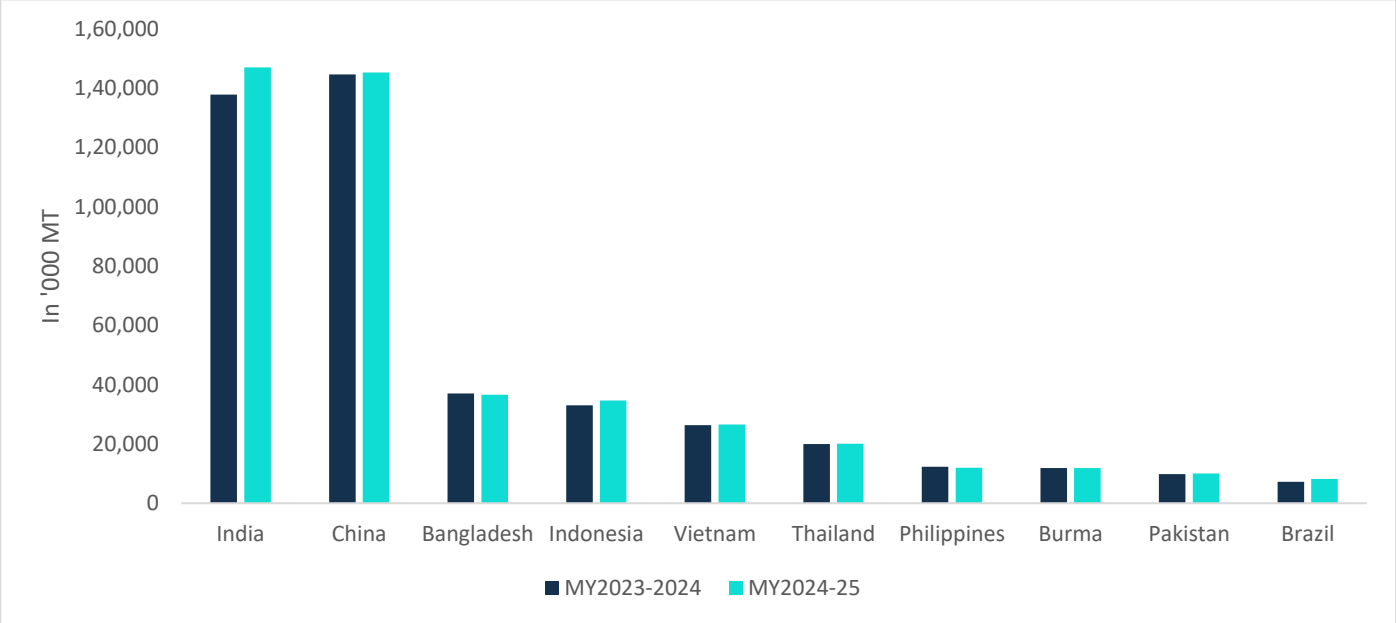
India's Dual Role as Top Exporter and Major Producer

The global rice industry is a critical segment of the agricultural sector, significantly contributing to food security, economic stability, and cultural traditions worldwide. Rice is a fundamental dietary staple which sustains millions of people globally. Asia is the epicentre of global rice production, with China, India, Indonesia, Bangladesh, Vietnam, and Thailand at the forefront. These countries benefit from favourable climatic conditions and have a long history of rice cultivation. Currently, India and China collectively account for more than half of the world's rice output. Supplying over half of the global population, nearly 83% of the world's rice is produced by just ten countries.

Global rice consumption is steadily increasing, driven by population growth, particularly in Asia and Africa. The productivity of rice cultivation has been significantly enhanced by technological advancements, including the development of high-yield varieties and improved irrigation techniques. These innovations have been pivotal in meeting the rising demand for rice, ensuring food security for a substantial portion of the global population. Rice is a staple food for over half of the world's population, especially in Asia, where it forms a substantial part of the daily diet. In this region, rice constitutes a significant portion of the daily diet, reflecting its importance not only as a food source but also as a cultural cornerstone.

Climate change and extreme weather events threaten rice production, impacting harvests and causing price fluctuations. Trade disruptions due to geopolitical tensions and logistical hurdles can affect rice availability and affordability. International trade in rice is crucial for food security. While other countries may cultivate basmati rice varieties, Indian basmati has Protected Geographical Indication (PGI) status, ensuring its authenticity and quality.

Chart 7: China and India dominate the Global Rice Production during MY 2023-24 and MY 2024-25

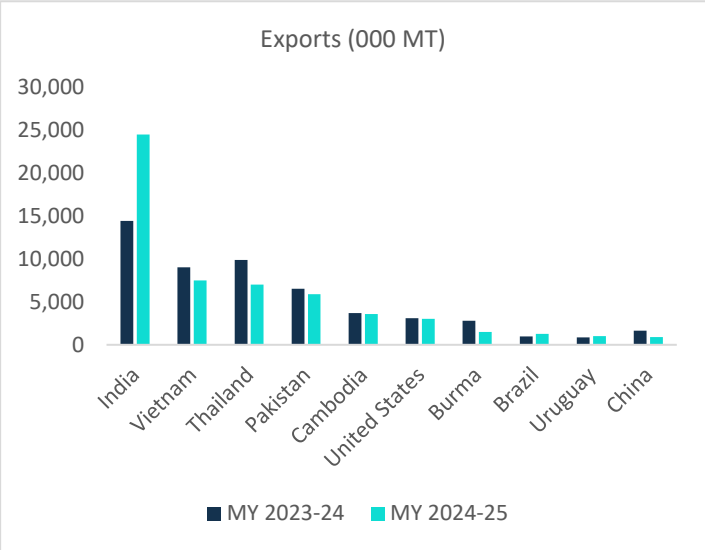


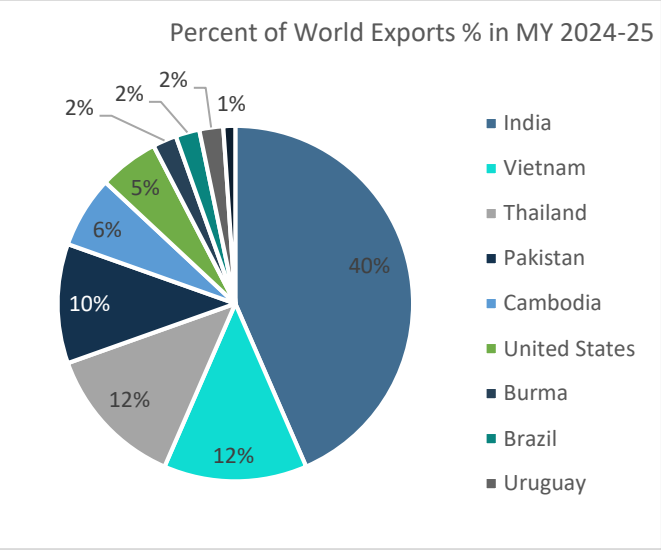
Source: United States Department of Agriculture (USDA)

Note: MY represents Marketing Year, and MY 2024-25 estimates are as of April 2025

Over 522.08 million metric tonnes (mmt) of milled rice were produced in the 2024-25 harvesting year at a global level, with India accounting for most of the upward revision. As per the USDA 2025 report, global rice production is estimated to surge by 3% (3.1 mmt) to a new record of 535.8 mmt in the marketing year 2024-25 as against the previous 2023-24 estimates of 522.08 mmt. Therefore, production is projected to reach a record high, driven by larger yields in Asia, particularly with increased production forecasts for Bangladesh, India and Pakistan, more than offsetting reductions for countries like Indonesia, Thailand and Japan. Global consumption is also anticipated to hit a new record, mainly fuelled by strong growth in India, Bangladesh, Indonesia and China as per MY 2024-25 estimates.

Chart 8: India is the largest exporter of rice to the world (MY 2024-25)

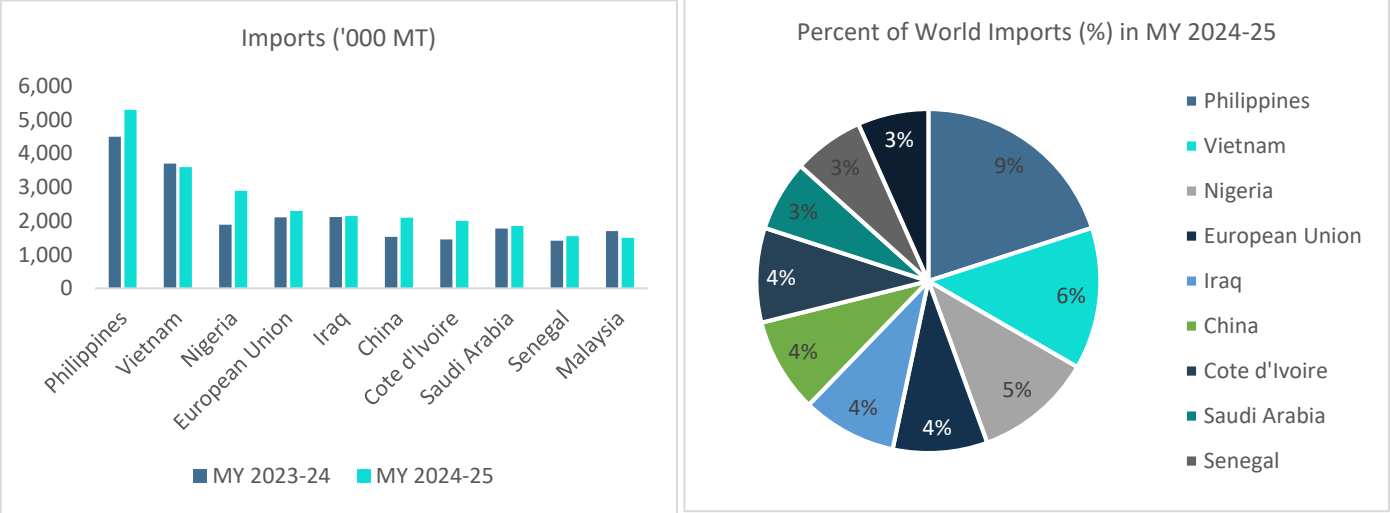




Source: United States Department of Agriculture (USDA)

During MY 2023-24, out of total global exports of 56.8 million metric tonnes, India had the highest export volume of rice worldwide, at 14.4 million metric tonnes, followed by Thailand with about 9.9 million metric tonnes of rice. Whereas during MY 2024-25, out of total global exports of 60.6 million metric tonnes, export volume from India increased by 70% with retained the highest share amongst other countries and Pakistan offsetting the same. Collectively, India, Thailand, Vietnam and Pakistan accounted for 74% of the dominant share globally during MY 2024-25. These countries export various types of rice, including jasmine, basmati, and parboiled rice. Thailand and Vietnam are known for their high-quality aromatic rice varieties, while India is famous for basmati rice. Rice cultivation is highly susceptible to changes in weather patterns, including droughts, floods, and rising temperatures, which can affect yields. Efforts are being made to develop climate-resilient rice varieties and sustainable farming practices.

Chart 9: Top Global Importers of Rice with % Share (MY 2024-25)



Source: USDA

Philippines, Vietnam, Nigeria, European Union and Iraq collectively accounted for 28% share of imports at global level during MY 2024-25 and import substantial quantities to supplement their domestic supply. Countries in Africa and the Middle East are significant importers of rice due to their insufficient domestic production to meet demand. Price fluctuations in the global rice market can have significant impacts on food security and economic stability, particularly in rice-dependent countries.

3.2 Trend in Indian Rice Industry

A better monsoon forecast will propel rice production in India.

Rice is the primary staple food crop cultivated nationwide, forming an essential part of the diet. India is the world's second-largest producer of rice and has witnessed significant growth in rice production over the years. According to the Ministry of Agriculture & Farmers Welfare, the total food grain production in the country as per the 2nd advanced estimates for 2024-25 was estimated at 3,309.19 lakh tonnes, which is lower by 13.8 lakh tonnes than the production of food grains of 3,322.98 lakh tonnes achieved during 2023-24. Out of total food grain production, production of rice constituted a 41.2% share amongst the other crops (Wheat, Maize, Pulses, Tur, etc).

Predominantly harvested in the fall, rice is a kharif season crop, requiring substantial water during transplanting and early growth stages, which the southwest monsoon (June-September) provides. The planting and growth of rice in India depend heavily on the arrival of the southwest monsoon in June or early July and its continuation through September.

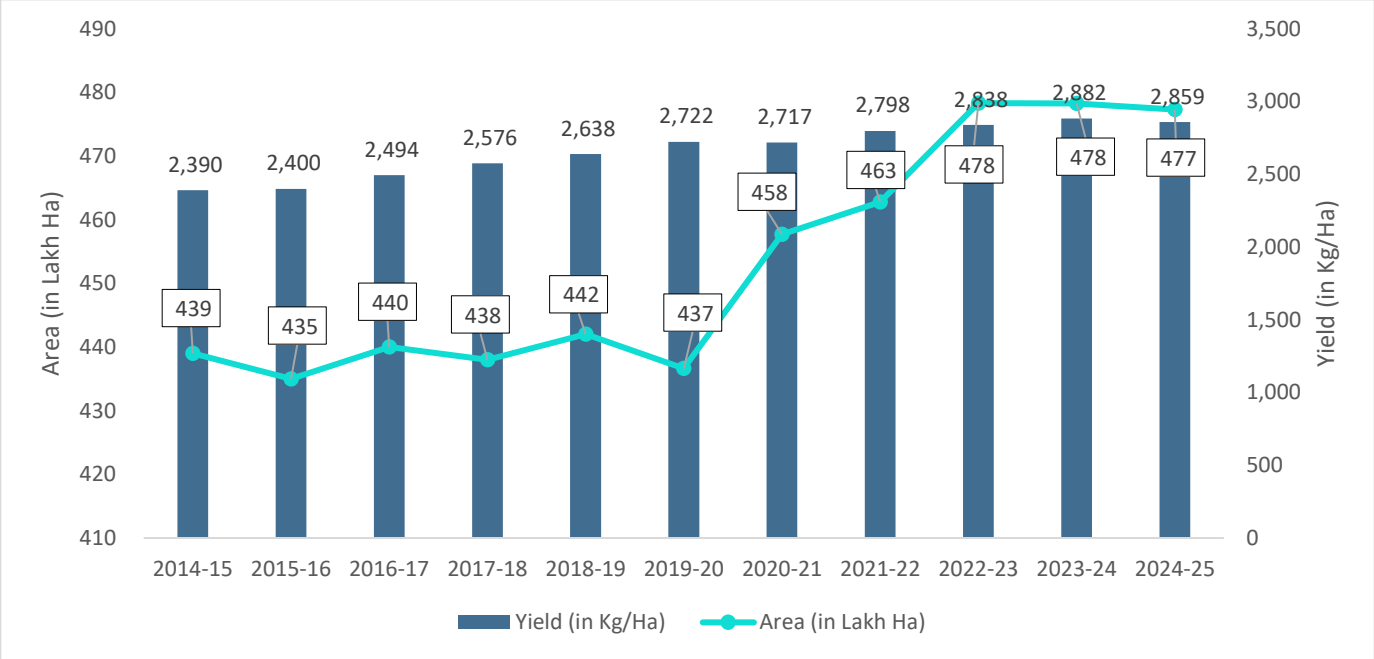
3.2.1 Trend of Rice Area under Cultivation and Yield

India has a total rice cultivation area of approximately 477 Lakh Hectares as of 2024-25, hovering under irrigated as well as rainfed conditions, which signifies a substantial portion of the country's total agricultural land, indicating the crop's importance in the agrarian economy. The rainfed areas are often affected by vagaries of the monsoon, making them more fragile. The area under rice cultivation has seen a consistent increase with minor fluctuations over the years, growing by 9% to 477 Lakh Hectares both in 2024-25 as against 435 Lakh Hectares in 2014-15. Similarly, yield is increased by 20% to 2,859 Kg per Hectare in 2024-25 as against 2,390 Kg per Hectare in 2014-15.

Supportive government policies, including subsidies and minimum support prices (MSP), along with the expansion of irrigation facilities through initiatives like the Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), ensuring a reliable water supply, have significantly enhanced the suitability of land for rice farming. Investments in rural infrastructure,

such as improved storage facilities, transportation, fertiliser management, and crop insurance, have further encouraged farmers to dedicate more land to rice cultivation. The introduction and adoption of high-yield and hybrid rice varieties have provided farmers with the potential for higher returns, motivating them to expand their cultivation areas. Additionally, enhanced transportation networks and market access enable farmers to sell their produce more efficiently, supporting their economic stability.

Chart 10: Trend of Rice Area under Cultivation and Yield



Source: Department of Agriculture and Farmers Welfare
Note: Data for the year 2024-25 are based on 2nd Advance estimates published in April 2025

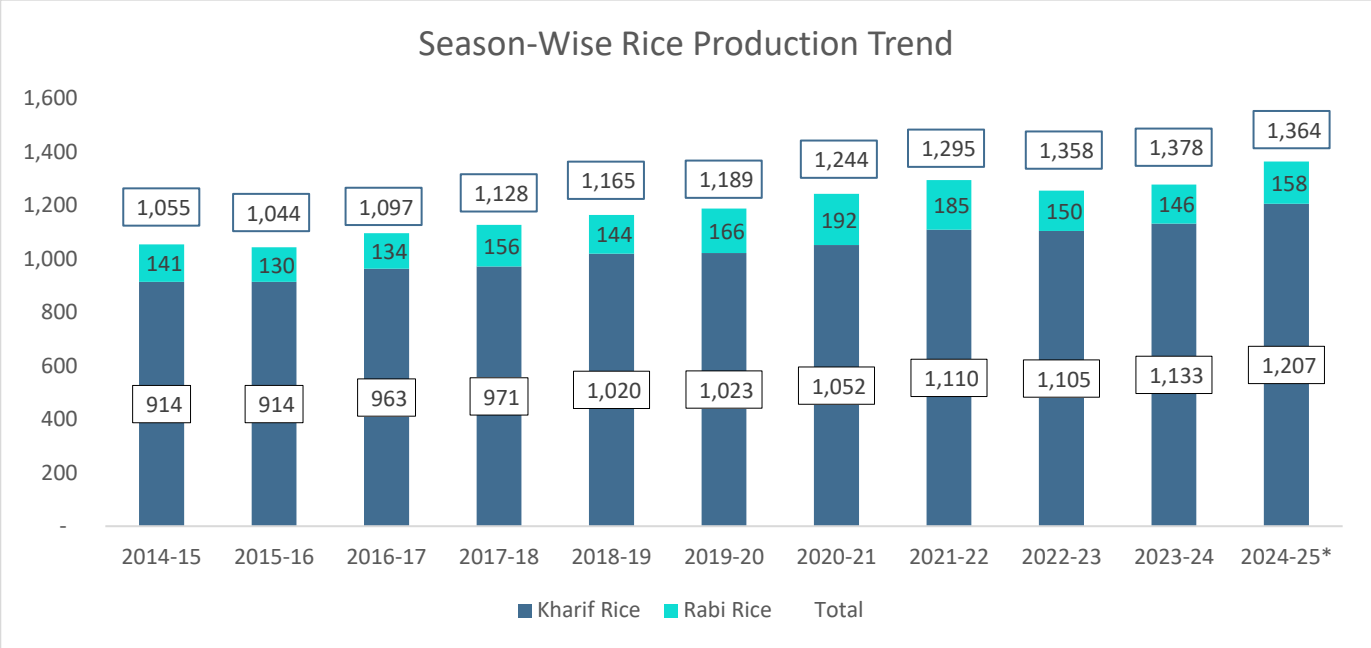
The Indian government has implemented several policies to promote rice cultivation, including subsidies for seeds, fertilisers, and machinery. Schemes such as the National Food Security Mission (NFSM) and the Rashtriya Krishi Vikas Yojana (RKVY) focus on boosting rice production. These combined efforts have led to the expansion of rice cultivation areas and increased yields, effectively meeting the growing demand for this staple crop and ensuring not only food security for the nation but also contributing to the livelihoods of millions of farmers across the country.

3.2.2 Trends in Supply of Rice

The production of rice is influenced by factors like climate, soil quality, and agricultural practices. The trend in rice supply in India has shown a consistent increase in both Kharif and Rabi seasons, with a few fluctuations. From 2014-15 to 2020-21, rice production in both seasons rose steadily. During this period, Kharif rice production increased from 914 lakh tonnes to 1,052 lakh tonnes, while Rabi rice production increased from 141 lakh tonnes to 192 lakh tonnes.

The total production of rice during the 2023-24 marketing season (October-September) was estimated at record levels of 1,378 lakh tonnes, higher by 0.14% than the previous year's production of 1,358 lakh tonnes. This remarkable surge in rice production can be attributed to various factors, including improved agricultural practices, increased acreage, government support for the rice farming sector and favourable weather conditions during the season.

Chart 11: Rice production in India commands a significant 41% share among other crops during 2024-25

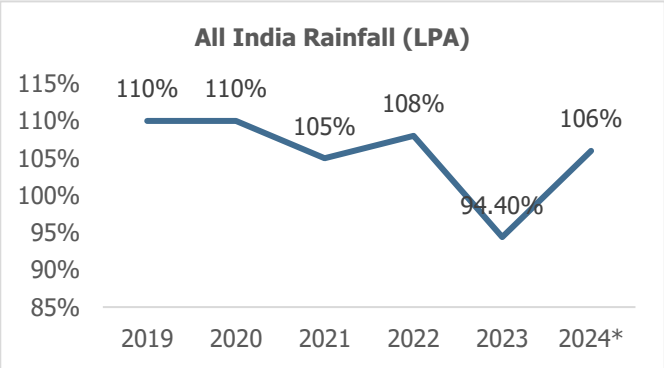


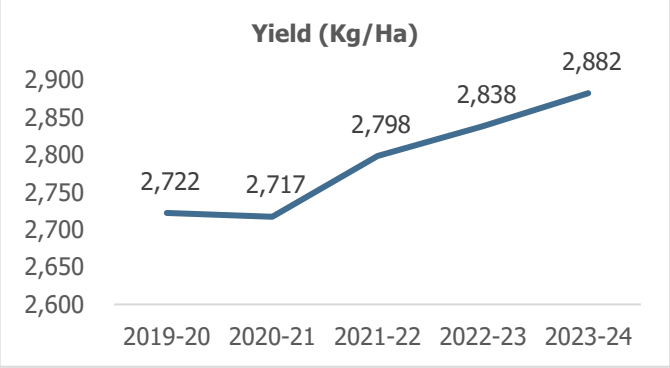
Source: Department of Agriculture & Farmers Welfare (DA&FW)
*: Second Estimates of Production of Food Grains for 2024-25 published on 11.03.2025

As per the latest final estimates for 2024-25 by the Department of Agriculture and Farmers Welfare, the total rice production is estimated at 1,364 lakh tonnes (marginally down by 1% as against the previous year) in the sowing season, coupled with the erratic monsoon affecting crop yields in some regions. As per the USDA, farmers realised higher returns from rice over other crops on relatively firm prices on strong domestic demand as well as higher government MSP for paddy rice. Timely and well-distributed 2025 monsoon rains will be critical for the forecast area planted and yields, as more than 40% of the rice acreage is largely rainfed (unirrigated). The growing likelihood of higher summer temperatures going forward will be a key monitorable.

An above-normal southwest monsoon for the current year has brightened the prospects of agriculture and rural demand. However, much would depend on geographical spread of rainfall during the southwest monsoon season which has been uneven over the past few years.

Chart 12: IMD Forecasts for All India Rainfall (in LPA) and Yield (Kg/Ha)



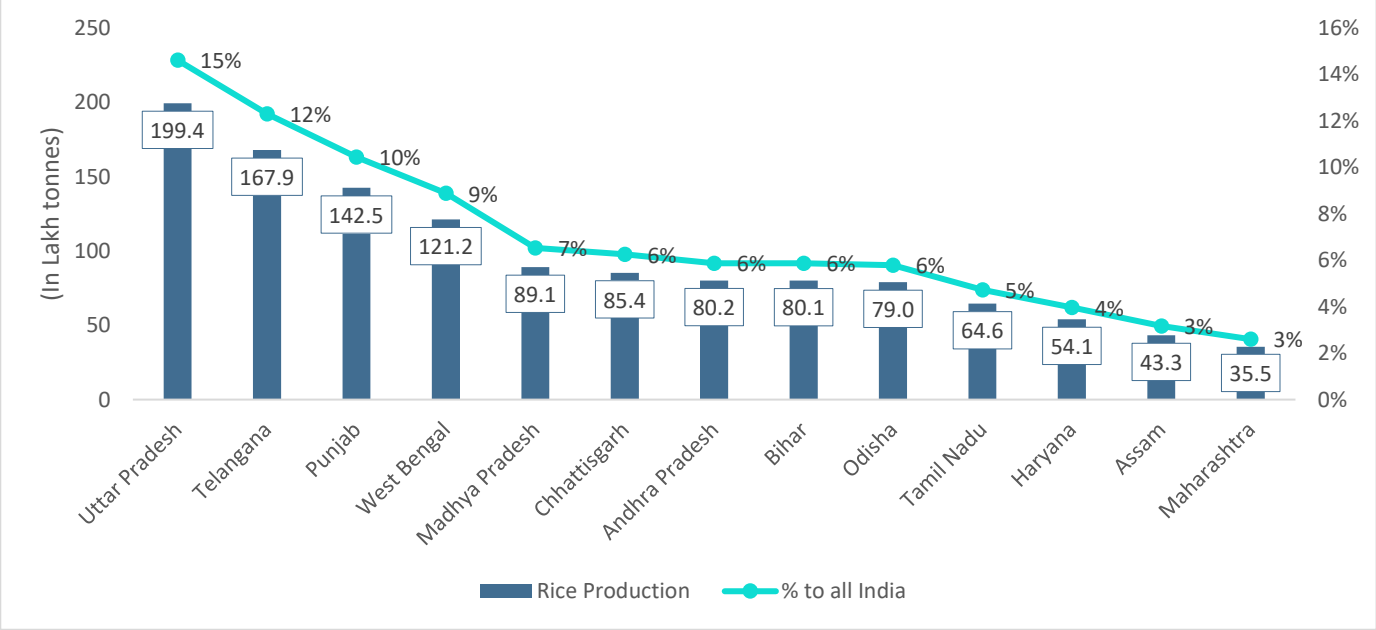


Note: 2023 and 2024 covers Southwest Monsoon season (June-September) and 2019-2022 covers All India Rainfall Pattern (in LPA)

Final estimates of Yield have been taken from Ministry of Agriculture and Farmers Welfare and years considered here from October-September.
*: Estimates

The rainfall for the country as a whole, for the Annual (Jan-Dec)-2022 has been recorded as 1257 mm which is 108% of its Long Period Average (LPA). The rainfall in India in the past decade has decreased marginally owing to extreme weather conditions. This resulted in huge crop losses, adversely impacting the industry. Monsoons are integral to India, providing about 70% of annual rainfall and affecting major crops such as rice, wheat, sugarcane, soybeans, and groundnuts.

Chart 13: Rice Production in Major Indian Producing States during 2024-25



Source: Ministry of Agriculture & Farmers Welfare
Note: Data for the year 2024-25 is as per 2nd Advanced Estimates

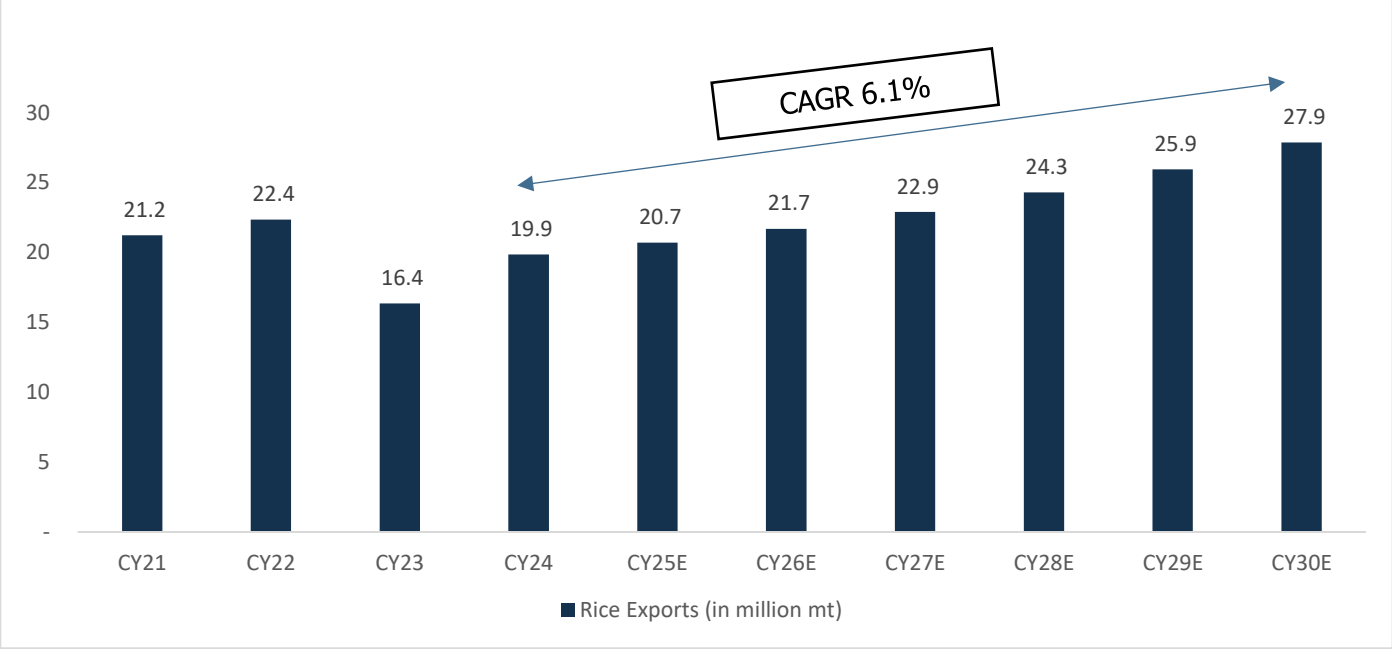
The major rice cultivating states in India are Uttar Pradesh, Telangana, Punjab, West Bengal, Madhya Pradesh, Chhattisgarh, Andhra Pradesh, Bihar, Odisha, Tamil Nadu, Haryana, Assam and Maharashtra etc. As per Ministry of Agriculture & Farmers Welfare, Uttar Pradesh, Telangana, Punjab, West Bengal, Madhya Pradesh, Chhattisgarh and Andhra Pradesh together accounted to 65% of all India level production in marketing year 2024-25.

3.2.3 India's Export Destinations

India's rice in recent years has shown significant growth. In CY24 the total rice exports of India increased by 21% y-o-y to 19.9 million metric tonnes as compared to 16.4 million metric tonnes in CY23. Basmati rice exports surged to a record 5.9 million metric tons in CY24, driven by strong demand from key markets such as Saudi Arabia, Iraq, and the United Arab Emirates. Non-Basmati rice also showed strong growth of 25% y-o-y in CY24.

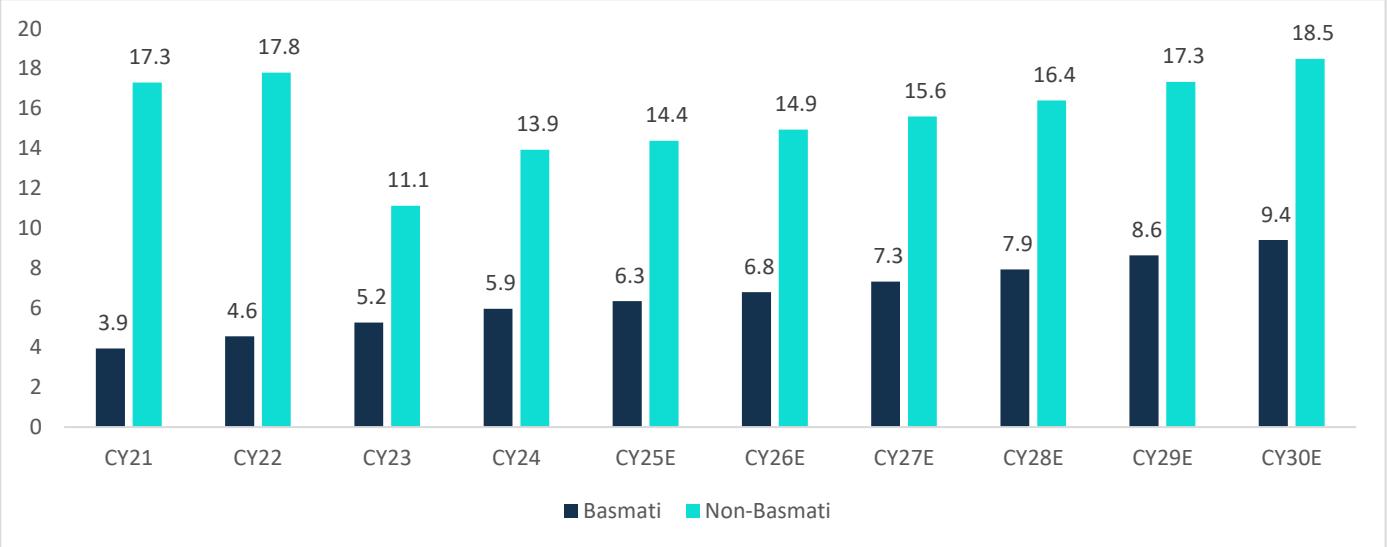
Going forwards with favourable monsoon conditions as well as favourable trade policies expected to boost crop yields, the rice exports are expected to grow at a CAGR of 6% between CY24 to CY30. This expansion is seen as vital for the country's economy, where agriculture plays a critical role for 42% of the population. Further Basmati rice and non-basmati rice are expected to grow at a CAGR of 8% and 5% respectively between CY24 to CY30.

Chart 14: Total World Exports from India (in million mt)



Source: CMI, CareEdge Research.
Note: E indicates Estimated

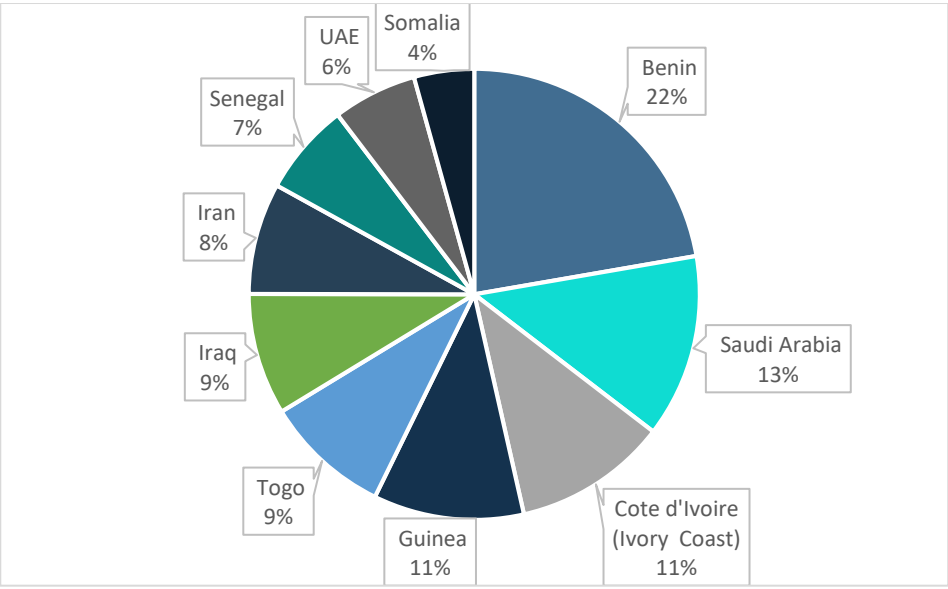
Chart 14 Basmati and Non-Basmati World Exports from India (in million mt)



Source: CMI, CareEdge Research.
Note: E indicates Estimated

Top countries contribute up to 50% of India's Rice Exports in 2024-25. India's robust domestic demand, strategic reserves, and significant rice production contribute to its role as a major player in the global rice market. Any reduction in India's rice exports could impact global food inflation. India, being leading exporter of rice, caters to a wide range of global markets. The country's rice export destinations span across various continents, which highlights its significant role in the international rice trade.

Chart 15: Middle East Accounting for Larger Share in India's Total Exports of Rice in FY25*, (In Tonnes)



Source: CMIE
Note: FY25 includes data from (April-February 2025)

The Middle East stands out as one of the largest importers of Indian rice, especially Basmati rice, with major markets including Saudi Arabia, Iran, the UAE, Iraq, and Kuwait. Africa, on the other hand, is a vital market for non-Basmati rice from India, with countries like Benin and Senegal meeting local staple food needs. In Asia, countries such as Bangladesh, Nepal, China, and Malaysia import both Basmati and non-Basmati rice varieties.

Trade policies, geopolitical factors, and the domestic production levels of importing countries significantly influence export patterns. Consequently, Basmati rice predominates in exports to the Middle East, Europe, and North America, while non-Basmati rice is more favoured in Africa and parts of Asia. India mainly exports parboiled rice to West Africa and basmati rice to the Middle East.

In March 2024, as part of India's outreach to the Global South, the government has approved the export of 110,000 tonnes of rice to three African countries to support their food security needs. According to a notification by the Directorate General of Foreign Trade (DGFT), 30,000 tonnes of non-basmati white rice will be exported to Tanzania, while Djibouti will receive 30,000 tonnes of broken rice, and Guinea-Bissau will receive 50,000 tonnes.

El Nino is a climatic phenomenon characterised by warmer-than-average ocean surface temperatures in the central and eastern tropical Pacific Ocean. Hence, El Nino-led drought severely impacted Asian rice fields, reducing yields and tightening global rice supplies.

Although the export of non-basmati white rice has been banned since July 2023, to ensure adequate domestic supplies and control inflation, some exports are being permitted to friendly nations to support their food security, which has been disrupted due to the ongoing Russia-Ukraine war. Further measures included restricting non-basmati white rice

exports, a 20% duty on parboiled rice and specific milled rice varieties, effective from May 1, 2025. The government also removed the minimum export price (MEP) for basmati rice of USD 950 per metric tonne.

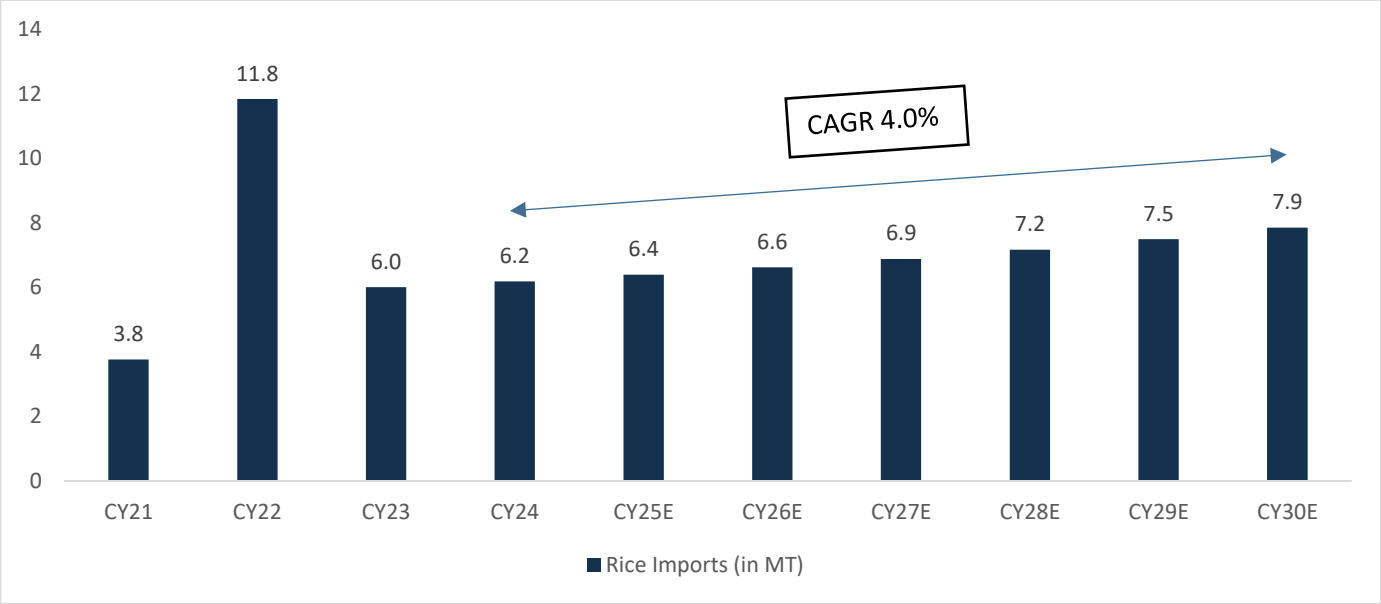
3.2.4 Import of Rice to India

India is one of the world's largest producers and exporters of rice and imports usually happen when there is a shortfall in domestic production due to adverse weather conditions, such as droughts or floods, or when specific varieties not widely grown in India are needed for niche consumption.

Most rice imports come from countries like the Philippines, Vietnam and Nigeria. In recent years, India's rice imports have remained minimal compared to its exports, as domestic production continues to meet most of the country's consumption needs. In CY24, the total rice imports of India increased marginally by 3% y-o-y to 6.2 million metric tonnes as compared to 6.0 million metric tonnes in CY23. Basmati rice imports remained stable, whereas non-basmati rice showed an uptick of 4% y-o-y in CY24.

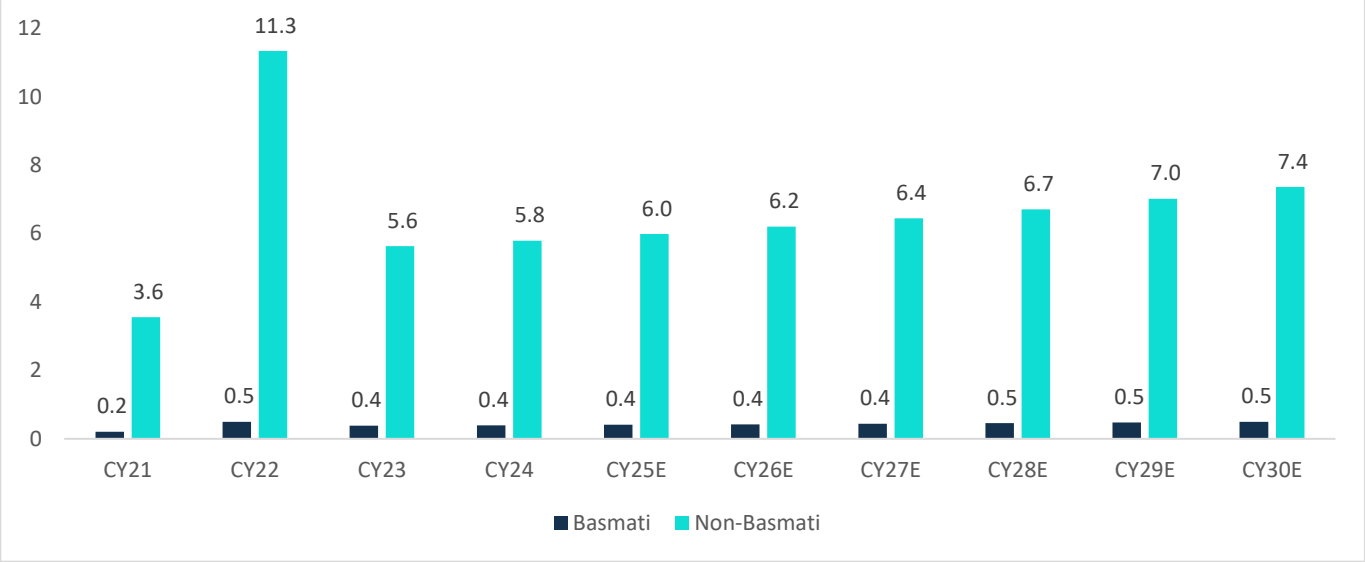
Going forward, the rice exports are expected to grow at a CAGR of 4% between CY24 to CY30. Further, Basmati rice and non-Basmati rice are expected to grow at a CAGR of 4% each, between CY24 to CY30.

Chart 16: Total World Imports to India (in mt)



Source: CMI, CareEdge Research.
Note: E indicates Estimated

Chart 17: Basmati and Non-Basmati World Imports to India (in mt)



Source: CMI, CareEdge Research.
Note: E indicates Estimated

3.2.5 MSP in Rice

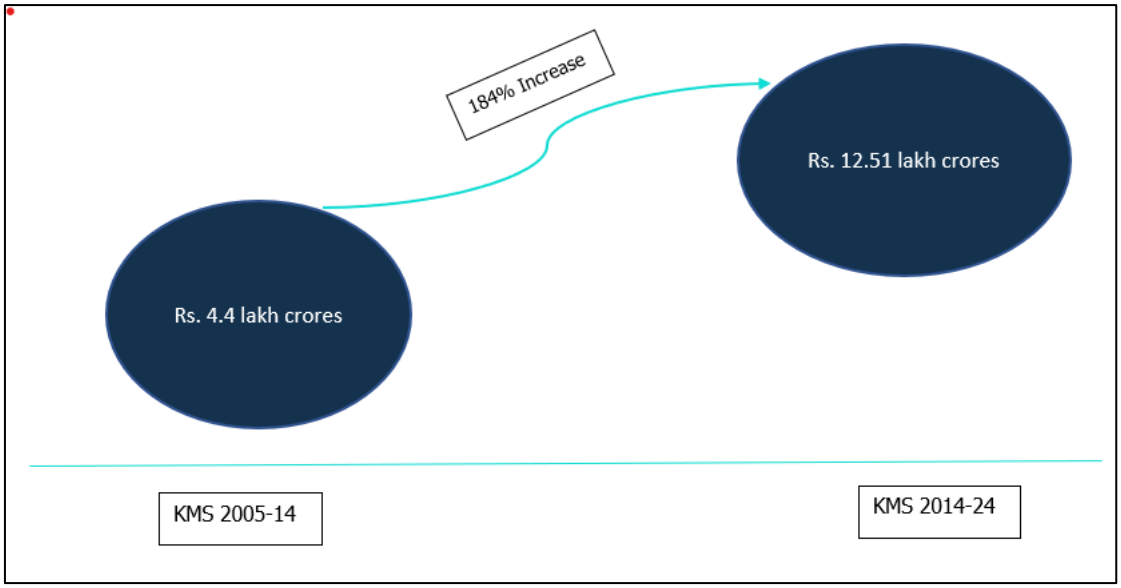
A key policy to support farmers, ensure food security, and stabilise prices in India

The Minimum Support Price (MSP) in the rice industry serves as a significant policy tool implemented by the Indian government to ensure price stability, income security, and food security for farmers. The MSP is a price floor set by the Indian government to safeguard farmers against market volatility and ensure a minimum level of income for their produce. This mechanism acts as a safety net, ensuring farmers receive a fair price for their crops, thereby fostering increased production and ensuring food security.

In the rice sector, MSP announced by the government is for various rice varieties, including common varieties like non-Basmati rice and premium varieties like Basmati rice. The MSP is determined based on factors such as production costs, market prices, input costs, and demand-supply dynamics. The announcement of MSP assures rice farmers that they will receive a fair price for their produce, irrespective of market fluctuations. This encourages farmers to invest in rice cultivation, adopt modern agricultural practices, and increase productivity.

MSP also plays a vital role in stabilising rice prices in the market, as it serves as a reference price for procurement agencies and private traders. The government often procures rice at MSP through agencies such as the Food Corporation of India (FCI) to build buffer stocks and fulfil public distribution system (PDS) requirements.

Chart 18 : MSP paid to farmers for paddy procurement increased which benefitted their lives



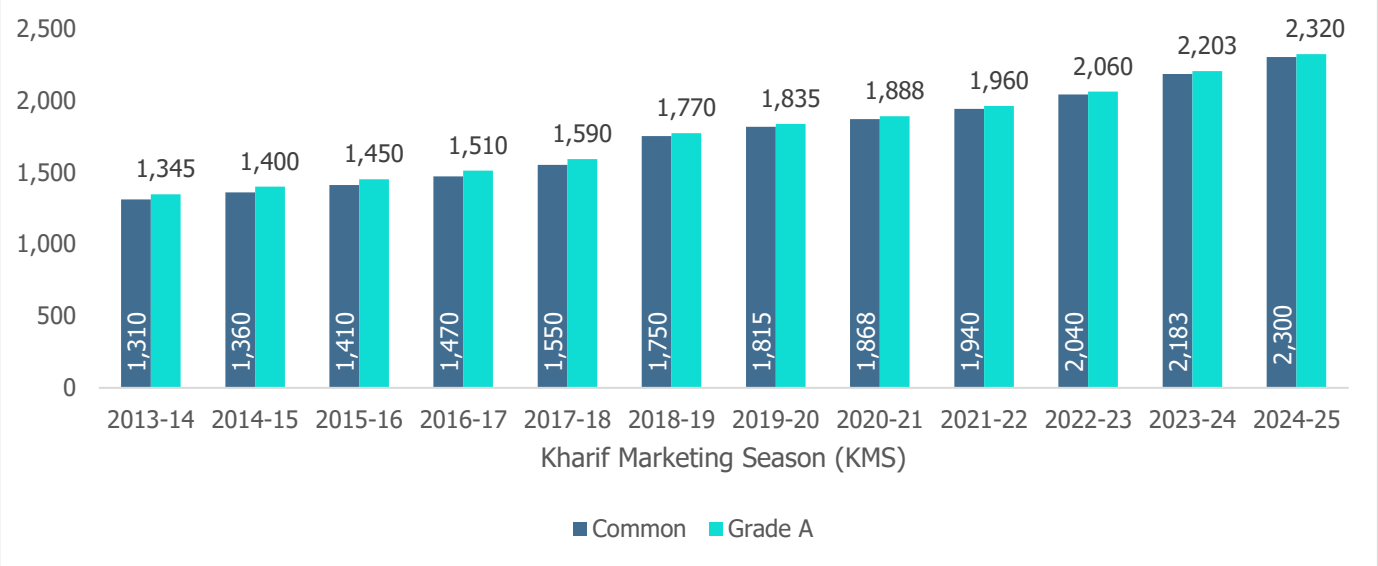
Source: Ministry of Consumer Affairs, Food and Public Distribution

The government has been urged to revise MSPs regularly based on comprehensive cost calculations and market dynamics to better support farmers' livelihoods, ensure fair and remunerative prices for farmers, enhance procurement efficiency, and promote sustainable agriculture.

Overall, MSP in the rice sector remains a critical policy instrument aimed at supporting farmers, ensuring food security, and stabilising rice prices in India. Its effectiveness, however, hinges on robust implementation, periodic revisions, and complementary policies to address structural challenges in agriculture.

The current MSP for rice in India applies to paddy, which is unmilled rice. The paddy procurement is undertaken by both state-owned FCI as well as private agencies. The paddy is procured at the MSP directly from the farmers and is utilised for meeting the demand under several welfare schemes. Basmati rice is the costliest product in the world since its price is mostly set and it commands high rates on the worldwide market.

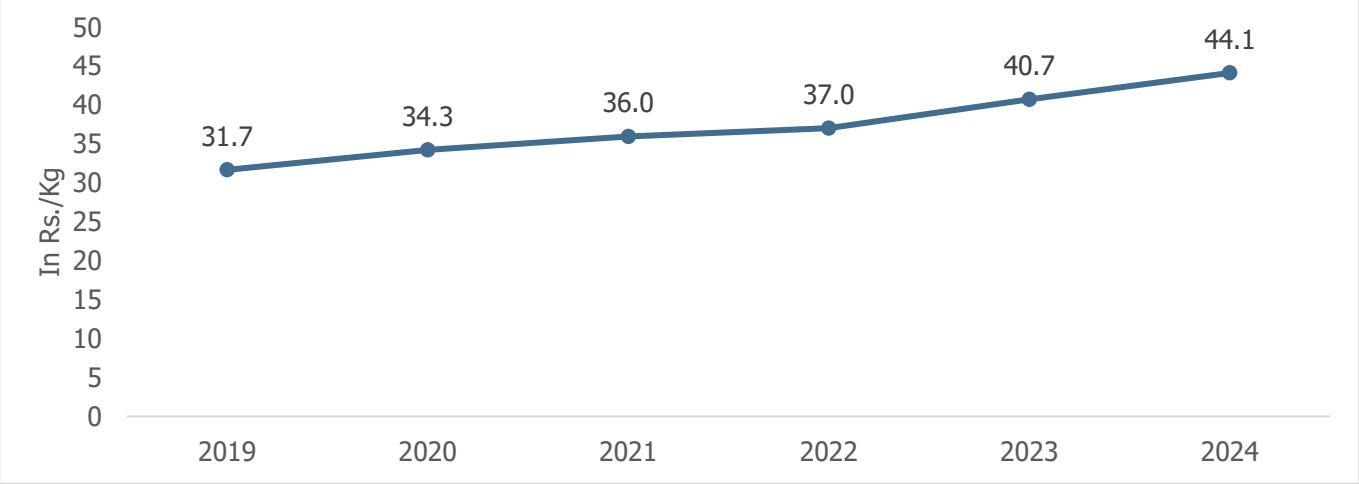
Chart 18: MSP for Paddy over Past Years



Source: CMIE, Ministry of Agriculture and Farmers Welfare, Press Information Bureau, dated June 2024

The MSP trend for both Paddy-Common and Paddy-Grade A has consistently increased over the last decade. This rise reflects the government's commitment to providing fair compensation to farmers and supporting their livelihoods. During Kharif Marketing Season (KMS) 2023-24, the government has announced higher MSP at Rs. 2,183 per quintal (increased by around 7% over last year) for Paddy- 'Common', while Rs. 2,203 per quintal (increased by 6.9% over last year) for Paddy-Grade 'A', demonstrating the government's ongoing efforts to enhance the income security of agricultural producers. This is the second steepest increase in the last decade, and the last highest increase of ₹200 per quintal in the paddy MSP was recorded in 2018-19. Similarly, for KMS 2024-25, the government has increased the MSP by 5% for both Common and Grade A to Rs. 2,300 per quintal and Rs. 2,320 per quintal, respectively, thereby ensuring remunerative prices to the farmers for their produce. Hence, as per the Ministry of Agriculture and Farmers Welfare, the decision of the government to increase the MSP of Kharif crops fulfils the commitment to the farmers to provide at least 50% return over the cost of production.

Chart 19: Price Trend of Rice (All India Average)



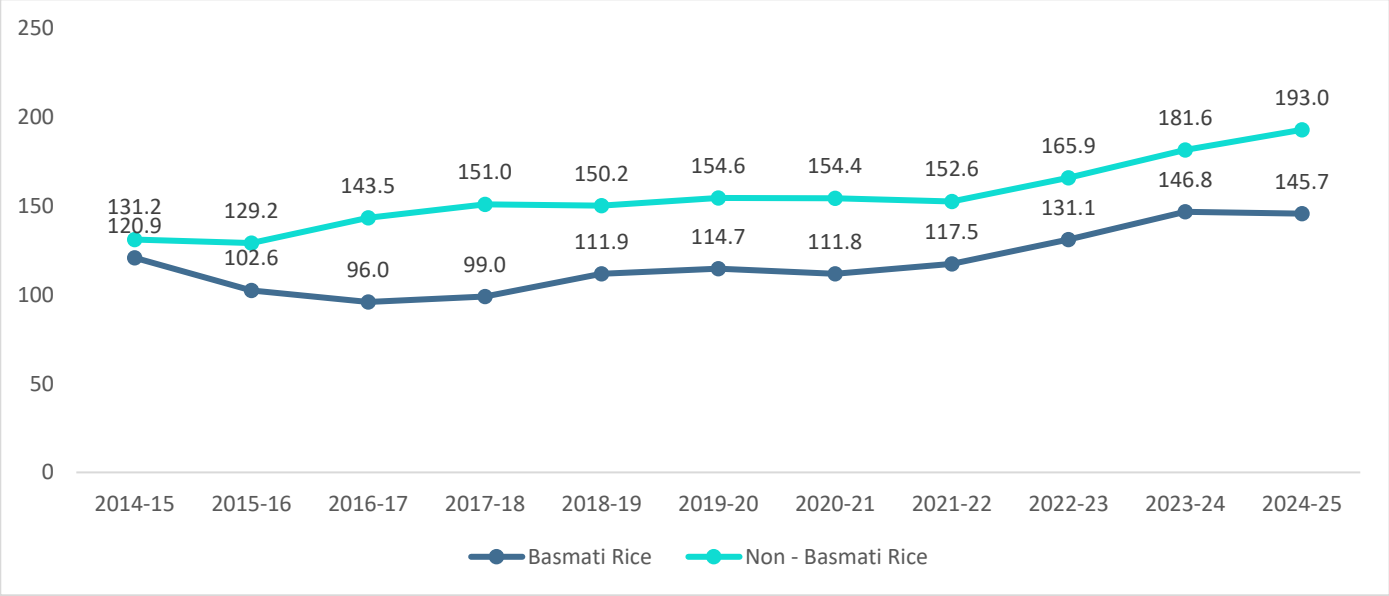
Source: Department of Consumer Affairs (Price Monitoring Division)

Globally, weather disruptions and El Nino risks in Southeast Asia, a major growing region, have impacted market dynamics. India, the largest rice exporter, imposed export restrictions in 2022 and 2023, leading to soaring prices. Amid the rise in rice prices, the Union government has extended the 20% export duty on parboiled rice indefinitely.

Government interventions, including Minimum Support Prices (MSPs), procurement policies, subsidies, and trade regulations, play a significant role in determining rice prices. MSPs act as a price floor, providing a guaranteed minimum price to farmers and influencing market prices. Government procurement operations, particularly for public distribution and buffer stocking, also impact market prices.

Rice prices are influenced by the balance between production and consumption within the country. Any disruptions in production due to adverse weather conditions, such as droughts, floods, or pest infestations, can lead to fluctuations in prices. Historically, rice prices in India have experienced fluctuations influenced by seasonal variations in supply, market demand, and international trade dynamics. Generally, prices tend to fluctuate throughout the year, with peaks often observed during the harvesting season due to increased supply and troughs during the off-season or lean periods. The prices have shown an increasing trend over the past years in India. It has increased by 10% in CY23 as against CY22. Moreover, the cost of storage and transport has also increased in the last few years.

Chart 20: Wholesale Price Index for Rice (Basmati & Non-Basmati Rice)



Source: CMIE

The Wholesale Price Index (WPI) for Basmati and Non-Basmati rice in India varied based on market conditions, demand-supply dynamics, and other factors. The WPI measures the changes in the average wholesale prices of goods and services over time and serves as an important indicator of inflationary trends in the economy.

For Basmati rice, the WPI reflects the wholesale prices of different Basmati rice varieties, including traditional Basmati, Pusa Basmati, 1121 Basmati, and others. Similarly, for Non-Basmati rice, the WPI covers various non-Basmati rice varieties commonly consumed in India, such as Sona Masuri, Ponni, IR-64, and others.

The WPI for Basmati and Non-Basmati rice can fluctuate based on factors such as production levels, market demand, government policies, international trade dynamics, and seasonal variations. During periods of high demand or supply

shortages, rice prices tend to increase, leading to a corresponding rise in the WPI. Conversely, during periods of oversupply or weak demand, rice prices may decrease, resulting in a decline in the WPI.

3.3 India Dominating Global Basmati Rice Market

Basmati rice production in India is a significant component of the country's agricultural sector and economy. It occupied a special status in rice cultivation. As per the Crop Survey conducted for basmati rice during 2023, India is one of the largest producers of Basmati rice globally, producing around 75% of the world's basmati rice, renowned for its premium quality grains. According to the Directorate General of Commercial Intelligence and Statistics data, India exported basmati rice to 150 countries in 2023-24, while in the 2022-23 fiscal year, it had sold the commodity to 149 countries. Iran, Saudi Arabia, the United Arab Emirates, and Iraq are the main importers among them. The areas of Basmati Rice production in India are the states of Jammu and Kashmir, Punjab, Haryana, Delhi, Uttarakhand, Himachal Pradesh, and western Uttar Pradesh. These regions provide the ideal agro-climatic conditions, including fertile soil, temperate climate, and access to water from rivers like the Ganges and Yamuna, essential for the cultivation of Basmati rice.

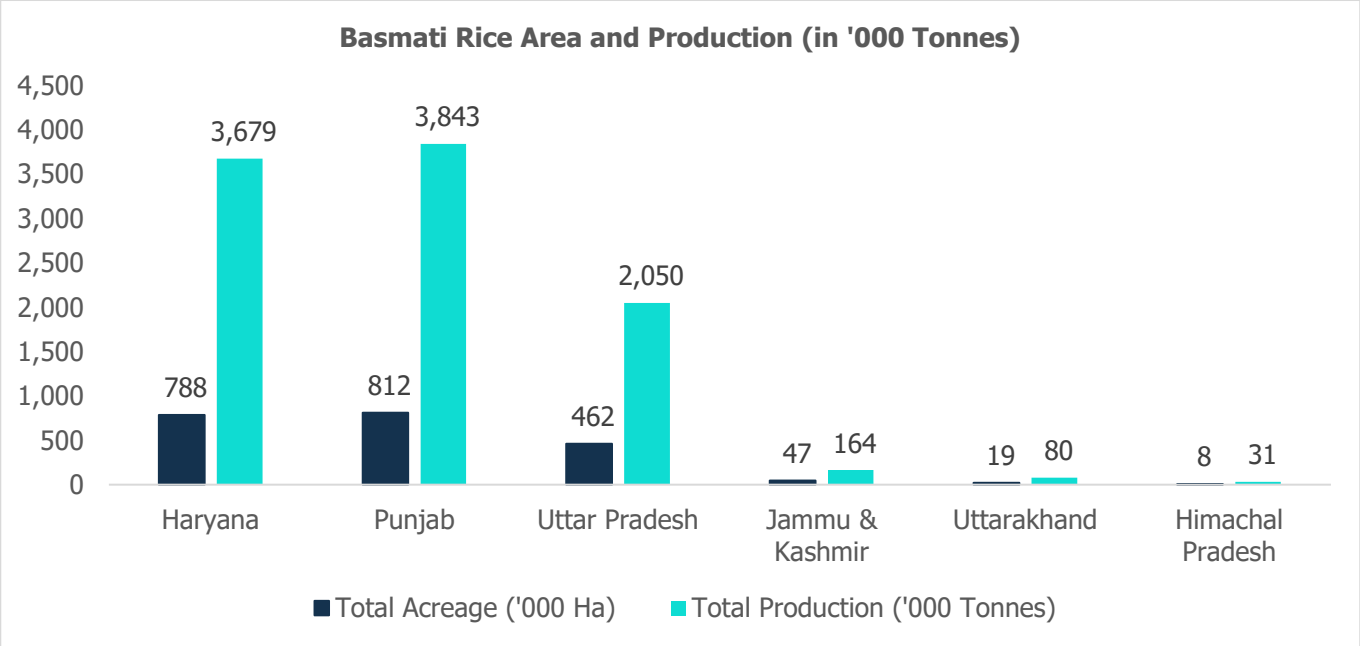
The basmati rice is also stated to be the Pearl of Rice. Basmati rice cultivation in India typically follows traditional methods, although modern agricultural practices are also being adopted to enhance productivity and quality. Farmers employ techniques such as direct seeding or transplanting of seedlings, controlled irrigation through canal systems or tube wells, and the use of organic fertilisers and pesticides to maintain soil health and minimise environmental impact. Basmati rice is harvested manually to ensure minimal damage to the delicate grains. The harvested paddy undergoes milling and processing to remove the husk, bran, and other impurities, resulting in polished white grains ready for consumption.

The basmati rice industry in India is predominantly unorganised, with approximately only 30-40% of production being managed by major players. The necessity for extensive branding and marketing efforts deters small and medium-sized enterprises (SMEs) from entering the basmati segment. Additionally, rice millers must be situated near paddy-growing areas to ensure continuous milling operations year-round.

In the non-basmati segment, less than 10% of the market is organised, given the presence of small players and their limited advertising and sales-related spends.

The Agricultural and Processed Food Products Export Development Authority (APEDA), Ministry of Commerce and Industry, had conducted a Basmati crop survey to estimate acreage, assess crop health and expected yield of aromatic and long grain rice during the 2022-2023 kharif crop season using climate-based yield modelling. As per the survey model, field-based as well as satellite imagery surveys is being carried out based on a sample group of farmers selected at district level in seven Basmati producing states of Punjab, Haryana, Himachal Pradesh, Uttarakhand, Delhi, western Uttar Pradesh (30 districts) and three districts in Jammu & Kashmir. As per the survey model, field-based as well as satellite imageries survey is being carried out based on sample group of farmers selected at the district level in seven Basmati producing states of Punjab, Haryana, Himachal Pradesh, Uttarakhand, Delhi, western Uttar Pradesh (30 districts) and three districts in Jammu & Kashmir. The cultivation and quality of Basmati rice are influenced by various factors, including geographical conditions and agricultural practices. Efforts have been made to promote the use of certified seeds and good agricultural practices to meet international standards.

Chart 21: State-level Satellite data and Field-based Basmati Rice Area and Production during Kharif 2023

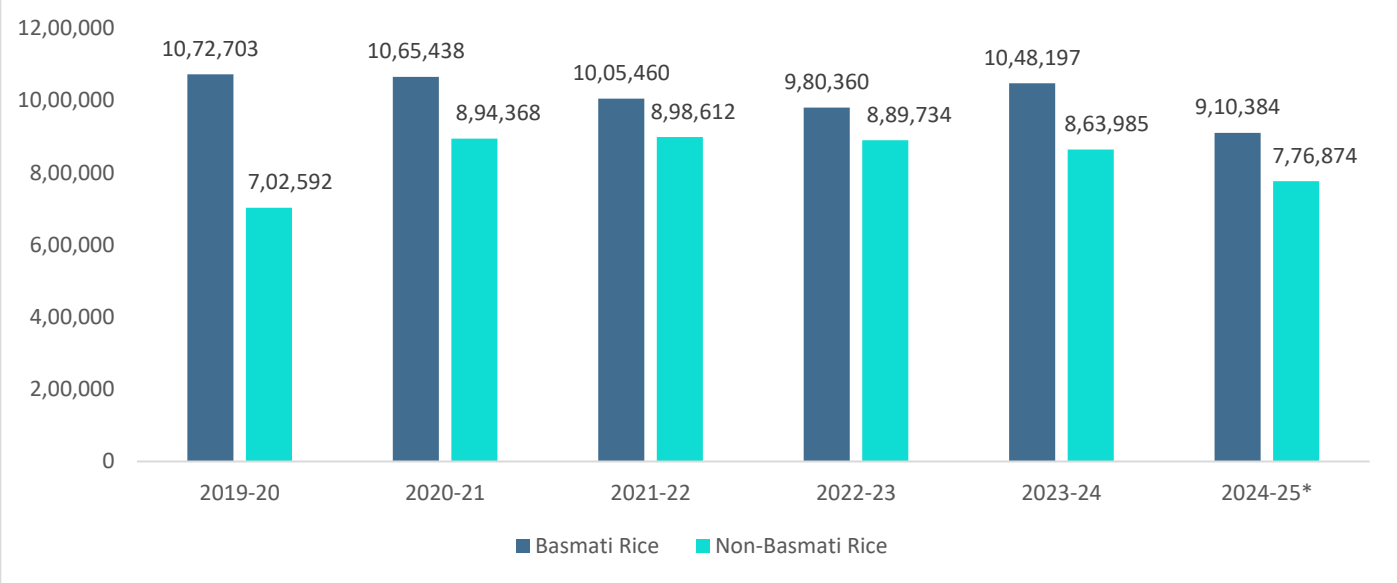


Source: Crop Survey Report by APEDA as of Dec 2023

Note: The above estimates for basmati acreage and production were done from July-October covering the majority of districts in the above states.

The above chart depicts that states like Punjab, Haryana and Uttar Pradesh collectively constitute over 97% share in the total production of basmati rice amongst the above regions with the highest acreage. The acreage of Basmati Rice in Haryana is 788 thousand hectares, with the highest acreage of sown varieties being PB 1121, PB 1718 and PB 1885 collectively. In Punjab, the cropping pattern suggests that Basmati varieties are consistently spread across the state and found in almost all the districts. The highest acreage of sown varieties is PB 1121, PB 1718 and PB 1885 collectively, followed by PB 1509, PB 1692, PB 1847 and PB 1401, PB 01, PB 06, and PB 1882. The total production figures of the Basmati Varieties in Punjab are 3,843 thousand tonnes, which is 4% higher than production in Haryana with 3,679 thousand tonnes. In Uttar Pradesh, the acreage of sown varieties of PB 1509, PB 1692, and PB 1847 collectively is higher than PB 1121, PB 1718, and PB 1885. The estimated production of Basmati Rice is 2,050 thousand tonnes.

Chart 22: Production Trend of Basmati and Non-Basmati Rice in India (In Tonnes)



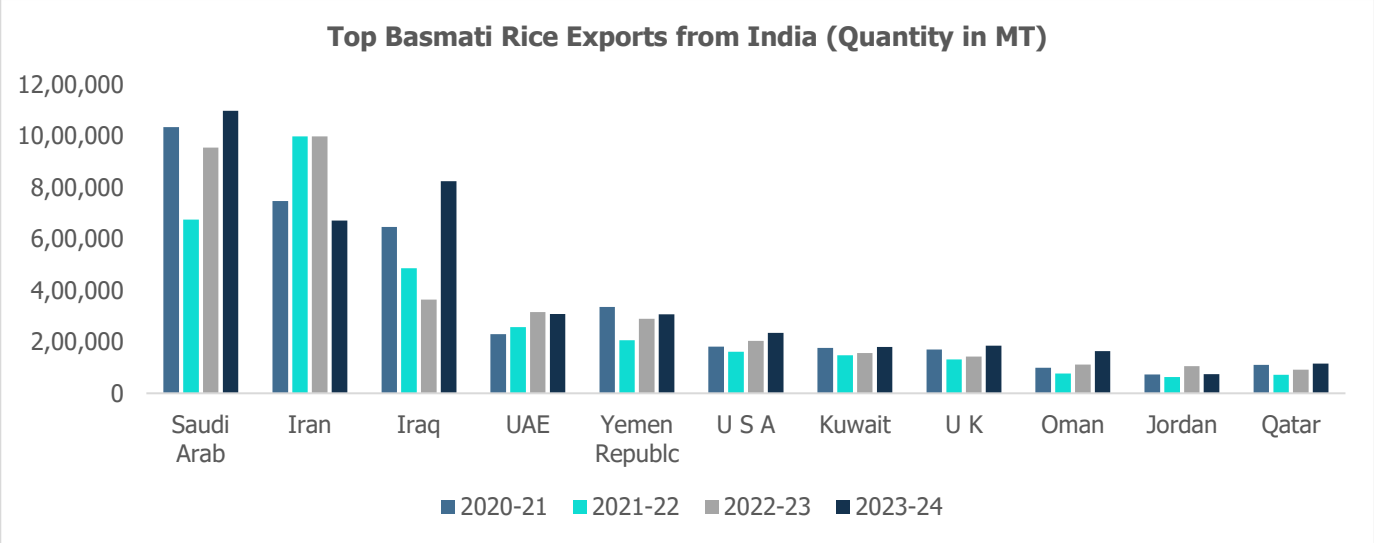
Source: CMIE
Note: 2024-2025* includes data from (April-February 2025)

India remains one of the largest producers as well as exporters of basmati rice in the world, with the largest area under basmati rice in the state of Haryana, followed by Punjab and then Uttar Pradesh. Basmati rice production in India has shown significant trends and developments over the past five years. In terms of production volumes, India has consistently produced substantial quantities of Basmati rice with few fluctuations. Overall, basmati rice production in India has been robust, contributing significantly to both domestic consumption and international trade. The production witnessed a decline of ~5% y-o-y in marketing year 2024-25 as against the previous year. The decline in rice production is mainly attributed to the uneven distribution of monsoon rains despite an increase in the area dedicated to paddy production, coupled with certain geopolitical developments impacting rice trade further. Moreover, with Pakistan's increased production, it has allowed Pakistan to offer lower prices than India, impacting India's market share.

Despite facing challenges such as water scarcity and climate change, India remains the leading producer and exporter of Basmati rice, catering to the diverse palates of consumers worldwide.

During marketing season 2024-2025, Basmati growers continued obtaining higher prices and profit margins on strong export demand.

Chart 23: Basmati Rice: India's Pride on the International Stage



Source: APEDA Agriexchange

India, being the leading exporter of Basmati Rice to the global market, had exported 52,42,048 MT of Basmati Rice (increased by 15% as against the previous year) to the world for the worth of Rs. 48389.18 crores/ USD 5837.12 million) During the year 2023-24, according to APEDA, with increased y-o-y percentage share of countries like Iraq, Oman, UK, Qatar, Saudi Arabia, USA and Kuwait. Similarly, during the year 2022-23, the country exported 45,58,972 MT of Basmati Rice (increased by 15% as against the previous year) to the world for the worth of Rs. 38524.11 crores/ USD 4787.50 million. During the same period, Iran, the largest importer of Indian basmati rice, decreased its purchases in 2022-23, but higher shipments to other countries offset the shortfall. The countries like Jordan have increased their exports by 69%, Oman by 45%, Saudi Arabia and Yemen Republic both increased their exports by 41% as against the previous period 2021-22.

During the past five years, India exported rice to about 150 countries globally. The major export destinations were Saudi Arab, Iran, Iraq, UAE, Yemen Republic, etc. Having the largest share of the global rice trade, India is expanding its rice export footprint in Asia, Africa, and the European Union.

India’s basmati rice exports will be key monitorable in 2024 after nearing a record high in 2023 due to increased competition from Pakistan, which is offering the grain at more competitive prices. The increased production in Pakistan this year, coupled with lower prices, is impacting India’s export prospects as buyers opt for Pakistan’s lower-priced rice. Moreover, the depreciation of the Pakistani rupee has made Pakistan’s exports more attractive, which may result in a surge in total rice exports from the country in 2023-24. Therefore, the fall in basmati rice production estimates in India in MY 2023-24 is influenced by competition, currency dynamics, as well as market demand.

In September 2022, India, which is the largest rice exporter globally, accounting for about 40% of global rice export currently, implemented an export ban on broken rice and imposed a 20% export tax on some non-basmati rice varieties due to reduced expected production caused by erratic monsoons.

Overall, basmati rice production in India plays a crucial role in the country's agricultural economy, providing livelihoods to millions of farmers and contributing to food security and export earnings. With its premium quality grains and global popularity, Basmati rice remains a flagship agricultural product that symbolises India's rich culinary heritage and agricultural excellence.

3.4 Key players in the Indian Rice Industry

3.4.1 KRBL Limited

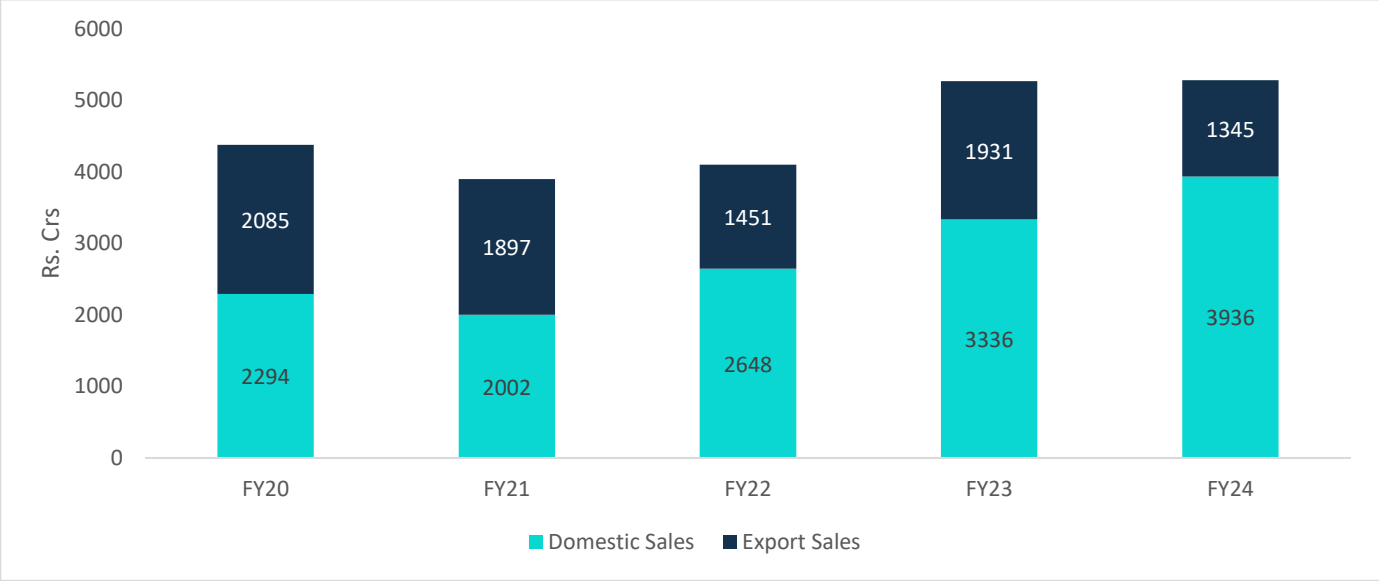
Product profile:

- India Gate Basmati – premium & daily regular basmati
- India Gate Golden Sella Range
- Zabreen Basmati Range
- India Gate White Sella Range
- Doon Basmati Range
- Bemisal Basmati Range
- Unity Basmati Rice
- Nurjahan Basmati Range
- Non-basmati rice - Sona Masoori, Surti Kolam, Wada Kolam, Jeera Rice, Ambemor, Gobindobhog

Product Portfolio

- India Gate Brown Rice
- Quinoa
- Flax Seeds
- Chia Seeds
- India Gate Weight Watchers Special

Chart 24: KRBL Limited's Domestic and Export Sales (in Rs. Crores)



Source: Company Reports, CareEdge Research

KRBL Limited's domestic sales are higher than their export sales. The domestic sales stood at Rs. 3,936 cr. with a y-o-y growth of 18%. The domestic sales grew at a CAGR of 14% between FY20 to FY24.

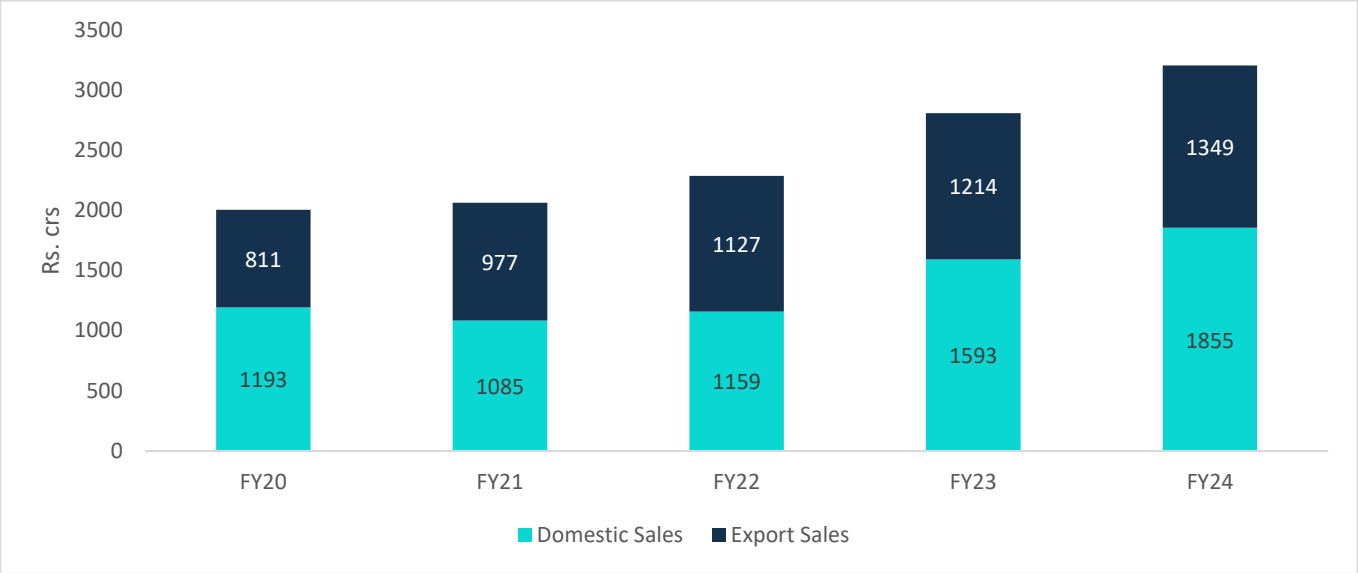
The exports for FY24 stood at Rs. 1,345 cr. showing a negative trend, decline of -30% y-o-y for FY24, and a negative CAGR of -10% between FY20 to FY24.

3.4.2 LT Foods Limited

Product profile:

- Dawaat Finest Rice
- Royal Basmati Rice
- Others - Heritage, Devaaya, Gold Seal Indus Valley, 817 Elephant, Leev, Golden Star, Kari, and Rozana

Chart 25: LT Foods Limited's Domestic and Export Sales (in Rs. Crores)



Source: Company Reports, CareEdge Research
Note: These are standalone numbers.

LT Foods Limited's domestic sales for FY24 stood at Rs. 1,855 cr. with a y-o-y growth of 17%. The domestic sales grew at a CAGR of 12% between FY20 to FY24.

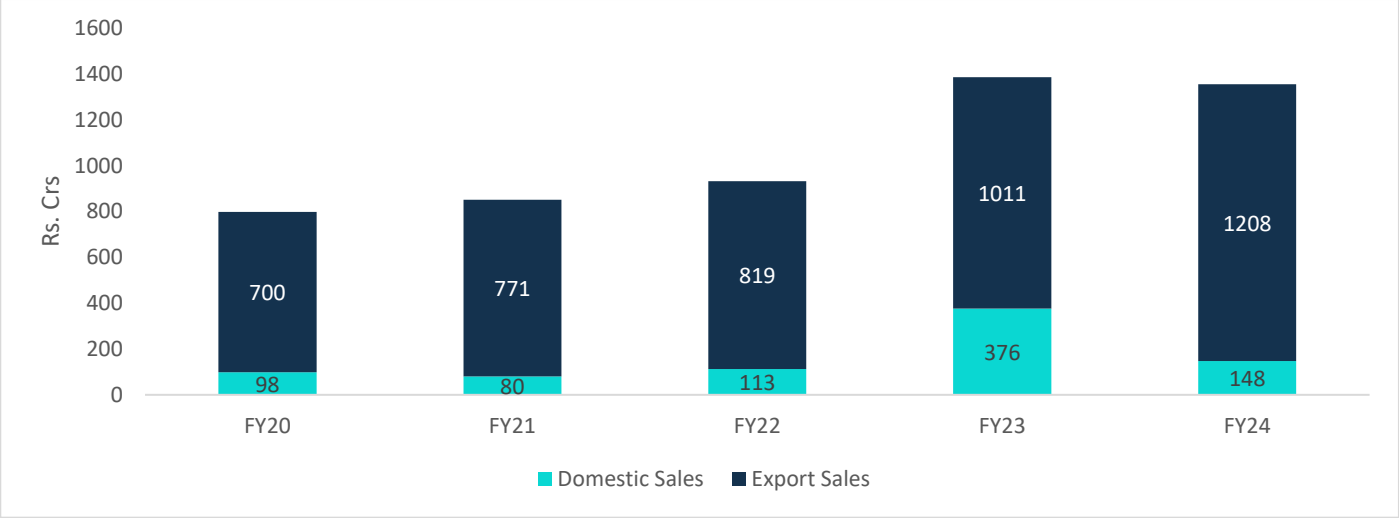
The exports for FY24 was Rs. 1,349 cr., a y-o-y growth of 11% for FY24, the exports for LT Foods Limited Exports have grown at a CAGR of 14% between FY20 to FY24.

3.4.3 Chaman Lal Setia

Product profile:

- Maharani Supreme Basmati Rice
- Maharani Basmati Rice 1121
- Maharani Classic Reserve Basmati Rice
- Maharani Royal Basmati Rice (Golden Sella)
- Maharani Brown Basmati Rice
- Maharani Everyday Basmati Rice
- Maharani Rice is suitable for Diabetics
- Begum Basmati Rice (Golden)
- Roopalee 1121 Basmati Rice
- Mithas Rice

Chart 26: Chaman Lal Setia Exports Limited's Domestic and Export Sales (in Rs. Crores)



Source: Company Reports, CareEdge Research

Chaman Lal Setia Exports Limited's domestic sales stood at Rs. 148 cr. for FY24, with a y-o-y decline of -61%. The domestic sales grew at a CAGR of 11% between FY20 to FY24.

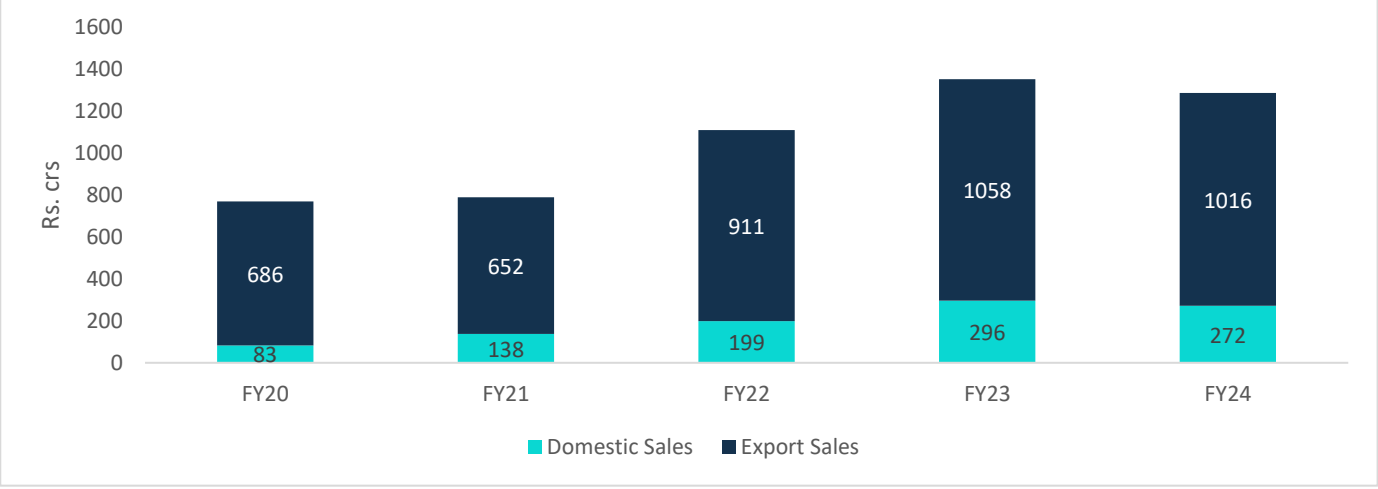
The exports for FY24 are Rs. 1,208 cr., a y-o-y growth of 20% for FY24, reflecting a CAGR of 15% between FY20 to FY24.

3.4.4 GRM Overseas

Product profile:

- Basmati Rice - 10X Basmati Rice
- 10x Steam Rice
- 10 X Long Grain Basmati Rice
- 10x Cream Sella
- 10x Golden Sella

Chart 27: GRM Overseas Estimated Production Domestic and Export Sales (in Rs. Crores)



Source: Company Reports, CareEdge Research

GRM Overseas domestic sales stood at Rs. 272 cr., a -8% y-o-y decline. The domestic sales grew at a CAGR of 35% between FY20 to FY24.

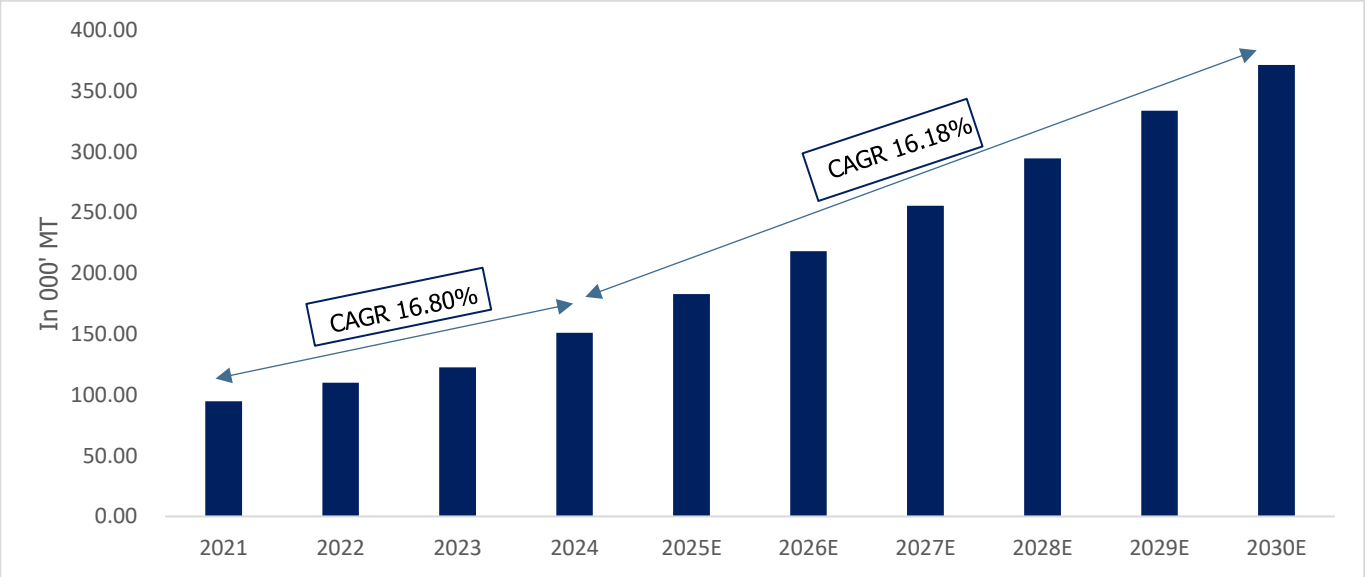
The exports for FY24 stood at Rs. 1,016 cr., a y-o-y decline of -4% for FY24, reflecting a CAGR of 10% between FY20 to FY24.

3.4.5 Sarveshwar Foods Limited

Product profile

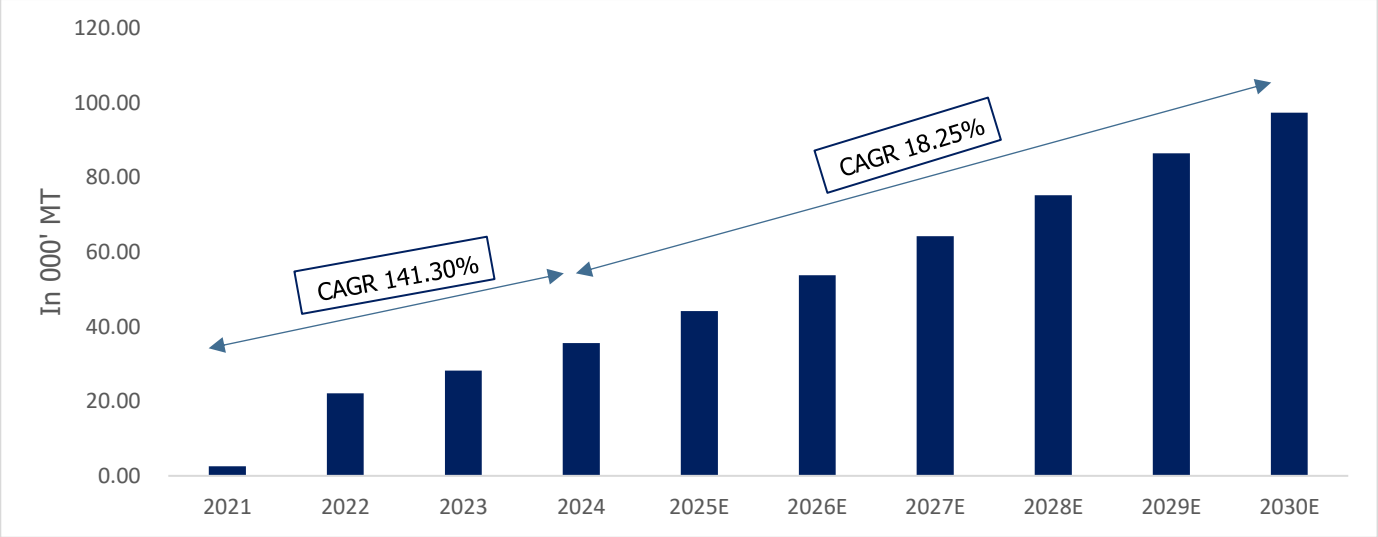
- Basmati Rice – Jammu Traditional Rice
- Super Basmati Rice
- Utra XL Basmati Rice

Chart 28: Sarveshwar Foods Limited (SFL)'s Estimated Production



Source: CMI, CareEdge Research.
Note: E indicates Estimated

Chart 29:Sarveshwar Foods Limited (SFL)'s Exports



Source: CMI, CareEdge Research.
Note: E indicates Estimated

Sarveshwar Foods Limited has shown robust growth in the past years with a CAGR of 16.80% from 2021 to 2024, and a projected growth estimated CAGR of 16.18% from 2025 to 2030.

Similarly, the exports for Sarveshwar Limited Exports have grown at a CAGR of 141.30% from the 2021–2024 period and the outlook for exports are projected with a CAGR of 18.25% from 2025 onwards.

4. Aata

India's consumer food business has thrived due to increased urbanisation, enhanced living standards, a growing number of supermarkets and malls, and the booming online food delivery sector. The industry is undergoing significant transformation driven by global events and evolving consumer dynamics. The pandemic-fuelled surge in online consumer goods demand has compelled Fast Moving Consumer Goods companies to adopt direct-to-customer strategies and prioritise digital distribution channels. Shifting consumer priorities and a heightened focus on conscious living post-pandemic are prompting the industry to offer products that meet nutritional needs and promote holistic health. Moreover, supportive government policies and subsidies for wheat cultivation, the primary raw material for sooji, is paving the way for the growth of this industry.

The demand for atta in India is deeply rooted in cultural and traditional preferences for staple foods like roti, chapati, and paratha. These atta-based foods hold immense cultural significance, ensuring consistent demand across diverse demographic segments. Urbanisation, rising disposable incomes, and changing lifestyles further contribute to the growing consumption of packaged atta products.

Increasing health consciousness among consumers plays a pivotal role in driving the demand for atta, as it is perceived as a healthier alternative to refined flour (maida) due to its higher fibre content and lower glycaemic index. The convenience offered by packaged atta products has also led to higher adoption among urban consumers with busy lifestyles. Ready-to-use atta packs save time and effort in the kitchen, making them attractive to modern consumers seeking convenience without compromising on nutrition.

Packaged atta is commonly used in chapatis, bread, pastries, and other wheat-based products. Its extended shelf life, due to protective packaging, makes it a practical choice for households. Moreover, the versatility of atta allows for a wide range of culinary creations beyond traditional Indian dishes, from bread and pizza bases to cakes and cookies. atta can be used in various international cuisines, expanding its potential usage among diverse consumer groups. Packaged Aata offers hygiene, ease of storage, and consistent quality compared to traditionally ground wheat (chakki atta). Traditionally, Indian households used chakki atta, freshly ground whole wheat flour from local mills.

The COVID-19 pandemic has significantly influenced consumer behaviour, increasing awareness and preference for packaged products, including atta. Heightened concerns about hygiene and safety have made packaged atta a preferred choice due to its perceived protection from contamination. With lockdowns and movement restrictions, consumers turned to packaged atta for its convenience, ensuring a reliable and readily available supply without frequent market visits. Additionally, packaged atta products often come with quality certifications and consistent standards, providing consumers with assurance regarding the quality and purity of the product.

On the supply side, India's abundant wheat production ensures a steady supply of raw material for atta milling. Additionally, technological advancements in milling processes and logistics infrastructure facilitate the efficient production and distribution of atta across the country. However, fluctuations in wheat prices, weather-related challenges affecting crop yields, and logistical constraints can impact the balance between demand and supply in the atta market.

Going forward, India is poised for a consecutive record wheat harvest in the marketing year (April-March) 2024-25 as per the USDA. Favourable weather in key wheat-growing regions is supporting record planting and normal crop growth. FAS (Foreign Agricultural Service) New Delhi forecasts a record production of 112.5 mmt from 31.8 million hectares, surpassing the current year's record of 110.6 mmt from 31.4 million hectares, assuming normal weather conditions hold from March end 2024 through the harvest at the end of April. The timely withdrawal of the 2023 monsoon, favourable weather conditions, and expectations of higher market prices and government support have driven record

wheat planting. Consistent weather and soil moisture, maintained through supplemental irrigation, are enhancing crop growth and yield prospects.

Established brands have leveraged their reputation to build trust among consumers, emphasising the safety, quality, and nutritional benefits of their packaged atta products. This trust, combined with the practical benefits and health advantages of atta, continues to drive its demand in the Indian market. The atta industry is dominated by established brands such as ITC's Aashirvaad Atta, MTR Foods, Hindustan Unilever's Shakti Atta and General Mills (Pillsbury), alongside regional players with a strong local presence.

5. Sooji

With sooji's use in a wide range of dishes, makes it a staple item in Indian households. Increasing urban population leads to higher consumption of processed and convenience foods. The demand for quick and easy-to-prepare food options like upma, rava dosa, and halwa is on increasing trend. The supportive government policies and subsidies for wheat cultivation, the primary raw material for sooji, is paving way for growth of this industry.

Sooji, or semolina, is a coarse flour derived from durum wheat, integral to various cuisines, especially Indian dishes like upma, halwa, and different breads. Its culinary versatility makes it a vital product in the food industry. The Indian sooji market is substantial and growing, fuelled by rising consumption of traditional and convenience foods. Increasing disposable incomes, urbanisation, and evolving dietary preferences drive this growth, with sooji seen as a nutritious alternative to refined flours.

Consumers are increasingly favouring branded and packaged sooji for its assured quality, hygiene, and convenience, especially after the COVID-19 pandemic spurred a rise in home cooking. Packaged sooji ensures consistent texture and quality, unlike loose, locally milled varieties.

Sooji is distributed through various channels, including traditional retail, modern trade, and online platforms. Supermarkets and hypermarkets play a crucial role in urban areas, while local kirana stores remain important in rural regions. E-commerce platforms are increasingly becoming popular, providing consumers with the convenience of home delivery and a wider range of choices.

The sooji market faces challenges such as fluctuating wheat prices and the need to maintain quality standards. However, there are significant opportunities for growth. The increasing demand for convenience foods, the trend towards healthier eating, and the expansion of retail networks are expected to drive market expansion. Innovations in packaging and the introduction of fortified and organic sooji products are also likely to attract health-conscious consumers.

The sooji market is populated by several key players, both large and small. Sooji production is primarily driven by major atta and wheat flour processing companies such as ITC (Sunfeast brand), MTR Foods, Hindustan Unilever (Shakti brand), along with several regional players. These companies typically operate dedicated sooji production lines within their larger wheat processing facilities. Additionally, numerous regional and local brands cater to specific geographic areas, maintaining strong local presence and loyalty.

Hence, the Sooji industry in India is poised to benefit from the overall growth of the packaged wheat products market, driven by urbanisation, convenience preferences, and product innovation. Therefore, from savoury upma to sweet halwa, this ingredient reigns supreme in Indian kitchens.

6. Maida

With growing popularity of bakery products it boosted the demand for maida. Additionally, increasing health consciousness is also leading consumers to seek alternatives to maida (refined flour), such as whole wheat flour, multigrain flour, and other healthier options etc. Urbanisation, as well as fast-paced and evolving lifestyles, drive the demand for quick and easy meal options, which are often made from maida, and the growth in e-commerce platforms and online grocery shopping increases the availability and sales of packaged maida.

Maida, a refined wheat flour, holds a significant place in the Indian culinary landscape, widely used in dishes ranging from bread to sweets. The maida market in India is robust and continually evolving, driven by changing consumer preferences, culinary innovations, and economic factors.

Maida is derived from wheat flour through a refining process, making its supply closely linked to wheat production and processing capacity in India. Government policies related to agriculture, trade, and food processing can influence the supply of maida by affecting wheat procurement, pricing, and distribution channels. The capacity and efficiency of maida mills and processing units play a crucial role in determining the supply of maida in the market.

One of the key drivers of the maida market is the growing demand for bakery and confectionery products in urban and semi-urban areas. Maida's versatility makes it essential in the production of bread, biscuits, cakes, pastries, and other baked goods, catering to the evolving tastes of Indian consumers. The expanding food processing industry in India also significantly contributes to the growth of the maida market. Food manufacturers utilise maida in snacks, ready-to-eat meals, instant noodles, and other processed foods, capitalising on its texture-enhancing and binding properties.

The maida market experiences seasonal fluctuations, with demand surging during festive seasons and special occasions due to increased consumption of sweets, snacks, and baked goods. Additionally, the popularity of Western-style bakery products and the rise of quick-service restaurants (QSRs) have further stimulated demand for maida-based items.

There is a noticeable shift towards branded and fortified maida products, driven by rising consumer awareness regarding food safety, quality, and nutrition. Established brands are focusing on product diversification, packaging innovations, and marketing strategies to cater to evolving consumer preferences and maintain competitiveness in the market.

Despite growth opportunities, the maida market faces challenges such as price volatility of wheat, fluctuations in raw material availability, and increasing competition from alternative flours and gluten-free products. Strategic initiatives, technological advancements, and collaborations across the value chain are expected to drive the continued expansion of the maida market in India.

7. Besan

Besan is one of the gluten-free, high in protein, and rich in vitamins as well as minerals, which makes it one of the popular choices for health-conscious individuals. Besan, being staple ingredient amongst Indian households, used in a variety of Indian cuisines, hence the growing trend towards vegetarianism and veganism is boosting the demand for plant-based protein sources like Besan and also consumers are moving towards value-based items. Government is also supporting for pulse farming and initiatives to promote the chickpea cultivation.

The besan market in India is robust, driven by its culinary versatility, health benefits, and changing consumer preferences. Strategic focus on innovation, quality control, and expanding distribution channels will continue to drive market growth.

Besan, also known as gram flour, is a staple in Indian cuisine, made from ground chickpeas. Its extensive use in traditional dishes, snacks, and sweets underpins its significant market presence in India. Integral to recipes such as pakoras, dhoklas, and laddoos, besan's versatility drives consistent demand across diverse demographic segments.

The growing health consciousness among consumers has boosted besan's popularity due to its high protein content, gluten-free nature, and lower glycaemic index compared to wheat flour. Additionally, rapid urbanization and evolving lifestyles have increased the demand for ready-to-cook and packaged food products, where besan is a key ingredient. The preference for branded and packaged besan is rising due to concerns about quality, hygiene, and convenience, with consumers trusting branded products for their consistency and safety.

Manufacturers are introducing organic besan products to cater to health-conscious consumers. Innovations in packaging to enhance shelf life and convenience are also on the rise. Local kirana stores continue to be major distribution channels, especially in rural and semi-urban areas. In urban areas, supermarkets and hypermarkets play a crucial role by offering a wide range of besan products. Online platforms are gaining popularity, providing consumers with the convenience of home delivery and access to various brands and products.

India is a major producer of chickpeas, the primary raw material for besan. The supply of besan is closely tied to chickpea production, which can be influenced by climatic conditions and agricultural practices. The efficiency and capacity of besan mills significantly affect the overall supply, making advances in milling technology and infrastructure development crucial for maintaining a steady supply of high-quality besan.

Fluctuations in chickpea prices can impact the cost of besan, affecting both producers and consumers. Ensuring consistent quality and purity of besan can be challenging due to varying standards among local and regional producers. However, the ingrained culinary usage of besan in numerous Indian dishes ensures a steady demand.

India's burgeoning population and rapid urbanisation will likely boost the demand for packaged and convenient food options, potentially increasing the demand for ready-made besan mixes. Besan's status as a good source of protein and fiber compared to refined wheat flour makes it a healthier alternative, driving demand as health consciousness grows. Additionally, besan's naturally vegan and gluten-free properties cater to dietary restrictions, expanding its consumer base.

8. Salt

The salt industry in India is a vital contributor to the nation's economy, meeting the significant domestic demand for this essential commodity. The Indian salt industry not only fulfils domestic needs but also supports exports to various countries. It provides significant employment opportunities, especially in coastal areas where salt production is a primary source of livelihood. Government policies, pricing mechanisms, and quality standards play crucial roles in shaping the industry's dynamics.

Salt production in India primarily occurs in coastal regions, where traditional methods involve solar evaporation of seawater in salt pans. Key production hubs include Gujarat, Tamil Nadu, Andhra Pradesh, and Maharashtra. This method, where seawater is evaporated under the sun to leave behind salt crystals, is a blend of small-scale operations and larger organized enterprises. While traditional practices remain widespread, modernization and mechanization have enhanced efficiency and productivity. Technological advancements in refining, iodization, and packaging have also improved quality standards, ensuring a steady supply of iodized salt to address public health concerns related to iodine deficiency.

The main sources of salt in India are Sea brine, Lake brine, Rock salt deposits and sub-soil brine. The types of salt produced in India is listed below:

Rock salt: Extracted from underground rock formations.
Marine salt: Produced by evaporating seawater.
Solar salt: Made by evaporating brine under the sun.
Iodized salt: Table salt fortified with iodine, a crucial mineral for cognitive development and thyroid function.

Efforts to promote iodized salt consumption, ensure sustainable production practices, and address environmental concerns are key to the industry's development.

Salt serves as a critical raw material in the chemical industry, essential in manufacturing chemicals such as sodium carbonate (soda ash), caustic soda, hydrochloric acid, chlorine, bleaching powders, chlorates, sodium sulphate (salt cake), and sodium metal. Beyond culinary and household uses, salt is extensively employed in industries like meat packing, fish curing, dairy processing, hide preservation, soap and detergent manufacturing, dyeing and fabric finishing, refrigeration, ceramics glazing, explosives production, leather tanning, bakery products, soil stabilization, artificial rubber manufacturing, and wood preservation.

India is the third largest salt (common) producing country in the world after China and the USA. The production of Rock - Salt in 2021-22 was 286 tonnes (down by 41%) as compared to that in the preceding year. However, significant increase of 250% was witnessed in 2022-23 to 1,002 tonnes as against 2021-22. As per Indian Minerals Yearbook 2022 (dated March 2024), the per-capita consumption of salt in the country is estimated to be about 14 kg, which includes edible and industrial salt. Moreover, the current annual requirement of salt in the country is estimated to be 63 lakh tonnes for edible use (including requirement for cattles) and 118 lakh tonnes for industrial use.

9. Sugar

India stands as the largest consumer and second-largest producer of sugar globally, commanding about 15% of global sugar consumption and 20% of global production as per Ministry of Consumer Affairs, Food and Public Distribution. These significant figures mean that trends in the Indian sugar industry heavily influence global markets.

In the Eastern Hemisphere, India is the leading player in the sugar market, with Brazil dominating the Western Hemisphere. Moreover, India's position as the third-largest ethanol producer, following the USA and Brazil, highlights its commitment to green energy. This strategic shift not only addresses the surplus sugar issue domestically but also reduces fossil fuel imports and helps meet India's COP 26 environmental targets.

The Indian sugar industry has undergone extensive modernisation and expansion, diversifying to exploit the potential of its by-products for additional revenue, thus ensuring a sustainable and profitable business model. Its resilience was particularly evident during the COVID-19 pandemic when it continued operations amidst lockdowns, even producing enough hand sanitisers to meet national demand.

Demand-Supply Scenario

For the sugar season (SS) October 2022- September 2023, the exact production after sugar diversion towards ethanol stood at 328 lakh tonnes (a decline of 8.4%), as compared to the record high production of 358 lakh tonnes in SS 2021-22. Falling sugar cane yields in states like Maharashtra and Karnataka due to the early maturity of the crop and adverse climatic conditions likely resulted in a decline in sugar outputs. However, for SS 2023-24 and SS 2024-25, Maharashtra and Uttar Pradesh will continue to be the major contributors with gross production at 111 lakh tonnes (down by 5% as against SS 2023-24) and at 113 lakh tonnes (up by 3% as against SS 2023-24) respectively as on July 2024.

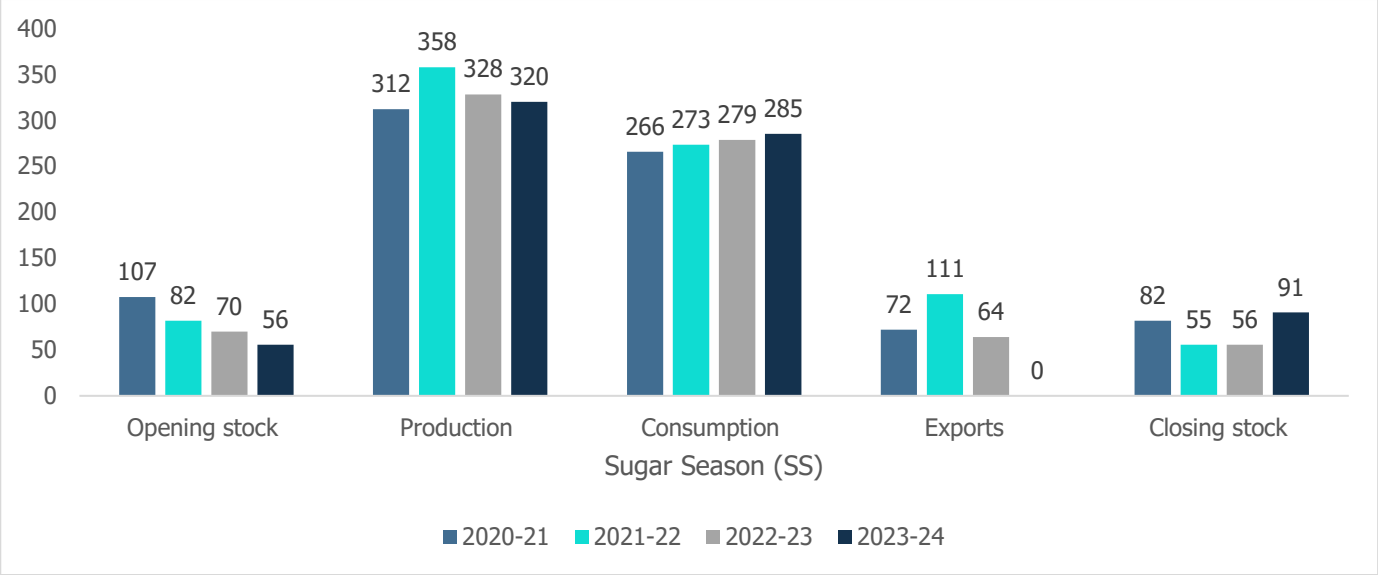
According to the Indian Sugar & Bio-Energy Manufacturers Association (ISMA), sugar production is estimated to decline for the current SS 2023-24 to 320 lakh.[2] tonnes after diversion. This is because India witnessed the weakest monsoon last year since 2018, resulting in reduced cane yields. However, favourable monsoon in the current SS will augur well for the sugar industry going forward. Moreover, the changing climatic patterns have led to a drop in per-hectare sugarcane yield and posed significant challenges for sugar production, especially in Maharashtra and Karnataka. Hence, the drop in Indian cane production due to El Nino, the global climate phenomenon, affected the ethanol production from syrup as well as B-Heavy Molasses from mid-December 2023.

The sugar season (SS) (October to September) 2021-22 registered a successful year with record-level sugarcane production, sugar exports, cane procured, ethanol production and cane dues clearance. With a consistent increase in surplus sugar production, the country's sugar exports grew by 105% CAGR to 112 lakh tonnes in the season. On account of increased exports and the rising ethanol diversion, the opening stock witnessed a reduction over the past three years. The opening stock of sugar was estimated at 56 lakh tonnes for the current SS 2023-24.

On the other hand, sugar consumption in the country has been healthy at an average of 265-275 lakh tonnes over the past three years and is estimated to increase to approximately 285 lakh tonnes in the current SS as per ISMA on the back of strong demand due to intense heatwave. Moreover, the growth has been recorded for production and consumption at a CAGR of 2.6% and 1.4%, respectively, in the past five years.

[2] Indian Sugar & Bio-energy Manufacturers Association (ISMA)

Chart 30: Past Trends in Sugar Stock, Production, Consumption, and Exports (in Lakh Tonnes)



Source: CMIE, ISMA, CareEdge Research

For the SS 2022-23, on account of lower sugar production amid buoyant demand, a quota system was introduced to curb exports. The export quota was set at 64 lakh tonnes for the SS 2022-23. As a result, exports of sugar declined to 64 lakh tonnes during the SS 2022-23, from 72 lakh tonnes and 112 lakh tonnes in SS 2020-21 and SS 2021-22, respectively.

At present, there are curbs on export of sugar for an indefinite period. However, these export curbs ensure adequate supply in the domestic market, affordable prices, sufficient availability for ethanol production, and adequate closing stock at the end of the season. The government is prioritising stock availability for domestic consumption and ethanol production.

Besides, in October 2023, the Director General of Foreign Trade (DGFT) extended the restrictions on the export of sugar (raw sugar, white sugar, refined sugar, and organic sugar) indefinitely beyond the previous cut-off date of October 31, 2023, under which all sugar exporters are required to secure permission from the food ministry for export. However, the restriction does not apply to sugar exported to the European Union and the US under the CXL concession and Tariff Rate Quota (TRQ).

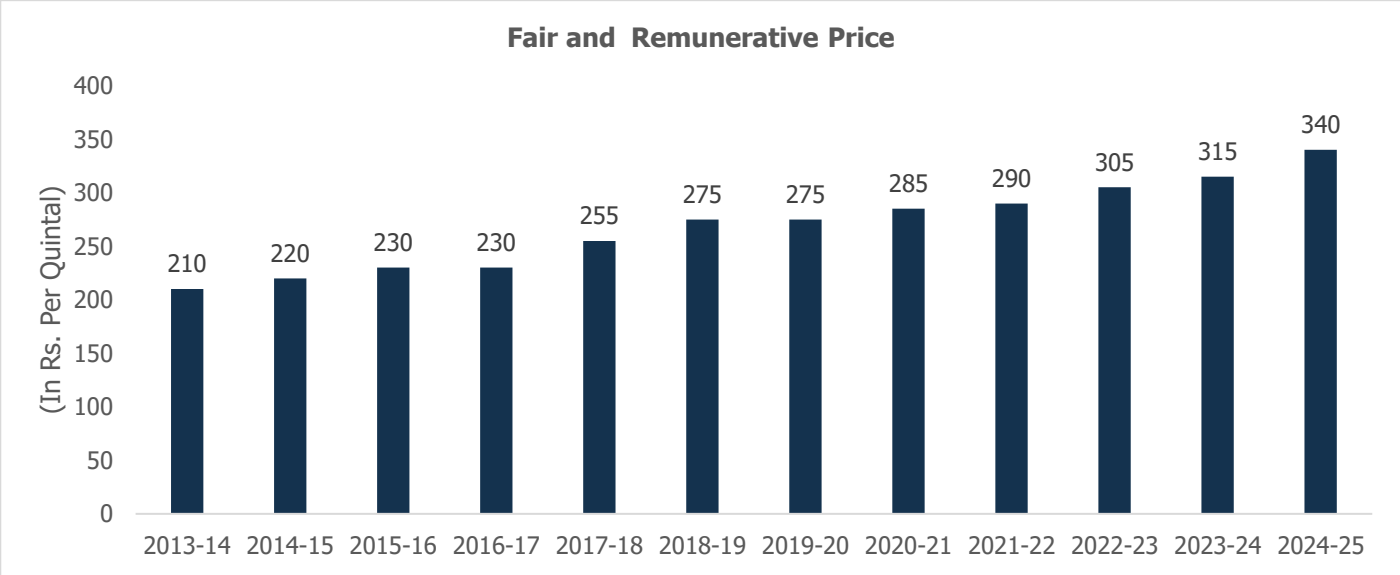
Another important factor for the sugar industry in India is the Ethanol Blended Petrol (EBP) program that aims to reduce the sugar glut situation in India. The EBP program supports the diversion of sugarcane and surplus sugar towards the manufacturing of ethanol, which became the key focus point for the future of the sugar industry. The government also releases procurement prices for ethanol to divert sugarcane towards the production of ethanol. Further ethanol blending helps sugar mills maintain their cash flow/profitability margin.

Whereas the Ethanol Blending Petrol (EBP) Programme was introduced in 2003 to increase the ethanol blend level with petrol. Under this, the ethanol blending percentage has increased from 5% in ESY 2019-20 to 12% in ESY 2022-23 and further 13.07% in ESY 2023-24 (as on July 2024). Driven by the significant progress in blending percentage, the government has advanced the target of 20% ethanol blending in petrol by 5 years to 2025 from 2030 previously. However, owing to government restrictions imposed in December 2023 on using sugar feedstocks for the production of ethanol, it will be key to monitor in the near term for achieving the ethanol-petrol blending target further.

India pays the Highest Cane Price in the World to Sugarcane Farmers.

The margin of sugar mills is largely impacted by the Fair and Remunerative Price (FRP) payable to farmers and the Minimum Support Price (MSP) fixed by the government. The government has increased the FRP at a CAGR of 4.1% over the past ten years in SS 2023-24. To benefit over 5 Crore sugarcane farmers and ensure their prosperity, CCEA has approved the FRP of sugarcane by Rs. 25 to Rs. 340 per quintal for the SS 2024-25, the highest-ever increase since 2014, on February 21, 2024. Additionally, in a boost for sugarcane farmers, the Uttar Pradesh government has announced an increase in the sugarcane State Advised Price (SAP) for SS 2023-24 to ₹20 per quintal. The SAP for early-maturing varieties of sugarcane has been increased from Rs 350 per quintal to Rs 370. However, the Minimum Support Price (MSP) was last fixed at Rs 31/kg in February 2019.

Chart 30: Sugar Season-wise FRP announced by the Central Government



Source: CMIE, Department of Food and Public Distribution

Payment of cane prices to sugarcane-growing farmers by the sugar mills is a continuous process. The cash flows for sugar mills have improved, resulting in prompt payment to cane farmers through the sale of ethanol. 99.9% of cane dues of sugarcane farmers for the sugar season up to 2021-22 have already been cleared by sugar mills. For the sugar season 2022-23, sugar mills have cleared about 99.5% of cane dues payments of farmers as per the latest announcement from the Ministry of Consumer Affairs, Food and Public Distribution.

Amidst a consistent rise in sugar production costs, adjustment of the MSP in line with the FRP for sugarcane will be a key monitorable going forward.

10. Impact of Pandemic and Other Geopolitical Developments on Indian Rice Industry

The COVID-19 pandemic and various geopolitical developments have significantly impacted the rice industry, affecting production, trade, and market stability. These events highlighted the vulnerability of global supply chains and the importance of resilience and adaptability in agricultural practices. Going forward, the rice industry is likely to see increased emphasis on sustainability, technological innovation, and strategic policy measures to ensure food security and economic stability.

Impact of Pandemic

The pandemic caused significant disruptions in global supply chains. Restrictions on movement, lockdowns, and reduced workforce availability affected the transportation and distribution of many products, including rice. Initial panic buying and stockpiling by countries for certain products led to price surges and concerns about food security in some regions, while subsequent stabilisation efforts and changing demand patterns caused fluctuations. Supply chain disruptions affected the availability of essential agricultural inputs such as seeds, fertilisers, and pesticides, potentially impacting crop yields. Lockdowns and border restrictions caused disruptions in the global rice trade. Labour shortages in harvesting and processing also affected production and exports. Restrictions and health concerns led to labour shortages in the farming, processing, and transportation sectors, impacting rice production and distribution efficiency. Restrictions on mobility, transport, and trade disrupted the rice value chain, which affected the movement of rice from farms to consumers. Moreover, declining income and reduced demand affected farmers' ability to invest in inputs and technology during the pandemic period.

This uncertainty and disruptions caused volatility in rice prices. This also led to delays and increased costs in shipping rice to international markets. Some rice-exporting countries imposed temporary export bans or restrictions to ensure domestic food security, leading to supply shortages in importing countries and increased global market volatility. For instance, countries like Vietnam temporarily restricted rice exports to ensure domestic food security during the pandemic.

With more people staying home, demand for packaged and ready-to-eat rice products increased. Lockdowns and restrictions on movement limited access to fresh groceries, including vegetables and meat. Rice, being a shelf-stable staple, became more appealing during that period.

Impact of Geopolitical Developments

The pandemic and geopolitical developments underscored the importance of food security, prompting many countries to reassess their rice reserves and dependency on imports. This led to increased efforts to boost domestic production capacities and diversify supply sources. The challenges faced during the pandemic accelerated the adoption of technological innovations in the rice industry. Digital platforms for market access, precision farming techniques, and improved supply chain management systems became more prominent post-pandemic.

1. Altered Weather Patterns Significantly Affect Crop Yields, Fisheries, and Food Security

Shifting rainfall patterns and extreme weather events pose significant risks to food security at local, regional, and global levels. Climate change disrupts the rice industry by affecting both productivity and stability. Rising temperatures cause heat stress in rice plants, particularly during the grain-filling stage, resulting in smaller, less viable grains. Disrupted temperature patterns affect planting and harvesting schedules, leading to mismatches with optimal climate conditions.

Increased frequency and severity of droughts limit water availability crucial for rice paddies, while excessive rainfall and flooding cause crop damage, soil erosion, and nutrient depletion. Extreme weather events such as cyclones and storms

physically damage rice fields, destroy infrastructure, and disrupt supply chains, resulting in economic losses for farmers and the industry.

These climate-related challenges significantly reduce rice production, impacting global food security and causing price hikes. For example, extreme heat during flowering stages can lead to spikelet sterility, reducing grain production. Geopolitical focus on climate change has heightened awareness and policy measures addressing the impacts of extreme weather on rice production. Global and regional initiatives promoting sustainable farming practices aim to reduce the environmental footprint of rice cultivation, influencing production methods and costs. Drought conditions in Southeast Asia exemplify how climate change can impact rice production.

In a recent scenario, an intensified El Niño has led to lower rainfall in rice-producing regions, negatively impacting yields. Additionally, climate change poses significant risks to rice production. El Niño influences the Indian monsoon, causing fluctuations in rainfall patterns. This results in mixed effects, with some districts experiencing reduced yields due to water scarcity, while others may see higher production.

Furthermore, in addition to war-related disruptions, an earlier-than-usual El Niño is expected to hamper rice production across Asia. The warmer and drier weather associated with El Nino can adversely affect rice crops, exacerbating food security concerns.

2. India’s Export Restrictions on Rice Continue to Disrupt Global Markets, Supplies and Prices

India's trade policies and tariffs for rice aim to balance domestic food security with the promotion of rice exports. Trade policies and tariffs play a complex role in India's rice industry, affecting both domestic production and exports. Export tariffs and bans can significantly reduce India's rice exports, impacting its global market share and potentially leading to a domestic surplus.

India's export restrictions can disrupt the global rice market, causing price fluctuations and raising food security concerns for rice-importing countries. Trade disputes and tariff impositions between major rice-exporting and rice-importing nations influence market access and competitiveness. Additionally, declining yields in prominent rice-producing countries like Thailand have led to rising rice prices in Southeast Asia.

India, being the world’s top rice exporter, imposed export bans on non-basmati rice, exacerbating food insecurity in Southeast Asia. India imposed a 20% export duty on certain varieties of rice in September 2022, which aims to control domestic rice prices after a heatwave threatened harvests. Further, in July 2023, India imposed a complete ban on broken rice exports and restricted exports of non-basmati rice varieties. This aims to further ensure domestic food security.

Concerns over domestic food security may prompt countries to restrict rice exports. For instance, Myanmar joined India and Vietnam in limiting rice exports in 2023, potentially tightening global rice availability. Recently, in May 2025, the government has again imposed a 20% export duty on parboiled rice exports.

3. Impact of the Ukraine Conflict on the Global Rice Industry

Russia’s invasion of Ukraine threatened global grain supplies as it reduced production and exports while increasing trade costs. Although Ukraine is not a major rice producer, the conflict has disrupted global wheat exports, leading some countries to substitute rice, thereby increasing demand and putting pressure on prices. The ongoing war in Ukraine has significantly affected the global food supply, including the rice industry, coupled with India’s decision to ban rice exports due to erratic monsoons. Before the conflict, Ukraine was a key exporter of agricultural commodities, including grains. The disruption of Ukrainian grain exports has had a ripple effect on global markets.

The impact extends to rice production across South and Southeast Asia, which may face additional challenges, especially with the anticipated El Nino event. The conflict has tightened global cereal and oilseed markets, resulting in prices nearly doubling over the past two years and adversely affecting food security in importing countries.

Therefore, the war in Ukraine has caused significant supply chain disruptions, increased trade risks, and affected global food security. As a result, rice production, along with other cereals and oilseeds, faces considerable challenges due to these combined factors.

Hence, international conflicts disrupt trade routes and create uncertainty, affecting the global supply of commodities like rice.

4. Disruption in the Red Sea impacting the Cost of Transport

Due to increased shipping costs, shipping companies are hesitant to use the Red Sea route due to security concerns. This forces them to take longer routes, significantly increasing shipping costs for rice exports. The uncertainty and delays caused by the Red Sea situation can disrupt established supply chains, making it difficult to fulfil export commitments on time. The increased costs and logistical challenges may make Indian basmati rice less competitive in some export markets, which may affect demand from traditional buyers in the Middle East, Europe, and North America. Exporters are exploring alternative routes, though they are often longer and more expensive. The higher shipping costs might be passed on to consumers in the form of increased rice prices in importing countries.

These disruptions created a temporary but significant challenge for Indian basmati rice exports. While the industry has adapted through alternative routes and price adjustments, the ongoing situation adds uncertainty and could potentially hinder future export growth.

Hence, this current disruption in the Red Sea is impacting cost of transport which is key monitorable as the Indian players may expect the actual impact to be in Q1FY25 going forward.

5. Impact of Israel-Hamas Conflict on the Rice Industry

Israel's economy heavily relies on imports for major agricultural commodities like cereals and oilseeds from world. Palestine imports cereals in major share from India as compared to other items like oilseeds, fruits, vegetables etc.

Israel and India share a significant bilateral trade relationship, with India primarily dealing in non-agricultural products, which reduces the direct impact on Indian farmers and exporters in agriculture.

On the other hand, the Israel-Hamas war added to the uncertainty in the global economy, involving significant chances of disruptions and economic slowdown worldwide if the war escalates further. The war could also affect the oil-producing countries, including Iran and Saudi Arabia, in West Asia, even while Israel and Gaza are not major oil producers themselves, as the India-Middle East-Europe Economic Corridor (IMEEC), which connects India and Europe through West Asian countries, can face delays or challenges owing to war. Hence, the trade between the two countries may not be immediately impacted, but could suffer if the entire region becomes involved in the conflict.

However, Indian exports could get heavily impacted if the region turns into a conflict zone with countries such as Iran, Israel and other adjoining nations being major consumers of Indian basmati rice.

India's basmati rice trade, which was hit by supply chain disruption caused by the Houthi attacks on the Red Sea, is still at ongoing risk, posing a rise in shipping costs. The Red Sea is an important lifeline for the global supply chain and international trade. Over one-third of production is shipped to countries like West Asia, North Africa and North America via the Red Sea, constituting the majority share of India's basmati rice exports, which got severely impacted as the alternative route via the Cape of Good Hope has raised shipping costs. Therefore, basmati prices will be key monitorable

in the near to medium term if disturbances on the Red Sea and Houthi attacks are sustained, posing a rise in shipping costs.

6. Impact of the Sudanese Civil War on the Rice Industry

Sudanese war, erupted in April 2023, between the Sudanese army and the paramilitary Rapid Support Forces (RSF) had created one of the world's largest hunger crisis spreading across the nation. The ongoing civil war threatened Sudan's food security and agricultural capacity as agriculture accounts for more than 50% share of total national exports and contributing massively in Sudan's GDP.

Despite the export duty on non-basmati white rice in India, its exports increased owing to international shortages amidst geopolitical tensions as well as unfavourable weather conditions.

Sudan's location on the Red Sea is crucial for trade flows through the Suez Canal, and any disruption could impact global trade, including that of India and other countries. However, the direct impact of the Sudanese war on the Indian rice industry may be limited currently, as it broadly affects regional dynamics.

11. Complexities Hindering the Growth of the Rice Industry in India

1. **Regulatory and Policy Challenges:** Complex and sometimes inconsistent government regulations can create barriers for farmers and exporters. Navigating these regulations requires significant effort and resources. While government subsidies support farmers, over-reliance on these can sometimes create inefficiencies and discourage innovation and sustainable practices.

   Moreover, any change on the policy front, such as a ban on rice exports or an increase in the minimum support price, affects rice mills in the country. These government-imposed export bans and tariffs, which aim at ensuring domestic food security, can reduce India's share in the global market and affect overall industry growth.

2. **Climate Change and Environmental Factors:** Rising temperatures, erratic rainfall patterns, and extreme weather events like droughts and floods significantly impact rice yields. The frequency and severity of these events have increased, leading to instability in production. Rice cultivation is highly water-intensive, and regions reliant on monsoon rains are particularly vulnerable to changes in rainfall patterns. Water shortages and inefficient irrigation practices exacerbate this challenge. Being an agricultural commodity, paddy is subject to risks in the form of vagaries of nature. Paddy cultivation requires a large amount of water; unseasonal rain and floods could seriously threaten the crop.

   Moreover, farming practices, including the overuse of chemical fertilisers and pesticides, have led to soil degradation, reducing the fertility and productivity of rice paddies.

3. **Economic and Market Factors/ Currency Fluctuations:** Strong currency fluctuations can lead to substantial changes in rice prices, thereby influencing global demand. The fluctuations in rice prices, influenced by both domestic policies and international market dynamics, create uncertainty for farmers. Export restrictions and tariffs can also disrupt market stability.

   Since India competes with countries, such as Vietnam and Thailand, in the export market therefore any significant appreciation of the rupee can lower the mills' profitability. Currency fluctuations, along with trade policies, play a crucial role in shaping rice exports. Volatile exchange rates make it challenging for exporters to predict and plan their pricing strategies effectively.

4. **Infrastructure Deficiencies:** Inadequate storage facilities, poor transportation infrastructure, and limited access to modern farming equipment hinder the efficiency and profitability of rice farming.

5. **Technological Gaps**: Although there have been advancements in agricultural technology, the adoption rate among small-scale farmers remains low. This limits potential productivity gains and the implementation of sustainable practices.

   Additionally, the use of traditional and outdated processing technology can lead to low-quality output, resulting in broken rice, discoloured grains, uneven shapes and sizes, and other impurities. Upgrading to modern equipment can mitigate these risks. Further, conducting extensive and regular quality checks can ensure the production of superior quality rice.

6. **Cost of Inputs**: Rising costs of seeds, fertilizers, and other agricultural inputs add to the financial burden on farmers, affecting their overall profitability.

7. **Research and Development:** Insufficient investment in R&D for developing resilient and high-yield rice varieties limits the industry's ability to cope with changing environmental conditions and pest resistance.

8. **Socio-Economic Factors:** The fragmentation of agricultural land into small holdings makes it sometimes challenging to achieve economies of scale, affecting productivity and profitability. Moreover, urban migration and the shift away from agriculture as a preferred occupation lead to labour shortages, particularly during peak planting and harvesting seasons.

12. Government Policies

The government of India has implemented various initiatives to support and enhance the rice industry. These initiatives aim to improve productivity, ensure food security, provide financial assistance to farmers, and promote sustainable agricultural practices.

1. Pradhan Mantri Krishi Sinchayee Yojana

The Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), launched in 2015-16, aims to improve water accessibility on farms, expand irrigated areas, enhance on-farm water use efficiency, and promote sustainable water conservation. PMKSY operates as an umbrella scheme with two major components managed by the Ministry of Jal Shakti: The Accelerated Irrigation Benefit Programme (AIBP) and Har Khet Ko Pani (HKKP). HKKP includes four sub-components: Command Area Development & Water Management (CAD&WM), Surface Minor Irrigation (SMI), Repair, Renovation and Restoration (RRR) of Water Bodies, and Ground Water (GW) Development (approved only until 2021-2022). Additionally, the scheme incorporates the Watershed Development component (WDC) by the Ministry of Rural Development and the Per Drop More Crop (PDMC) component by the Department of Agriculture and Farmers Welfare (Doa&fw), which is now implemented separately.

The AIBP component of PMKSY provides partial financial assistance from the Government of India for major and medium irrigation projects. In 2016-17, 99 ongoing projects were prioritised under PMKSY-AIBP, with funding from the long-term Irrigation Fund via NABARD. Out of these, 53 projects have been completed, while the rest are at various stages of implementation, with significant progress reported in many. Further, after approval of the implementation of PMKSY during 2021-2026, seven additional irrigation projects have been included under PMKSY-AIBP from 2021-22 onwards, which are presently under implementation.

2. National Food Security Mission (NFSM)

The National Food Security Mission (NFSM), launched in 2007-08, aims to boost the production of rice, wheat, and pulses through area expansion and productivity enhancement, while also restoring soil fertility, creating employment opportunities, and improving farm-level economies. Coarse cereals were added to the mission in 2014-15. During the 12th Five-Year Plan, the NFSM targeted an additional production of 25 million tonnes of food grains. Post-12th Plan, new targets included an additional 13 million tonnes of food grains from 2017-18 to 2019-20. Currently, NFSM includes sub-components such as NFSM-Rice, NFSM-Wheat, NFSM-Pulses, NFSM-Coarse Cereals, NFSM-Nutri-Cereals, and NFSM-Commercial Crops, and is implemented in 28 states and 2 union territories. The mission helps with cluster demonstrations, seed distribution, farm machinery, water application tools, and farmer training, covering about 80,000 hectares under food grain technology demonstrations from 2014-15 to 2019-20.

The NFSM focuses on improving seed replacement rates and varietal replacement by distributing mini kits of the latest varieties free of charge. Provides financial assistance for the adoption of improved technologies, high-yielding varieties, and better farm practices. The mission also focuses on enhancing productivity and sustainability in rice farming.

3. Rashtriya Krishi Vikas Yojana

Initiated in 2007, the Rashtriya Krishi Vikas Yojana (RKVY) is an umbrella scheme designed to ensure the holistic development of agriculture and allied sectors in India. It allows states to select their agricultural development activities in line with district and state plans. Initially, it operated as an Additional Central Assistance scheme with 100% central funding, which changed to a Centrally Sponsored Scheme in 2014-15. Since 2015-16, the funding pattern has been adjusted to 60:40 between the Centre and states, with a 90:10 ratio for North Eastern and Himalayan states, and 100% central funding for Union Territories. The scheme incentivises states to increase public investment in agriculture by

providing them with the flexibility and autonomy to plan and execute projects based on their specific needs and agro-climatic conditions.

In November 2017, the Cabinet approved the continuation of RKVY as Rashtriya Krishi Vikas Yojana-Remunerative Approaches for Agriculture and Allied Sector Rejuvenation (RKVY-RAFTAAR) for 2017-18 to 2019-20 with a financial allocation of Rs. 15,722 crores. RKVY-RAFTAAR aims to make farming a profitable activity by strengthening farmers' efforts, mitigating risks, and promoting agri-business entrepreneurship. Funds are allocated across three main streams: Regular RKVY-RAFTAAR (70% of the annual outlay) focusing on infrastructure, value addition, and flexi funds; special sub-schemes (20%) targeting national priorities; and innovation and agri-entrepreneur development (10%) for fostering innovation and entrepreneurship through skill development and financial support.

Basically, the scheme aims to incentivize states to invest more in agriculture and allied sectors. Funds are provided to states for various agricultural development projects, including those focused-on rice cultivations. This initiative promotes the development of infrastructure, research, and innovation in the rice sector.

4. Pradhan Mantri Kisan Samman Nidhi

PM-KISAN is a central sector scheme launched on 24th February 2019, which aims to provide financial assistance to all landholding farmer families nationwide, subject to specific exclusion criteria, to help cover expenses related to agriculture, allied activities, and domestic needs. Under this scheme, Rs. 6000 per year is transferred directly to farmers' bank accounts in three instalments of Rs. 2000 each, every four months. So far, benefits worth ₹2.81 lakh crore have been provided to over 11 crore farmers under the scheme.

5. Pradhan Mantri Fasal Bima Yojana (PMFBY)

Launched in 2016, the Pradhan Mantri Fasal Bima Yojana (PMFBY) aims to provide a simple and affordable crop insurance product that offers comprehensive risk coverage to farmers. This scheme protects crops against all non-preventable natural risks from pre-sowing to post-harvest, ensuring adequate claim amounts. In the year 2022-23, PMFBY enrolled 1174.7 lakh farmer applications with an allocated fund of Rs. 15,500 crores. The scheme supports sustainable agricultural production by providing financial aid for crop loss, stabilising farmers' income, encouraging modern agricultural practices, and ensuring the flow of credit to the agriculture sector, thereby contributing to food security and crop diversification.

The PMFBY utilises an integrated IT solution and a web-based ecosystem to streamline service delivery, unify databases, and eliminate manual processes. This digital approach enhances the speed and efficiency of providing insurance services to farmers. The government is working to integrate all stakeholders, including farmers, insurance companies, financial institutions, and government agencies, on a single IT platform to improve administration, coordination, and transparency. The Crop Insurance portal has digitised notifications for areas, crops, and schemes, facilitating easier access for farmers, especially those in remote and economically disadvantaged regions, to avail crop insurance services. This secure and seamlessly integrated ecosystem ensures that multiple stakeholders, including farmers, government functionaries, and financial intermediaries, can access real-time information and monitoring.

13. Threats & Challenges to Issuers and their Products

1. **Pest and Disease Infestation** – Rice crops are highly prone to various pests and diseases that can cause substantial yield losses. Harmful pests such as the brown planthopper, rice hispa, stem borer, armyworm, and rice bug can damage crops, while diseases like rice blast, bacterial leaf blight, sheath rot, tungro virus, and false smut pose serious threats to production. Ineffective pest and disease control strategies can further worsen the situation, leading to lower harvests and financial strain on farmers. However, when it comes to storage, parboiled rice provides an edge over raw rice as it is less prone to insect infestation, making it easier to store and maintain with minimal losses.
2. **Climate Change and Weather Uncertainty** – The increasing global temperature due to greenhouse gas emissions has disrupted traditional weather patterns, delayed monsoons and altered rainfall distribution. These climatic shifts make rice farming more unpredictable, as sudden droughts, unseasonal rainfall, and extreme weather events negatively affect crop growth and yield. Farmers face mounting challenges in adapting to these rapid environmental changes.
3. **Declining Soil Fertility and Degradation** – The overuse of chemical fertilisers and poor agricultural practices has severely impacted soil health. The excessive application of fertilisers has led to increased soil salinity, gradually reducing its fertility and negatively affecting crop yield. The destruction of the natural microbial ecosystem further weakens soil quality, leading to long-term sustainability issues for rice farming.
4. **Nutrient Imbalance and Poor Soil Management** – To achieve high yields, rice plants require an adequate supply of nutrients from the soil. However, indiscriminate use of chemical fertilisers disrupts the soil's natural microbiome, affecting nutrient absorption. The depletion of essential nutrients leads to weaker plants, lower resistance to pests and diseases, and a decline in grain quality, ultimately reducing the economic value of the harvest.
5. **Fluctuating Market Prices and Rising Costs** – Farmers are often caught in a cycle of price instability, where rice prices fluctuate due to global demand, government policies, and market speculation. Additionally, the rising costs of essential agricultural inputs such as fertilisers, pesticides, labour, and irrigation further strain profitability. Without stable pricing and cost-effective solutions, many rice farmers struggle to sustain their livelihoods.
6. **Policy Gaps, MSP Challenges, and Dependence on Subsidies** – The effectiveness of the Minimum Support Price (MSP) system remains inconsistent, as many farmers face obstacles in accessing fair prices due to middlemen and inefficient procurement mechanisms. Moreover, the industry's dependence on government subsidies raises concerns about long-term financial sustainability. Frequent policy shifts also create uncertainty, affecting investment and production decisions.
7. **Shifting Consumer Preferences and Health Concerns** – Growing health consciousness is influencing dietary habits, with an increasing preference for alternative grains like millets, quinoa, and wheat. Additionally, concerns over pesticide residues and arsenic contamination in rice have driven demand for organic and chemical-free varieties. Traditional rice farmers face the challenge of adapting to these evolving consumer trends while maintaining productivity and profitability.
8. **Quality Concerns**- Ensuring consistent quality in the rice industry is a major challenge, as variations in grain size, moisture levels, and purity can hurt consumer trust and damage a brand's reputation. Risks like contamination from pesticides, heavy metals, or harmful microbes pose serious health concerns and can lead to strict regulations. Additionally, mixing lower-quality or different varieties of rice weakens product values and reduces and shakes customer confidence. Poor handling after harvest, such as inadequate storage or transportation, can cause spoilage, insect damage, or broken grains, all of which harm quality.
9. **Rising Competition-** Competition in the rice industry is increasing both in the domestic market as well as in the global market. With more local producers and brands entering the market, prices and profit margins are under constant pressure. At the same time, changing consumer tastes toward other grains and superfoods are cutting into rice demand. Competition has increased by eating into established brands, as retailers offer private-label and low-cost rice options. Companies with proper milling, packaging, and marketing techniques also get an upper hand.
10. **Geopolitical Risks –** Geopolitical factors pose significant challenges for the rice industry. Trade barriers like tariffs, quotas, and export bans can disrupt supply chains and limit access to important markets. Political

instability or sudden policy shifts in key rice-producing regions can also hamper production and exports. Fluctuations in currency values add uncertainty to pricing and profitability, while international sanctions may restrict trade and complicate business partnerships. Additionally, evolving climate change policies often require farmers and producers to adopt more sustainable, but sometimes costlier, practices, adding further pressure on the industry.

14. Road Ahead

From Farm to Grid: Potential of Indian Rice Industry to contribute to renewable energy solutions with rice straw and husk

The rice industry in India presents a significant opportunity to contribute to the country's renewable energy goals. Rice cultivation generates a substantial amount of agricultural residue, such as rice husk and straw (by-products of rice production), which can be utilised as feedstock for renewable energy production through processes like biomass gasification and combustion for power generation.

Rice straw and husk are widely used for various purposes in India, such as animal feed, fuel, as well as a raw material in the paper and packaging industries. By harnessing rice residue for renewable energy, the industry can play a pivotal role in reducing greenhouse gas emissions, reducing the country's dependence on fossil fuels, mitigating air pollution caused by open-field burning of crop residues, and promoting sustainable agricultural practices. Furthermore, integrating renewable energy production with rice cultivation can create additional revenue streams for farmers and enhance rural livelihoods. Therefore, the availability of these by-products has opened up opportunities for producing renewable energy in the form of biofuels.

Implementing initiatives to promote the adoption of renewable energy technologies in the rice industry, coupled with supportive government policies and incentives, can unlock the industry's potential to contribute significantly to India's renewable energy targets while fostering rural development and environmental sustainability.

The consumption of basmati and non-basmati rice in the domestic market is evolving due to modern trade, D2C initiatives, e-commerce, premiumization, branding, and celebrity endorsements. Consumers are increasingly opting for organised, branded options that promise quality, convenience, and authenticity, with modern trade and e-commerce making rice shopping more accessible. Premium basmati rice is gaining popularity as consumers seek higher-quality products, and branding plays a key role in differentiating these offerings. Celebrity endorsements further elevate brand visibility and appeal. Meanwhile, even non-basmati rice is seeing a rise in share from organised players, driven by innovation in packaging, quality, and consumer-focused marketing. This shift is reshaping the market towards more premium and branded rice choices, enhancing competition and consumer engagement.

Outlook

The future of the rice industry in 2025 will be shaped by policy reforms, technological advancements, and sustainable farming initiatives. Governments worldwide are working to ease trade regulations, stabilise exports, and adopt climate-resilient agricultural practices to strengthen food security and enhance market competitiveness. A key focus is on developing high-yield and disease-tolerant rice varieties to counter the impact of climate change. Sustainable farming methods such as direct-seeded rice (DSR), precision agriculture, and organic cultivation are gaining prominence to conserve water, improve soil fertility, and boost productivity.

In FY25, India's rice production was approximately 149 million tonnes, and basmati rice production was approximately 1 million tonnes. Out of which, nearly two-thirds of the Basmati rice sold domestically is still distributed in loose form. As per industry sources, only around 19% of the urban households opt for packaged Basmati rice, indicating significant untapped potential for growth within the packaged segment of the domestic market.

The broader trend of shifting consumer preference from loose to packaged rice, across both Basmati and non-Basmati categories, presents a great opportunity for growth and capturing the unorganised market. This shift is driven by rising demand for convenience, food safety, and consistent quality, especially among the rapidly growing urban Indian population. As migration towards urban cities continues, the demand for packaged rice is expected to rise simultaneously

The Indian rice market is projected to experience steady growth, with its value expected to rise from USD 55,103 million in CY25 to USD 59,754 million in 2030, reflecting a Compound Annual Growth Rate (CAGR) of 1.8% between CY25 and CY30. The growth is driven by increasing domestic consumption, robust export demand, and the adoption of advanced agricultural practices. While the growth rate remains moderate, ongoing investments in technology-driven farming, improved irrigation systems, and policy reforms aimed at ensuring farmer welfare and trade stability will be critical in sustaining the industry's momentum. A balanced approach integrating innovation, sustainability, and efficient market strategies will be key to ensuring long-term resilience and expansion in India's rice sector.

15. Competitive Landscape

Business Profile

Table 5: Peer business profiling

Peers	Installed Capacity (MTPA)	Distributors	Rice Exports - Value in Rs. Million	Export Countries	Domestic Sales - Value in Rs. Million	Advertisement Expense - in Rs. Mn	Plant Location
KRBL Limited	12,14,720	850+	13,310	90+	39,362	806.80	5 Plants Uttar Pradesh Punjab Haryana Delhi Gujarat
LT Foods Limited	7,99,350	1,400+	13,487	80+	18,554	1315.30	2 plants Haryana Punjab
Amir Chand Jagdish Kumar (Exports) Limited	5,50,800	425+	8,240	37	7,255	71.43	3 Plants Punjab Haryana Delhi
GRM Overseas Limited	4,40,800	270+	10,160	58	1,453	4.48	3 Plants Gohana Road, Haryana Naultha (Panipat), Haryana Gandhidham, Gujarat
Chaman Lal Setia Exports Limited	2,40,900	440+	12,080	90	1,476	2.29	1 Plant Karnal
Sarveshwar Food Limited	1,27,750	75+	79	45	791	0.33	3 Plants Seora, Jammu Bharmna, Jammu Gujarat

Source: Company Financials, CareEdge Research

Table 6: Player Profile

Product portfolio	KRBL Limited	Amir Chand Jagdish Kumar (Exports) Limited	LT Foods Limited	Sarveshwar Food Limited	GRM Overseas Limited	Chaman Lal Setia Exports Limited
Basmati	✔	✔	✔	✔	✔	✔
Non-Basmati	✔	✔	✔	✔		
Regional rice	✔	✔	✔	✔	✔	
Edible Oil					✔	
Atta		✔				
Besan		✔				
Maida		✔		✔		
Sooji		✔				
Salt		✔				
Sugar		✔		✔	✔	
Masala packs / Spices	✔			✔	✔	

Source: Company Financials, CareEdge Research

15.1 LT Foods Limited

LT Foods Ltd, established in the year 1978, is the leading processor of rice and other speciality foods in India. The Company has a presence in over 60 countries and over 1,300 distributors worldwide. The Company's key brands include 'Daawat' and 'Royal'. Apart from basmati, it also sells speciality rice (Arborio Rice, Jasmine Rice, Fortified Rice, etc.), organic food and ingredients, and health products. The total manufacturing capacity of the company is 3,00,000 MTPA, while operational capacity utilisation was recorded at 80% as of March 2023. Furthermore, the company plans to expand its product portfolio in the ready-to-cook division both domestically and internationally.

Table 7: Financials of LT Foods Limited

Financial indicators	FY22	FY23	FY24	9MFY25
Net Sales (Rs. Crores)	5427	6936	7772	6453
Operating profit (EBITDA) (Rs. Crores)	592	729	987	743
Operating margin (in %)	10.90%	10.50%	12.70%	11.52%
EBIT Margin (in %)	9.07%	9.30%	11.37%	10.35%
Net profit (Rs. Crores)	309	423	598	451
Net profit margin (in %)	5.70%	6.10%	7.69%	6.99%
Total debt (Rs. Crores)	1061	938	526	NA
Debt-to-Equity	0.49	0.34	0.15	NA
Current Ratio	1.78	1.96	2.08	NA
Return on Capital Employed (ROCE) (in %)	15.58%	18.78%	23.26%	NA
Return on Equity (ROE) (in %)	15.31%	17.10%	19.20%	NA
Return on Assets (ROA) (in %)	7.30%	8.73%	10.52%	NA
Debtor Days	37	34	32	NA
Creditor Days	65	68	79	NA

Inventory Days	154	143	154	NA
Working Cycle	126	109	107	NA

Source: Company Financials, CareEdge Research

15.2 KRBL Limited

KRBL Ltd, incorporated in 1889, is one of the largest rice millers in India with the combined capacity of 195 tonnes per hour for milling and 191 tonnes per hour for packaging as of March 2023. It is also a leading basmati rice manufacturer and exporter with shipments to more than 90 countries spanning across 6 continents. The company has established manufacturing facilities in 5 states, namely, Punjab, Gujarat, Delhi, Haryana and Uttar Pradesh, and it has over 800 distributors and 44,000+ retail outlets in India. The company has a wide range of products and is mainly into basmati rice (India Gate Classic Basmati Rice, Dubar Basmati Rice, India Gate Select Basmati Rice, etc) and non-basmati rice (India Gate Sona Masoori, India Gate Surti Kolam, India Gate Jeera Rice) products. They also have another segment among health food where a variety of products are crafted to address health concerns. These include Quinoa, Flax seeds, Chia seeds, etc. KRBL Ltd has plans to expand into the edible oils business by selling rice bran oil. Along with this, it aims to grow its premium non-basmati segment by establishing 2 processing facilities in Karnataka and Madhya Pradesh.

Table 8: Financials of KRBL Limited

Financial indicators	FY22	FY23	FY24	9MFY25
Net Sales (Rs. Crores)	4211	5365	5385	4152
Operating profit (EBITDA) (Rs. Crores)	662	939	802	435
Operating margin (in %)	15.72%	17.50%	14.90%	10.47%
EBIT Margin (in %)	14.97%	17.82%	15.23%	10.60%
Net profit (Rs. Crores)	459	701	596	322
Net profit margin (in %)	10.91%	13.07%	11.07%	7.75%
Total debt (Rs. Crores)	89	201	507	NA
Debt-to-Equity	0.02	0.04	0.10	NA
Current Ratio	7.38	6.35	5.40	NA
Return on Capital Employed (ROCE) (in %)	16.57%	22.19%	16.10%	NA
Return on Equity (ROE) (in %)	11.83%	16.00%	12.48%	NA
Return on Assets (ROA) (in %)	9.80%	13.56%	10.35%	NA
Debtor Days	21	20	20	NA
*Creditor Days	24	14	10	NA
Inventory Days	251	238	293	NA
Working Cycle	248	244	303	NA

Source: Company Financials, CareEdge Research

*Note: Creditors days is calculated using COGS due to limited disclosures

15.3 Amir Chand Jagdish Kumar (Exports) Limited

Amir Chand Jagdish Kumar (Exports) Ltd, established in 2003, is one of India's leading producers and exporters of basmati rice. Operating under the flagship brand Aeroplane Rice, the company also markets a diverse portfolio of rice varieties through other in-house brands such as La-Taste, Alibaba, World Cup, Hani, Palm Tree, Budallah, Sophia, and Al Jazira. The company ranks 3rd among the peer set in terms of revenue, it is also among the few Indian branded rice players that have ventured into FMCG staples. With a robust installed production capacity of 550,800 MTPA, Amir Chand Jagdish Kumar (Exports) Ltd has the 3rd largest installed production capacity as compared to select players. Amir Chand Jagdish Kumar (Exports) Limited is a key player in the organised rice market, and with organised players slowly taking over the unorganised market, the company may also benefit from the changing trend going forward.

Operations and Infrastructure

The company is engaged in rice milling and processing, with modern manufacturing facilities located in Amritsar (Punjab) and Safidon (Haryana). These strategic locations offer several competitive advantages, including easy access to high-quality paddy, favourable procurement pricing, and lower logistics costs. Its state-of-the-art, fully automated rice milling and processing units produce a wide range of rice, including aromatic rice, sweet rice, long grain rice, medium grain rice, and rough (paddy) rice. These products are distributed across both domestic and international markets.

Global Presence

Amir Chand Jagdish Kumar (Exports) Ltd exports to over 37 countries, including key markets like Iraq, Saudi Arabia, Oman, Jordan, U.A.E, Palestine, Azerbaijan, Bahrain and South Africa, among other major rice-importing nations.

Distribution and Market Reach

The company has established a strong distribution network, comprising over 425 partners in India and more than 50 international partners.

Financial and Brand Strength

In terms of domestic performance, the company reported Rs. 7,255 million in revenue from domestic sales in FY24. They have a cost-effective branding strategy, with an advertising expenditure of Rs. 71.4 million, substantially lower than its peers.

Manufacturing Footprint

The company operates across three key locations: Punjab, Haryana, and Delhi, reinforcing its presence in major agricultural and commercial hubs.

Table 9: Financials of Amir Chand Jagdish Kumar (Exports) Limited

Financial indicators	FY22	FY23	FY24	9MFY25
Net Sales (Rs. Crores)	1220	1316	1550	1421
Operating profit (EBITDA) (Rs. Crores)	68	80	110	124
Operating margin (in %)	5.58%	6.06%	7.08%	8.70%
EBIT Margin (in %)	5.79%	5.65%	6.73%	8.46%
Net profit (Rs. Crores)	17	17	30	49
Net profit margin (in %)	1.41%	1.33%	1.96%	3.43%
Total debt (Rs. Crores)	725	668	778	NA
Debt-to-Equity	2.75	2.38	2.50	NA
Current Ratio	1.20	1.25	1.23	NA

Return on Capital Employed (ROCE) (in %)	7.72%	7.82%	10.41%	NA
Return on Equity (ROE) (in %)	6.62%	6.43%	10.27%	NA
Return on Assets (ROA) (in %)	1.57%	1.58%	2.56%	NA
Debtor Days	69	86	75	NA
Creditor Days	26	34	19	NA
Inventory Days	207	176	168	NA
Working Cycle	250	228	224	NA

Source: Company Financials, CareEdge Research
Note: FY21 is standalone financials

15.4 Chaman Lal Setia Exports Limited

The company was established back in 1974 in Amritsar, in Punjab and went on to become one of the largest manufacturers and exporters of basmati rice. Situated amidst the richest paddy fields at Amritsar in Punjab, Karnal in Haryana and Delhi, the company gets an eagle eye view over 140 grain markets spread over 4000 sq. kms of rice producing area. The company is engaged in the business of milling and processing basmati rice. The company has been involved in export operations since 1982 and was recognised as an export house by the Ministry of Commerce in 1989; at present, the company is a 'Star Export House'. It holds a strong nationwide presence along with its substantial presence in more than 83 countries and packaging private label brands in numerous markets around the world.

Table 10: Chaman Lal Setia Exports Limited

Financial indicators	FY22	FY23	FY24	9MFY25
Net Sales (Rs. Crores)	932	1387	1356	1128
Operating profit (EBITDA) (Rs. Crores)	95	159	162	108
Operating margin (in %)	10.21%	11.45%	11.94%	9.55%
EBIT Margin (in %)	10.06%	11.86%	12.13%	9.73%
Net profit (Rs. Crores)	65	118	116	78
Net profit margin (in %)	6.97%	8.48%	8.53%	6.95%
Total debt (Rs. Crores)	118	121	175	NA
Debt-to-Equity	0.29	0.20	0.25	NA
Current Ratio	3.64	4.27	3.80	NA
Return on Capital Employed (ROCE) (in %)	20.78%	27.31%	21.84%	NA
Return on Equity (ROE) (in %)	17.12%	23.12%	17.49%	NA
Return on Assets (ROA) (in %)	11.40%	16.35%	12.76%	NA
Debtor Days	57	43	48	NA
*Creditor Days	11	6	3	NA
Inventory Days	110	98	123	NA
Working Cycle	156	135	168	NA

Source: Company Financials, CareEdge Research
Note – Consolidated data is not available, *Creditors days is calculated using COGS due to limited disclosures

15.5 GRM Overseas Limited

The group was established as a Partnership Firm in 1974 as "Garg Rice & General Mills". In 1995, the company was incorporated into a public company with the name of GRM Overseas Limited. The company is engaged in the milling and processing of basmati rice with an installed capacity of 4,40,800 metric tonnes per annum at Panipat, Haryana. During the initial years, GRM exported rice to the Middle East, the United Kingdom and the United States. The main products which the company deals in are domestic and international Basmati rice, spices, classic atta and the biryani kit. Gradually expanding its reach, GRM has developed a market for its rice in more than 38 countries.

Table 11: GRM Overseas Limited

Financial indicators	FY22	FY23	FY24	9MFY25
Net Sales (Rs. Crores)	1134	1379	1312	1057
Operating profit (EBITDA) (Rs. Crores)	93	98	72	47
Operating margin (in %)	8.17%	7.12%	5.50%	4.44%
EBIT Margin (in %)	11.16%	7.77%	7.70%	6.20%
Net profit (Rs. Crores)	85	63	61	41
Net profit margin (in %)	7.45%	4.56%	4.63%	3.86%
Total debt (Rs. Crores)	338	413	393	NA
Debt-to-Equity	1.64	1.50	1.18	NA
Current Ratio	1.37	1.45	1.67	NA
Return on Capital Employed (ROCE) (in %)	29.63%	17.55%	14.43%	NA
Return on Equity (ROE) (in %)	49.46%	26.10%	19.98%	NA
Return on Assets (ROA) (in %)	15.63%	8.66%	7.81%	NA
Debtor Days	106	107	123	NA
Creditor Days using COGS	23	23	21	NA
Inventory Days	49	67	74	NA
Working Cycle	132	151	176	NA

Source: Company Financials, CareEdge Research

15.6 Sarveshwar Foods Limited

Sarveshwar Foods Ltd, founded in 1890, has a rice processing and packaging capacity of more than 350 MT per day. The Company has operations in over 25 countries. It serves customers in several Indian states, including Jammu and Kashmir, Punjab, Haryana, Rajasthan, Uttar Pradesh, Gujarat, Madhya Pradesh, West Bengal, Maharashtra, Telangana, Andhra Pradesh, Karnataka, Kerala, and Tamil Nadu. The company has two state-of-the-art Buhler German Rice milling plants in Jammu with parboiling plants, grain storage silos, warehousing capacity of 3,00,000 Sq. Ft. with ultra-modern packaging plants on 24 acres of land. The main products which the company deals in are Indian traditional Basmati rice (Sarveshwar Traditional Basmati Rice, Sarveshwar Ultra Long Basmati Rice) and non-basmati rice. They have also launched new millet-based products with an emphasis on organic farming. The company has plans to expand its global footprint over time. Recently, it acquired Green Point Pt. Ltd, a Singapore-based firm, for USD 31,39,471 to increase its international presence.

Table 12: Financials of Sarveshwar Foods Limited

Financial indicators	FY22	FY23	FY24	9MFY25
Net Sales (Rs. Crores)	605	689	870	787
Operating profit (EBITDA) (Rs. Crores)	28	32	56	55
Operating margin (in %)	4.66%	4.71%	6.42%	7.00%
EBIT Margin (in %)	4.68%	5.48%	7.12%	7.23%
Net profit (Rs. Crores)	3	8	17	18
Net profit margin (in %)	0.56%	1.13%	1.93%	2.33%
Total debt (Rs. Crores)	265	287	297	NA
Debt-to-Equity	1.63	1.34	1.18	NA
Current Ratio	1.65	1.77	1.74	NA
Return on Capital Employed (ROCE) (in %)	6.64%	8.14%	11.79%	NA
Return on Equity (ROE) (in %)	2.08%	4.14%	7.19%	NA
Return on Assets (ROA) (in %)	0.67%	1.44%	2.46%	NA
Debtor Days	128	104	76	NA
Creditor Days using COGS	125	46	124	NA
Inventory Days	140	131	133	NA
Working Cycle	143	189	85	NA

Source: Company Financials, CareEdge Research

15.7 KPIs Comparison

Peers	LT Foods Ltd				KRBL Ltd				Amir Chand Jagdish Kumar (Exports) Ltd			
	FY22	FY23	FY24	9MFY25	FY22	FY23	FY24	9MFY25	FY22	FY23	FY24	9MFY25
Net Sales (Rs. Crores)	5427	6936	7772	6453	4211	5365	5385	4152	1220	1316	1550	1421
Revenue Growth (%)	14.46%	27.79%	12.06%	-	88.32%	117.23%	4.51%	-	10.72%	7.82%	17.76%	-
Operating Margin (in %)	10.90%	10.50%	12.70%	11.52%	15.72%	17.50%	14.90%	10.47%	5.58%	6.06%	7.08%	8.70%
Net Profit Margin (in %)	5.70%	6.10%	7.69%	6.99%	10.91%	13.07%	11.07%	7.75%	1.41%	1.33%	1.96%	3.43%
Total Debt (Rs. Crores)	1061	938	526	NA	89	201	507	NA	725	668	778	819
Debt-to-Equity	0.49	0.34	0.15	NA	0.02	0.04	0.10	NA	2.75	2.38	2.50	2.23
Current Ratio	1.78	1.96	2.08	NA	7.38	6.35	5.40	NA	1.20	1.25	1.23	1.23
Return on Capital Employed (ROCE) (in %)	15.58%	18.78%	23.26%	NA	16.57%	22.19%	16.10%	NA	7.72%	7.82%	10.41%	10.84%
Return on Equity (ROE) (in %)	15.31%	17.10%	19.20%	NA	11.83%	16.00%	12.48%	NA	6.62%	6.43%	10.27%	14.37%
Return on Assets (ROA) (in %)	7.30%	8.73%	10.52%	NA	9.80%	13.56%	10.35%	NA	1.57%	1.58%	2.56%	3.41%

Source: Company Financials, CareEdge Research

Peers	Chamanlal Setia Exports Ltd				GRM Overseas Ltd				Sarveshwar Foods Ltd			
	FY22	FY23	FY24	9MFY25	FY22	FY23	FY24	9MFY25	FY22	FY23	FY24	9MFY25
Net Sales (Rs. Crores)	932	1387	1356	1128	1134	1379	1312	1057	605	689	870	787
Revenue Growth (%)	9.51%	48.78%	-2.29%	-	41.83%	21.62%	-4.86%	-	17.49%	13.99%	26.15%	-
Operating Margin (in %)	10.21%	11.45%	11.94%	9.55%	8.17%	7.12%	5.50%	4.44%	4.66%	4.71%	6.42%	7.00%
Net Profit Margin (in %)	6.97%	8.48%	8.53%	6.95%	7.45%	4.56%	4.63%	3.86%	0.56%	1.13%	1.93%	2.33%
Total Debt (Rs. Crores)	118	121	175	NA	338	413	393	NA	265	287	297	NA
Debt-to-Equity	0.29	0.20	0.25	NA	1.64	1.50	1.18	NA	1.63	1.34	1.18	NA
Current Ratio	3.64	4.27	3.80	NA	1.37	1.45	1.67	NA	1.65	1.77	1.74	NA
Return on Capital Employed (ROCE) (in %)	20.78%	27.31%	21.84%	NA	29.63%	17.55%	14.43%	NA	6.64%	8.14%	11.79%	NA
Return on Equity (ROE) (in %)	17.12%	23.12%	17.49%	NA	49.46%	26.10%	19.98%	NA	2.08%	4.14%	7.19%	NA
Return on Assets (ROA) (in %)	11.40%	16.35%	12.76%	NA	15.63%	8.66%	7.81%	NA	0.67%	1.44%	2.46%	NA

Source: Company Financials, CareEdge Research

The strategic presence of the Amir Chand Jagdish Kumar (Exports) Ltd.'s (ACJKEL) rice mills in the regions gives a competitive advantage in terms of easy availability of paddy, lower freight, and favourable pricing terms. During FY24, the company achieved second second-highest y-o-y growth of about 18% amongst its peers and has demonstrated a CAGR of approximately 13% between FY22 and FY24.

On account of efficient utilisation of resources, the company has been able to generate a consistently improving return ratio in ROCE and ROE, for the last three Fiscals as compared to peers. This suggests that the company is effectively utilising its capital and equity to generate profits. This may be due to strong management practices, efficient cost control, and effective resource allocation.

Formulas used

Parameter	Formula
Revenue	Revenue from Operations
COGS	Cost of Material Consumed + Purchase Stock in Trade +/- Changes in inventories of finished goods, stock-in-trade and work-in-progress
EBITDA	Profit Before Tax + Depreciation & Amortization + Finance Cost – Other Income
EBIT	Profit Before Tax + Finance Cost
Debt	Long-term Borrowings + Short-term Borrowings
Cash	Cash and cash equivalents + Bank
EBITDA Margin	EBITDA/ Revenue from operations
EBIT Margin	EBIT/ Revenue from operations
PAT Margin	Profit after Tax/ Revenue from operations
Total Equity	Equity Share Capital + Other Equity reserves & surplus + Noncontrolling Interest
Capital Employed	Total Equity + Long term Debt + Short Term Debt – Cash and Cash equivalent
Debt to Equity	Debt/ Total Equity
Return on Equity (ROE)	PAT/ Average Equity
Return on Asset	PAT/ Average Total Asset
Return on Capital Employed (ROCE)	EBIT/ Average Total Capital Employed
Current Ratio	Current Assets/ Current Liabilities
Debtor Days	365/ (Revenue from Operations/Average Trade Receivables)
Creditor Days using Net Purchases	365/ (Net Purchases/Average Trade Payables)
Creditor Days using COGS	365/ (COGS/Average Trade Payables)
Inventory Days	365/ (Revenue from Operations/Average Inventory)
Working Cycle Days	Debtors Days + Inventory Days – Creditors Days

Contact

Tanvi Shah	Director – Advisory & Research	tanvi.shah@careedge.in	022 6837 4470
Nitu Singh	Associate Director	nitu.singh@careedge.in	+91 - 22 - 6837 4400

CARE Analytics and Advisory Private Limited

(Wholly-owned subsidiary of CARE Ratings Ltd.)
303B, B wing Times Square' Building Andheri - Kurla Rd, Gamdevi, Marol, Andheri East, Mumbai, Maharashtra 400059
Phone: +91-22-68374400

Connect:

About:

CareEdge is a knowledge-based analytical group offering services in Credit Ratings, Analytics, Consulting and Sustainability. Established in 1993, the parent company CARE Ratings Ltd (CareEdge Ratings) is India's second-largest rating agency, with a credible track record of rating companies across diverse sectors and holding leadership positions in high-growth sectors such as BFSI and Infra. The wholly-owned subsidiaries of CareEdge Ratings are (I) CARE Analytics & Advisory Private Ltd previously known as CARE Risk Solutions Pvt Ltd, and (II) CARE ESG Ratings Ltd, previously known as CARE Advisory Research and Training Ltd. CareEdge Ratings' other international subsidiary entities include CARE Ratings Africa (Private) Limited in Mauritius, CARE Ratings South Africa (Pvt) Ltd, and CARE Ratings Nepal Limited.

Disclaimer:

This report is prepared by CARE Ratings Limited (CareEdge Ratings). CareEdge Ratings has taken utmost care to ensure accuracy and objectivity while developing this report based on information available in public domain. However, neither the accuracy nor completeness of information contained in this report is guaranteed. CareEdge Ratings is not responsible for any errors or omissions in analysis / inferences / views or for results obtained from the use of information contained in this report and especially states that CareEdge Ratings has no financial liability whatsoever to the user of this report.